ABSTRACT

Buildings account for approximately 40% of total energy consumption in the United States, with HVAC systems representing the single largest energy end-use. Advanced control strategies such as model predictive control (MPC), reinforcement learning (RL), and differentiable predictive control (DPC) have demonstrated significant energy savings in individual studies, yet their widespread adoption remains limited by the lack of scalable, control-oriented dynamic models, the absence of consensus on control strategy selection, and the gap between single-zone research and multi-zone real-building deployment.

This dissertation addresses these barriers through an integrated research program spanning physics-informed machine learning (PIML) model development, advanced control evaluation, and scalable multi-zone deployment. For single-zone applications, a physics-consistent input convex neural network (PCICNN) is developed that embeds temporal causality through partial connectivity, ensures input convexity for optimization, and enforces physical consistency through architecture design and loss function augmentation. The PCICNN achieves prediction accuracy within sensor error bounds while maintaining physically consistent behavior under perturbed inputs. Using this model, the first long-term real-building comparison of MPC, RL, and DPC is conducted over five months in the same building, revealing that RL provides the most robust savings (up to 48%) across varying conditions, MPC achieves the highest peak savings (over 50%) but suffers from information gaps and non-optimal decisions, and DPC offers the fastest inference but the lowest overall savings (30.6%). A dual-phase benchmarking framework further establishes PCICNN as the most control-oriented PIML architecture among three representative models evaluated in both prediction and closed-loop control settings.

For multi-zone applications, a physics-consistent graph neural network (PCGNN) replaces manual adjacency-matrix neuron connections with group-shared multi-scale causal convolutions and heat diffusion graph layers, enabling scalable multi-zone temperature prediction. A κ-neighborhood truncated critic design for multi-agent RL reduces critic input dimensionality from scaling with the total number of zones to scaling with the local neighborhood size. The framework is validated through a 6-zone simulation and an 18-zone real-building deployment achieving 35–70% energy savings, representing one of the largest real-building multi-agent RL deployments for HVAC control in the literature.

As a complementary exploration, a feasibility assessment using the WHO's comprehensive airborne infection risk model examines whether HVAC-based clean air delivery warrants inclusion as a primary control objective. The analysis demonstrates that when infector presence is known, behavioral interventions outperform ventilation by 3–10 times, making ventilation a supportive rather than a dominant strategy; when unknown, ventilation is the only continuously available strategy and serves as the primary defense.

This research contributes to the field through: 1) a control-oriented PIML architecture (PCICNN) for single-zone building dynamics; 2) the first long-term real-building comparison of three advanced control strategies; 3) a dual-phase PIML model benchmarking framework; 4) a scalable PIML architecture (PCGNN) for multi-zone dynamics; 5) a scalable model-based multi-agent RL framework with neighborhood-truncated critics validated through real-building deployment; and 6) an exploratory evidence-based assessment of HVAC's role in infection risk mitigation.

SCALABLE ADVANCED CONTROL OF BUILDING HVAC SYSTEMS USING PHYSICS-INFORMED MACHINE LEARNING

by

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To my wife, Xintian, and my father in heaven.

致谢

我想了很久，不知从何写起。在逐渐接近毕业的日子里，我常幻想自己会是怎样的心情。回想这一路，我猜我大概会哭得很厉害——除了喜悦，更多是因为这一路并不容易。当我完成这篇毕业论文时，泪水确实在眼眶里打转；但答辩当天，我的心情反而无比平静，就像人生中再平常不过的一天。而今天，当我真正坐下来写致谢时，鼻子和眼睛又是酸酸的。

为什么会有这样的波动呢？我想要从小时候讲起。每当父亲问我的理想，我都说想成为工程师或发明家。虽然当时并不真正理解这两个词的含义，但冥冥之中似有注定——在求学的道路上，我一直朝着理工科的方向前进。除了这份天注定的浪漫说辞，还有一部分原因：我在娇生惯养下不谙世事，对文科一窍不通。凭借着偏科和一些小聪明，我度过了愉快的中学阶段。但一切都在高中时父亲确诊脑部胶质瘤的那一刻改变了。十七岁丧父，对很多人来说都是巨大的打击，对我也是如此。但对我而言，更深刻的影响是让我意识到，我需要像一个男人一样去面对生活，而不是像从前那样吊儿郎当。改变的过程是痛苦的。在社会化的道路上没有父亲引路，本应精彩的本科阶段充满了迷茫。在这种不安与对应试教育的不适应中，我并没有取得很好的成绩，最终只拿到了雪城大学的录取。我一直坚信，从最底层的原理出发去学习，才能到达顶峰。这种坚持在本科阶段并未奏效，但幸运的是，它让我在雪城大学名列前茅。突如其来的优异成绩让我不由得飘飘然，于是在 2020 年研究生春季学期结束后的暑假，我给董老师发了一封不同寻常的邮件：我对学习仍有激情，所以我想读您的博士。这封没有附带简历、看起来狂妄的套磁邮件当然适得其反。董老师

击碎了我的自负，但还是给了我进组的机会，并最终录取了我。这样看来，董老师似乎比我更有勇气，录取了毫无经验的我。科研的道路并不容易，不扎实的本科基础让我费了很大力气才做出一些成果。每当遇到挑战，我都会想：能怎么样呢？这难道比丧父时还要参加高考更难吗？难道会比本科时的迷茫痛苦更深吗？在这样的自我追问下，我的发表刊物一步一个台阶地上升。但人总是秉性难移，在这个过程中我不免又有自负的时候。虽然董老师并没有对我重拳出击，但每当我自负，同组的朋友姜子信就会在我的舒适领域做出更好的成果。这种良性的同侪压力鞭策着我，让我在挑战更高极限的同时保持清醒的认知，最终完成了这篇毕业论文。当然，这个过程中少不了董老师的鞭策与支持。我对董老师的态度，也从仰慕到质疑，再到理解，最终又回到了仰慕。我想，这就是教育的意义吧。

我一直在想，父亲的去世对我意味着什么。这一路有很多迷茫和痛苦，我曾把它们统统归结为父亲的缺失。博士最后一年，我去了哈佛大学交换。房东是一对来自中国的母女。有一天房东告诉我，她的女儿进了一个实验室，但由于沟通不畅，被老板指责了。我本以为以她们优渥的条件，女儿会甩担子不干，但让我吃惊的是，她快速消化了负面情绪，并用更加努力的实际行动化解了误会。我很钦佩这种品质。相处中我也见识过母女的激烈争论，以及女儿表现出的一些偏执。这些都让我有一种熟悉又陌生的感觉。临别时，房东告诉我，她的女儿在十几岁时就丧父了。那一瞬间，我全都理解了。我明白了为什么她会有与年龄和家境不符的韧性，也明白了那些偏执。那种熟悉又陌生的感觉，或许就是从前的我。我也终于明白，迷茫和痛苦带给了我一些特殊的品质。"天将降大任于斯人也，必先苦其心志"——这是父亲为了鼓励骨折的我坚持上学时引用的典故。我想，我的韧性便是从这些"苦其心志"中获得的吧。父亲最后留给我的这句话，或许就是他的离去对我的意义。

就像波士顿房东的女儿一样，我也有偏执的一面，这使我并不好相处。这种偏执，我觉得更像是对创伤的自我保护，但带刺的保护也伤害了一些亲近的人。有时我会从梦中惊醒，泪流满面。我的妻子张鑫恬始终陪伴在我身边，让我的不稳定稳定下来。我会对宠物米粒和比利发脾气，但他们始终对我亲近。我的导师和同组的朋友们，在我展现出偏执的时候包容了我。我的答辩委员会和合作者给予了我很大的支持。我的母亲、姥姥姥爷、姨父姨妈，更是给予了我各方面的托举。这来自四面八方的关爱，让我能在博士的最后阶段于这篇致谢中说出自己的心结。在答辩通过的这一刻，我想我已经实现了当年对父亲说过的那个理想。我想，我已经成为了一个更好的自己——因为有你们。

ACKNOWLEDGEMENT

I thought about this for a long time, not knowing where to begin. As I gradually approached graduation, I often imagined what I would feel. Looking back on this journey, I guessed I would probably cry hard — not just from joy, but more because this road was never easy. When I finished this dissertation, tears did well up in my eyes; but on the day of my defense, I felt unexpectedly calm, as if it were just another ordinary day. And today, as I truly sit down to write these acknowledgments, my nose and eyes sting again.

Why these emotional fluctuations? I think I need to start from childhood. Whenever my father asked about my dreams, I always said I wanted to become an engineer or an inventor. Though I did not truly understand what those words meant at the time, it seems fate had already decided — throughout my education, I kept moving toward science and engineering. Beyond this romantically fatalistic narrative, there was another reason: sheltered and spoiled, I knew nothing of the world and was hopeless at anything in the humanities. Riding on this lopsided aptitude and some quick wits, I had a pleasant time through middle school. But everything changed the moment my father was diagnosed with a brain tumor during high school. Losing a father at seventeen is a devastating blow for anyone, and it was for me too. But for me, the deeper impact was the realization that I needed to face life like a man, rather than coasting as I had before. The process of change was painful. Without a father to guide me through growing up, what should have been a vibrant undergraduate experience was filled with confusion. Amid this anxiety and my poor fit with exam-oriented education system, I did not achieve strong grades, and in the end received only an admission from Syracuse University. I have always believed that learning from first principles is the only way to reach the top. This conviction did not pay off during my undergraduate years, but

fortunately, it allowed me to excel at Syracuse University. The sudden academic success made me buoyant, so in the summer after my spring semester of graduate coursework in 2020, I sent Prof. Dong an unusual email: I still have passion for learning, so I want to pursue a PhD under your supervision. This audacious cold email, sent without even attaching a résumé, naturally backfired. Prof. Dong shattered my arrogance, but still gave me a chance to join the group and ultimately admitted me. In hindsight, perhaps it was Prof. Dong who showed more courage than I did, taking a chance on someone with no research experience. The path of research was not easy; my weak undergraduate foundation meant I had to expend great effort before producing any meaningful results. Whenever I faced a challenge, I would think: how bad can this be? Is this harder than taking the college entrance exam while grieving my father's death? Could this be more painful than the confusion of my undergraduate years? Driven by this self-questioning, my publications rose steadily, one step at a time. But old habits die hard, and I inevitably became arrogant again along the way. Although Prof. Dong never came down on me hard, whenever my ego swelled, my labmate and friend Zixin Jiang would produce better results right in my comfort zone. This healthy peer pressure spurred me on, keeping me clear-headed even as I pushed toward higher limits, and ultimately enabled me to complete this dissertation. Of course, none of this would have been possible without Prof. Dong's constant push and support. My attitude toward Prof. Dong evolved from admiration, to doubt, to understanding, and finally back to admiration. I believe that is what education is meant to be.

I have long wondered what my father's death truly meant for me. Along this journey there was much confusion and pain, and I once attributed it all to his absence. In the final year of my PhD, I went to Harvard University as a visiting researcher. My landlord was a mother and daughter from China. One day the mother told me her daughter had joined a lab but was reprimanded by her

supervisor due to a miscommunication. I assumed that, given their comfortable circumstances, the daughter would simply quit. But to my surprise, she quickly processed the negative emotions and resolved the misunderstanding through redoubled effort. I deeply admired this quality. During our time together, I also witnessed fierce arguments between mother and daughter, and a certain stubbornness in the daughter. All of this gave me a feeling that was at once familiar and foreign. When we parted, the mother told me her daughter had lost her father as a teenager. In that instant, I understood everything. I understood why she possessed a resilience that belied her age and background, and I understood the stubbornness. That familiar yet foreign feeling — perhaps it was a reflection of who I once was. I also finally understood that confusion and pain had given me some extraordinary qualities. "Heaven, when it is about to place a great burden on a man, must first harden his resolve" — this was the allusion my father cited to encourage me to keep attending school with a broken bone. I believe my resilience was forged from these trials of hardened resolve. This last message my father left me is perhaps the meaning of his passing.

Like my Boston landlord's daughter, I too have a stubborn side, which makes me difficult to get along with. This stubbornness, I think, is more of a self-protective response to trauma, but this thorny armor has also hurt those close to me. Sometimes I wake from dreams with tears streaming down my face. My wife, Xintian Zhang, has always been by my side, steadying my instability. I have lost my temper at my pets Milly and Billy, yet they have always stayed close to me. My advisor and my labmates embraced me even when my stubbornness showed. My dissertation committee and collaborators gave me tremendous support. My mother, my maternal grandparents, and my aunt and uncle lifted me up in every way. This love from all directions allowed me, in the final stretch of my PhD, to lay bare what has long weighed on my heart in these acknowledgments.

At the moment my defense was passed, I believe I have fulfilled the dream I once told my father. I believe I have become a better version of myself — because of all of you.

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Chapter 1. Introduction

This chapter establishes the foundation for the dissertation by presenting the background and rationale that drove this investigation. Section 1.1 introduces the dual challenge of building energy consumption and indoor environmental quality, including the heightened importance of the energy-health nexus following the COVID-19 pandemic. Section 1.2 then articulates the three primary research gaps that this dissertation aims to address, spanning physics-informed modeling, advanced control strategy evaluation, and multi-zone scalability, as well as a complementary exploration of airborne infection risk mitigation through HVAC systems.

1.1 Buildings at the Intersection of Energy and Health

Buildings are a cornerstone of modern society, yet they are also one of the largest contributors to energy consumption and environmental impact. In the United States, the building sector accounts for approximately 40% of total primary energy consumption and 74% of electricity usage [1]. Within this sector, heating, ventilation, and air conditioning (HVAC) systems represent the single largest energy end-use, consuming more than 410 billion kWh annually [2] and generating 16% of all U.S. carbon dioxide emissions [3]. The operation of building energy systems therefore presents both a significant challenge and a tremendous opportunity for improving energy efficiency and reducing greenhouse emissions [4].

A substantial portion of building energy consumption is attributable to suboptimal controls. Research has shown that 30% of building energy is wasted [5], with 15% directly attributable to poor control strategies [6]. One of the fundamental reasons for such waste is that the indoor environment is influenced by numerous factors, including outdoor temperatures, solar radiation, occupant behavior, and thermal inertia, yet most conventional control strategies react only to current sensor readings without anticipating future conditions [7]. Different operation control methods may lead to energy consumption differences of up to 65% [8], underscoring the critical importance of control optimization for building energy systems.

Simultaneously, indoor environmental quality (IEQ) is of paramount importance to human health, well-being, and productivity. People spend more than 90% of their time indoors [9], making thermal comfort and indoor air quality (IAQ) essential considerations in building operation. To maintain acceptable IEQ, indoor temperatures must be kept within comfortable ranges, and indoor contaminant concentrations must remain below established thresholds [10].

The COVID-19 pandemic has fundamentally transformed the understanding of indoor air quality and its relationship to public health. Since the World Health Organization declared COVID-19 a global pandemic in March 2020, growing evidence has established that SARS-CoV-2 transmits predominantly through inhalation of virus-laden aerosols in enclosed indoor spaces [11]-[15], driving urgent efforts to develop effective mitigation strategies for indoor environments. Extended time indoors during the pandemic intensified public attention to the indoor environment, leading to more stringent ventilation recommendations from CDC and EPA [16][17]. This heightened awareness has created a new research frontier: the energy-health nexus, in which building controls must simultaneously address energy efficiency and occupant health.

At their core, energy efficiency and indoor environmental quality present competing objectives. Increasing ventilation to improve IAQ requires additional energy for conditioning the outdoor air, while reducing energy consumption by minimizing ventilation may compromise air quality. Advanced control strategies that can intelligently balance these dual objectives are urgently needed but remain under-deployed in practice. This tension motivates the development of sophisticated, data-driven control approaches that can navigate the complex trade-offs between energy and health in real building environments.

1.2 Motivation

Advanced control strategies for building HVAC systems, including model predictive control (MPC), reinforcement learning (RL), and differentiable predictive control (DPC), have individually demonstrated 10–70% energy savings across various building types and HVAC configurations in research studies. Yet the adoption of these strategies in practice remains remarkably limited. Only few real-building deployments of data-driven advanced controls have

been reported in the literature, and most operate on a single thermal zone for short evaluation periods. The gap between research demonstrations and widespread deployment is not due to a lack of interest, but rather to a set of interconnected technical barriers that must be addressed systematically. This dissertation is motivated by the need to overcome these barriers and advance physics-informed machine learning and advanced control from laboratory-scale proof-of-concept to scalable, real-building deployment.

Lack of scalable and control-oriented building dynamic models: The first and most fundamental barrier is the lack of scalable, control-oriented building dynamic models. Advanced controls universally depend on dynamic models, whether for online optimization in MPC, offline training in RL, or differentiable backpropagation in DPC. Yet developing such models remains prohibitively labor-intensive: researchers have reported that model development for MPC alone typically takes 79 person-days, accounting for 33% of the total implementation effort. Physics-based models require extensive domain expertise, while purely data-driven models lack the physical consistency needed for reliable control under operating conditions beyond their training distribution. Physics-informed machine learning (PIML) models have emerged as a promising alternative, but existing PIML architectures are predominantly recurrent, have not been benchmarked against one another, and have rarely been validated in closed-loop real-building control. Without a PIML model that is simultaneously accurate, physically consistent, computationally efficient, and scalable to multi-zone buildings, advanced control strategy can be challenging to be deployed at scale.

Lack of consensus on different advanced control strategies: The second barrier is the absence of consensus on which advanced control strategy to deploy in practice. MPC, RL, and DPC each carry distinct theoretical advantages: MPC offers optimality guarantees through online

optimization, RL provides model-free deployment with low online computational cost, and DPC enables fast inference through offline-trained neural control laws. However, these advantages come with correspondingly distinct limitations in computational cost, robustness, training requirements, and implementation complexity. Existing comparison studies are almost exclusively simulation-based and have produced conflicting conclusions, with some finding RL inferior to the baseline and others finding it comparable to MPC. These inconsistencies arise from different assumptions, building types, and evaluation criteria, leaving practitioners without reliable guidance. A long-term comparison in the same real building, using the same dynamic model and evaluation metrics, is essential to resolve these contradictions and provide actionable recommendations.

Lack of multi-zone real-building deployment: The third barrier is the substantial gap between single-zone research and multi-zone real-building deployment. Real commercial buildings comprise tens to hundreds of thermally coupled zones, yet the vast majority of advanced control studies consider only a single zone or a small number of thermally independent zones. For single-agent RL, the joint state-action space grows exponentially with the number of zones, creating a curse of dimensionality that degrades learning efficiency and control performance. Multi-agent RL (MARL) offers a natural decomposition by assigning one agent per zone, but existing MARL studies for building control face their own scalability limitations: centralized critics that become unwieldy with many agents, model-free training that ignores available dynamics knowledge, and near-total absence of real-building validation. On the modeling side, extending PIML from single zone to multi-zone has only been demonstrated for up to 6 thermally coupled zones. Both the dynamic model and the control framework require fundamental redesign to bridge the gap from research-scale demonstrations to building-scale deployment.

A complementary question: the role of HVAC in airborne infection risk: Beyond the three primary barriers above, the COVID-19 pandemic raised a complementary question: whether indoor infection risk should be incorporated as an additional objective in advanced building control. Widespread recommendations to maximize ventilation and clean air delivery were largely based on simplified Wells-Riley models that may overestimate ventilation's relative effectiveness. With the availability of comprehensive epidemiological frameworks from the World Health Organization, this dissertation includes an exploratory assessment to determine whether HVAC-based interventions justify their energy cost relative to behavioral measures such as masking and distancing.

This dissertation addresses these three barriers through an integrated research program. On the modeling front, it develops a family of physics-consistent neural networks progressing from single-zone (PCICNN) to scalable multi-zone (PCGNN) architectures, with systematic benchmarking against competing PIML models. On the control front, it provides the first long-term real-building comparison of MPC, RL, and DPC, and subsequently develops a scalable multi-agent RL framework with neighborhood-truncated critics validated through real-building deployment at up to 18 zones. As a complementary exploration, it conducts a WHO-model-based feasibility assessment examining the role of HVAC-based clean air delivery in airborne infection risk mitigation. Together, these contributions aim to advance the field from isolated single-zone demonstrations toward practical, scalable building energy optimization.

Chapter 2. Literature Review and Contribution

This chapter provides a comprehensive literature review examining the current state of research across four areas central to this dissertation: physics-informed machine learning for building dynamics modeling, advanced control strategies for building HVAC systems, the scalability challenge of extending control from single-zone to multi-zone buildings, and the effectiveness of HVAC-based interventions for airborne infection risk mitigation. Each section systematically identifies research gaps in the existing literature, establishing the foundation and justification for the contributions of this dissertation.

2.1 Physics-Informed Machine Learning for Building Dynamics

The effectiveness of advanced building control strategies depends fundamentally on the quality of its underlying dynamic model. Whether the control approach is model predictive control (MPC), reinforcement learning (RL), or differentiable predictive control (DPC), a model of indoor environment dynamics is essential either for online optimization, offline agent training, or both. The development of such models has been a persistent bottleneck for the widespread adoption of advanced building controls.

Traditional building dynamic models can be categorized as white-box, grey-box, and black-box [18]. White-box models, such as EnergyPlus and TRNSYS, are developed based on physical laws including heat transfer, energy balance, and mass balance. While they induce the most physical detail and can predict indoor dynamics accurately, they require substantial domain expertise and computational resources that are impractical for real-time control applications [19]. For control-oriented applications, grey-box models, particularly resistance-capacitance (RC) network models, have been the predominant choice. These models leverage simplified physics to describe building thermal dynamics with relatively few parameters [20][21]. However, RC models have well-documented limitations: buildings are fundamentally different from each other, and the types of HVAC systems also vary significantly. Parameters of physics-based models vary greatly across buildings and require domain expertise to develop, even with auto-tuning processes [7]. Researchers have reported that the model development process for MPC typically takes 79 person-days, accounting for 33% of the total effort required for implementation and operation [19]. This substantial modeling burden impedes the adoption of model-based predictive control on a large scale.

As more buildings incorporate building automation systems (BAS) and IoT sensor technologies, data-driven (black-box) models have attracted growing interest. Once provided with good-quality data, these models can predict indoor dynamics with high accuracy while requiring minimal domain-specific expertise [22]. However, most data-driven models merely capture statistical correlations between inputs and outputs, without necessarily learning the underlying causal relationships or physical laws governing building dynamics [23]. As a result, purely data-driven models often exhibit physically inconsistent behaviors. For example, predicting indoor temperature to increase when additional cooling is applied. Such inconsistencies, while perhaps rare during interpolation, become particularly problematic in the control-relevant regime where the model must generalize for operating conditions beyond its training distribution.

Achieving high prediction accuracy, minimal development effort, swift computation, and consistent physical fidelity remain unattainable with any single traditional modeling approach. In pursuit of these combined objectives, researchers have increasingly integrated prior physical knowledge into data-driven models, leading to the emergence of physics-informed machine learning (PIML). As articulated by [24], there are three principles of physics-informed learning: observational biases, inductive biases, and learning biases. Architecture modifications that embed physical structure correspond to inductive biases, while physics-informed loss functions correspond to learning biases. These principles are not mutually exclusive, and hybrid approaches combining multiple biases have shown promise [25].

The use of PIML for indoor temperature dynamics modeling began around 2021, [26] integrated a state-space model structure into a neural network to model multi-zone thermal dynamics. This innovation has spurred a growing number of subsequent studies in the field. Despite this, it is still a nascent field with a limited number of studies. For some representative studies, [27] developed

an encoder-LSTM-decoder model with separate modules for unforced dynamics and energy accumulation. [28] deployed input convex neural networks to ensure that the resulting optimization problems remain convex. [29] proposed physics-constrained graph modeling for multi-zone building thermal dynamics. Despite these advances, most existing PIML models rely on recurrent architectures that are prone to gradient vanishing and explosion for long prediction horizons [26]. According to [28], as the prediction horizon increased from 1 hour to 6 hours, the mean squared prediction error increased from less than 0.2°C to more than 1°C. More diverse model architectures remain underexplored.

Beyond the diversity in model structures, most PIML studies on indoor temperature dynamics have focused primarily on the modeling itself, with their applications in advanced control remaining underexplored. As shown in Table 1, which summarizes recent PIML studies for indoor temperature modeling, 14 of the 22 studies did not apply the developed model to control implementation. Among those that did real building implementation, apart from the studies comprising this thesis [49][61], only one involved real-building deployment.

Table 1. review of PIML models for indoor temperature dynamics modeling. Comm. – commercial, Res. – residential.

Study	PIML Approach	Building Type	No. of Modeled Zones	Control Implementation
[47]	Structure	Lab	3	-
[48]	Hybrid	Lab	1	-
[49]	Hybrid	Lab	1	Real Building
[7]	Structure	Lab	1	-
[26]	Hybrid	Comm.	20	-
[27]	Structure	Lab	1	-

[50]	Structure	Lab	1	-
[51]	Loss	Comm.	1	-
[23]	Loss	Comm.	1	-
[52]	Hybrid	Lab	5	-
[53]	Hybrid	Lab	1	Simulation
[54]	Loss	Lab	1	-
[55]	Hybrid	Res.	3	Simulation
[56]	Hybrid	Comm.	4	Simulation
[29]	Hybrid	Comm.	6	-
[57]	Hybrid	Comm.	2	Simulation
[58]	Loss	Res.	1	-
[25]	Structure	Lab	6	Simulation
[59]	Structure	Comm. + Lab	5	Simulation
[60]	Structure	Res.	1	-
[61]	Hybrid	Comm.	6	Real Building
[62]	Hybrid	Lab	1	Real Building

In summary, while PIML models represent a significant advance over purely physics-based and purely data-driven approaches, several gaps remain:

- **Limited diversity of PIML architecture**: most of existing studies using recurrent network structure, which could suffer from gradient vanishing and high computational cost. More diverse model structures of PIML should be explored.
- **No benchmark of PIML models**: existing PIML models have not been systematically benchmarked against one another under consistent conditions, making it difficult for practitioners to select the most appropriate architecture for a given application.
- **Lack of control implementations**: most critically, the vast majority of PIML studies stop at the modeling stage without demonstrating their utility in closed-loop control, let alone real-building deployment.

These gaps motivate the development and evaluation of control-oriented PIML architecture in this dissertation, as well as the systematic comparison of PIML models in both prediction and real-building control settings.

2.2 Advanced Control Strategies: MPC, RL, and DPC

Among advanced control strategies for buildings, model predictive control is the most extensively studied. As its name suggests, MPC requires plant models to solve optimization over a prediction horizon. Many studies have experimentally demonstrated MPC with energy saving capabilities. [63] implemented gray-box MPC to optimize the energy usage of a solar house equipped with radiant floor systems and achieved more than 30% energy saving. [64] implemented gray-box MPC in an airport with air handler units and demonstrated 13% of energy reduction. [65] used adaptive MPC controllers in residential buildings and achieved 28% energy reduction. [66] used MPC to control AHU with variable airflow volume (VAV) systems, in which 20% of energy saving was achieved. [22] implemented black-box MPC in an office building and achieved 58% energy saving at maximum. Depending on different building and HVAC systems, the energy saving from MPC varies from 10% to more than 70% [28] [30][67]-[72].

Depending on whether optimization is solved online, MPC can be categorized as implicit or explicit [31]. Implicit MPC, which solves optimization online, requires more computational resources but can handle large systems. Explicit MPC, which derives feedback control laws offline, requires less online computation but scales poorly [73]. The weakness of implicit MPC is not obvious with gray-box models as the resulting linear programming is well developed, but due to the nonlinear nature of building dynamics and black-box models, which significantly increases the online computational burden, explicit MPC is also being paid attention to. To address the

scalability of the explicit MPC, using neural networks as control laws in building controls has proposed and developed [32][42][43][74]. Both black-box implicit MPC and explicit MPC with neural control law result in data-driven MPC. To differentiate them, we refer to the former as data-driven predictive control (DDPC) and the latter as differentiable predictive control (DPC). Differentiable predictive control bridges MPC and RL by training neural control laws offline using differentiable plant models. The control law network provides control sequences, and the PIML model provides environmental responses for training. Upon implementation, only the trained neural network is used for decision-making, requiring milliseconds rather than minutes for online computing. As differentiable PIML models can be used in training, DPC is supposed to have less offline training time. Despite these promising advantages of DPC, only limited number of studies used it for building control, and only [74] tested it in the real building.

Different from MPC which relies on the predictive models for decision-making, RL agents sample actions and learn the optimal actions by observing and interacting with the environment, which leads to model-free decision-making. Such interactions are described by the Markov process [75]. For better scalability, the process and the agent are typically represented by neural networks [76]. For the application of RL in building controls, most research focused on simulation studies [33][34] [35][77][78][79][80]. For some examples, [81] used tabular Q-learning and batch Q-learning to control photovoltaic-thermal modules in a low-exergy building and achieved 10% of energy saving. [36] used soft actor-critic (SAC), proximal policy gradient (PPO), trust-region policy gradient (TRPO), and twin-delayed deep deterministic policy gradient (TD3) to control a simulated data center in EengyPlus. The results showed around 10% of energy saving. [82] used a deep Q network (DQN) to control a radiant floor system in a simulated testbed and achieved 19% energy saving. Inspired by fitted Q-iteration, [83] proposed PhyQ to train RL agents and achieve 9% energy saving.

As for experimental RL studies in real buildings, [84] proposed and implemented LightlLearn to control the lighting in an office building for occupant comfort. [37] used asynchronous advantage actor-critic (A3C) to control a radiant heating system in an office building and reduced heating demand by 16.7%. [38] used PPO to control an office with an AHU-VAV system and reduced the demand by 16.7%. [40] used deep deterministic policy gradient (DDPG) to control 3 AHUs and achieved 30% of energy saving. [39] used branching dueling Q-network (BDQ) to control the personalized environmental control system in an office and showed a 14% cooling energy reduction. [85] controlled the thermal storage system in a commercial building by DQN and reduced the cost by 7.6%. [41] tested the performance of an RL-based controller on a hybrid variable refrigerant flow system for 2 months. The RL-based controller showed 15–50% energy saving potential compared with a rule-based controller.

Across different studies in real buildings, MPC, RL, and DPC can reduce HVAC energy usage, but the reduction varies significantly even for the same approach, making direct comparison of different control methods difficult. A limited number of simulation-based comparison studies exist [31][44][45][46], but their conclusions are inconsistent, particularly regarding RL performance. In some studies, RL cannot even outperform the baseline control, whereas in others it is comparable to MPC. Such inconsistency may be attributed to simplifying assumptions in simulations, where perfect knowledge of disturbances may conceal the potential of RL and DPC. How different controls perform under the uncertain environment of real buildings provides critical insight but is notably lacking in the current literature.

In summary, MPC, RL, and DPC have each demonstrated energy savings for building HVAC control, but several gaps remain:

- **Inconsistent comparison results from simulations**: existing simulation-based comparisons yield conflicting conclusions, particularly regarding RL performance. Simplifying assumptions such as perfect disturbance knowledge may conceal or exaggerate the relative strengths of different strategies, making simulation-only comparisons unreliable for practical guidance.
- **Limited real-building implementations**: the majority of RL studies remain simulation-based, and DPC has seen only one real-building deployment. The practical challenges of real-world implementation are underexplored.
- **No real-building comparison**: no existing study has deployed MPC, RL, and DPC in the same real building over a sufficiently long period to enable a fair and comprehensive comparison under consistent conditions, dynamic models, and evaluation metrics.

These gaps motivate a long-term, real-building comparison of the three advanced control strategies in this dissertation, using the same physics-informed dynamic model and testbed to provide actionable guidance on control strategy selection.

2.3 The Multi-Zone Scalability Challenge

Single-agent RL has been extensively studied for building energy system control in recent years. Figure 1 (a) shows the number of zones and scale of some single-agent RL studies. Simulation studies have demonstrated 10-50% energy savings across a range of HVAC configurations, with the majority controlling only a single zone or two zones [77]-[79][33][34][36][80]-[85]. Experimental deployments have validated these savings on real buildings, with studies addressing up to 6 controlled zones [37]-[40][49][61][62][84]. Despite the success of single-agent RL for building system control, its application is still limited to 6 zones at maximum. The biggest

challenge is the curse of dimensionality: for a building with N zones, each with d_s state variables and d_a action variables, the joint state-action space scales as $O(d_s^N \times d_a^N)$. For example, a 20-zone building with 7 states and 1 action per zone yields a joint state space on the order of 10^{16}, a scale that requires significant amount of samples to learn an optimal control policy, which increase offline training time significantly and therefore is challenging for critic and policy networks to learn state and action values and optimal control. Besides the challenges from increased dimensions, using single-agent RL to control multiple systems and zones creates monolithic architecture in which the failure of any single component can compromise the entire system, thereby increase deployment complexity and reduce operational robustness.

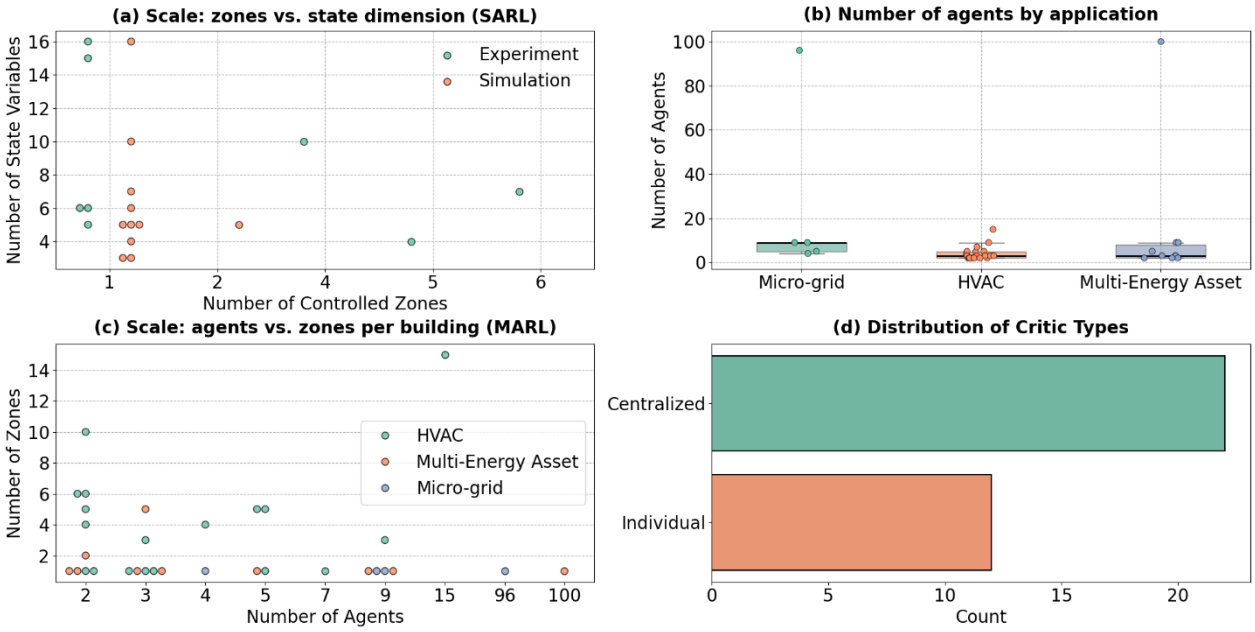


Figure 1. Single-agent (a) and multi-agent RL (b-d) for building energy systems.

Multi-agent reinforcement learning is naturally a good candidate for optimal building system control, where each zone or subsystem can be treated as an individual agent that learns a local control policy while coordinating with others. MARL has been studied extensively in recent years

for the control of building HVAC systems, multi-energy assets, and microgrids. Figure 1 (b) – (c) summarizes the key characteristics of 34 recent studies based on Table 2. MARL has been demonstrated to achieve 3.95% to 41.7% energy reduction for HVAC systems, improved self-sufficiency for multi-energy assets, and near-optimal performance for microgrid control. Despite these promising results, four common limitations can be identified that prevent scaling from scaling to larger, more complex buildings.

Limited scale of thermally coupled zones: Most existing MARL studies involve only 2–5 agents representing distinct subsystems such as an AHU with VAV zones, or a chiller and a battery. Among the 34 studies reviewed, 11 consider multi-zone indoor thermal dynamics, of which 9 model no more than 6 zones [89][90][91][94][96][100][110][115][116]. Only two studies address 10 and 15 thermally connected zones [105][114], respectively. Two additional studies scale to 96 and 100 agents [98][117], but in both cases each agent controls a separate building treated as a single node rather than multiple thermally coupled zones within one building. A key factor limiting this scaling is the modeling effort required to capture multi-zone dynamics. All reviewed MARL studies rely on physics-based simulation environments, where developing a detailed thermal model for each building demands extensive effort due to differences in building layouts, envelope properties, and HVAC configurations. Although physics-informed machine learning has emerged as a promising and more scalable alternative [120], existing studies applying this approach to building thermal modeling have only been demonstrated for up to 6 zones [25][47][59][61]. Extending such models to buildings with tens of zones remains an open challenge, which in turn limits the scale at which MARL can be deployed for coordinated multi-zone control.

Scalability of the critic architecture: Even when the number of zones is modest, the critic design poses a scalability concern. Among the 34 studies, 22 employ a centralized critic that receives the

concatenated states and actions of all agents. Usually, centralized training and decentralized execution (CDTE) are used. In this setting, the policy networks make control decisions based on local observations, while the global value is learned by critic networks using the concatenation of local observations. At the current scale of 2–5 agents with 3–15 state variables each for building applications, this concatenation yields critic inputs on the order of 20–50 dimensions, which is manageable for standard neural network function approximators. However, the input dimension grows with the number of agents. For a large building with hundreds of zones, the concatenated input to critic networks would reach thousands of dimensions. High-dimensional critic inputs degrade learning efficiency in two compounding ways: first, the number of samples required for accurate value function approximation grows exponentially with input dimension, as the critic must cover an increasingly high-dimensional space; second, the credit assignment problem becomes more severe, as the critic must disentangle each agent's contribution to the global return from a joint signal that mixes information from all zones, many of which have negligible influence on the agent in question. Together, these effects slow convergence and destabilize training, making centralized critics challenging for buildings with tens or hundreds of zones. The remaining 12 studies use independent per-agent critics that rely only on the local states of the agent itself. While this avoids the dimensionality problem, it sacrifices inter-agent coordination entirely: each agent optimizes its local objective without awareness of neighboring zones' states or actions. In thermally coupled buildings, this can lead to conflicting control decisions, for example, one zone increasing its heating load while an adjacent zone simultaneously cools, resulting in wasted energy and degraded comfort. In fact, the indoor dynamics of one zone is usually affected by its direct adjacencies, which can be leveraged to reduce the learning dimension of centralized critic

networks. However, none of the existing studies exploit this property to design a critic whose input dimension scales with the local neighborhood size rather than the total number of zones.

Underutilization of dynamic models: Beyond the critic design, how the agents learn also presents an efficiency gap. A widely recognized advantage of model-free RL is that it requires no explicit dynamics model, learning optimal policies directly from interaction with the environment. However, in real building control practice, directly training agents through interaction with the physical system is cost-prohibitive due to occupancy comfort requirements, seasonal constraints, and the risk of equipment damage from exploratory actions. Consequently, nearly all existing studies, 31 of 34, train their model-free agents in simulation environments built upon physics-based models that encode thermodynamic equations such as energy balance, heat conduction, and convective heat transfer. This creates a paradox: the simulation environment already contains an explicit dynamics model, yet the model-free algorithm treats it as a black box that only returns states and rewards after each interaction. The structural knowledge embedded in the governing equations, the causal relationships between control actions, disturbances, and temperature responses, is entirely discarded by the learning algorithm, which must rediscover these relationships through extensive trial and error. A model-based approach, by contrast, can internalize a learned dynamics model to generate synthetic trajectories, enabling agents to anticipate the consequences of their actions without additional environment interactions. This substantially reduces the sample complexity of policy learning and accelerates convergence—advantages that become even more significant when fine-tuning policies on real buildings, where every interaction is costly.

Lack of real-building validation: Finally, 33 of the 34 studies evaluate their MARL algorithms exclusively in simulation. While simulation is essential for safe and reproducible experimentation,

it cannot fully capture the disturbances, sensor noise, and occupant behavior encountered in practice. Real-building deployment of MARL for HVAC control remains rare, limiting the practical credibility of existing approaches.

Table 2. Representative studies of MARL for building systems from 2018 to 2025. S: Simulation. E: Experiment. MF: model-free. MB: model based. C: Centralized. I: Individual. Bldg: Building.

Ref.	Application	Type	No. of Bldg	No. of Zone	No. of Agent	Critic	States (Max)	Actions (Max)	M
[86]	Microgrid	S	NA	NA	5	C	13	7	MF
[87]	Microgrid	S	4	1	4	I	3	1	MF
[88]	HVAC/ Thermal	S	1	1	2	I	9	1	MF
[89]	HVAC/ Thermal	S	1	4	2	C	9	1	MF
[91]	HVAC/ Thermal	S	1	5	3	C	15	1	MF
[93]	HVAC/ Thermal	S	1	5	5	I	2	1	MF
[92]	HVAC/ Thermal	S	1	1	3	C	3	1	MB
[90]	HVAC + Storage	S	1	1	5	C	10	3	MF
[94]	HVAC/ Thermal	S	1	4	4	I	6	2	MF
[95]	HVAC + Storage	S	1	1	3	C	9	1	MF
[96]	HVAC/ Thermal	S	1	5	5	I	4	1	MF
[97]	HVAC + Storage	S	9	1	9	C	27	3	MF
[98]	HVAC + Storage	S	96	1	96	C	18	1	MF
[99]	Energy Storage	S	1	1	2	C	7	2	MF
[100]	HVAC + Storage	S	1	2	2	C	5	1	MF
[101]	Energy Storage	S	9	1	9	C	14	1	MF

[102]	HVAC/ Thermal	E	3	1	3	C	10	1	MF
[103]	HVAC/ Thermal	S	2	1	7	C	3	1	MB
[104]	Energy Storage	S	9	1	9	C	N/A	3	MF
[105]	HVAC/ Thermal	S	3	15	15	C	6	1	MF
[106]	HVAC + Storage	S	1	1	3	C	6	1	MF
[107]	HVAC/ Thermal	S	1	1	3	I	6	1	MF
[108]	HVAC + Storage	S	1	1	3	C	6	1	MF
[109]	HVAC + Storage	S	9	1	9	I	11	5	MF
[110]	HVAC/ Thermal	S	1	6	2	I	8	1	MF
[111]	HVAC/ Thermal	S	1	3	3	I	3	1	MF
[112]	HVAC/ Thermal	S	1	1	5	I	4	1	MF
[113]	HVAC/ Thermal	S	1	3	9	I	7	1	MF
[114]	HVAC/ Thermal	S	1	10	2	C	6	1	MF
[115]	HVAC/ Thermal	S	1	6	2	C	7	1	MF
[116]	Energy Storage	S	1	5	2	I	6	1	MF
[117]	Energy Storage	S	100	1	100	C	13	2	MF
[118]	HVAC/ Thermal	S	1	1	2	C	3	1	MB
[119]	HVAC/ Thermal	S	1	1	2	C	22	2	MF

2.4 Effectiveness of HVAC Control on Airborne Infection Mitigation

Beyond PIML models and advanced control for buildings, the COVID-19 pandemic intensified interest in mitigating its infection risk. Multiple mitigation strategies have been proposed, including engineering controls (ventilation, air cleaning), administrative measures (occupancy

limits, distancing policies), and personal protective equipment (masks, respirators). Among engineering controls, increasing clean air delivery—through enhanced outdoor air ventilation, improved filtration, and portable air cleaners—has been widely promoted as a primary mitigation strategy [121][122]. Public health agencies recommended maximizing outdoor air intake, upgrading filtration, and deploying portable HEPA cleaners. These strategies share a common mechanism: removing or diluting infectious aerosols, often unified under the concept of clean air delivery as an equivalent outdoor air ventilation rate [123].

Clean air delivery is intuitively appealing. By reducing airborne pathogen concentrations, it should lower inhalation exposure and infection probability. Classical Wells-Riley models predict monotonic risk decrease with increasing ventilation [124], leading pandemic-era guidance to emphasize "more clean air" as a universally effective solution [125][126].

However, early COVID-19 infection risk models employed significant simplifications that may overestimate ventilation's relative impact. Many studies used Wells-Riley-based models where ventilation was the only controllable variable [129][130], with other factors like masking and distancing either fixed or ignored. Proposed modifications on Wells-Riley model incorporated time-dependent emissions [130], respiratory tract clearance [131], biological decay [132], social distancing indices [133], and transient occupancy [134][135], but ventilation consistently remained the sole controllable parameter.

These models often oversimplified critical factors: constant breathing rates ignoring activity variations, neglected particle size distributions following evaporation, overlooked respiratory deposition fractions across nasopharyngeal, tracheobronchial, and alveolar regions, and assumed uniform aerosol emissions despite evidence that speaking and singing produce dramatically

different size distributions and rates than breathing [135]. Additionally, models frequently neglected short-range transmission via expiratory jets, which can dominate exposure at close distances before aerosols mix into room air.

COVID-19 transmission is multifactorial, involving biological, environmental, and behavioral variables. Airborne transmission occurs through short-range (within ~2 meters, dominated by expiratory jets) and long-range (beyond 2 meters, via well-mixed air) pathways [127]. Quantifying infection risk requires characterizing emission rate (respiratory activity, viral load, infector masking), removal rate (ventilation, air cleaning, decay, settling), exposure (occupancy, room volume, temporal dynamics), cumulative inhaled dose (deposition, viability, susceptible masking), and infection probability. Fixing other factors while varying only ventilation may exaggerate clean air delivery's impact in realistic settings where multiple factors vary simultaneously. This overestimation is problematic because increasing ventilation substantially increases heating and cooling loads, energy consumption, and operational expenses, particularly in extreme climates without energy recovery [128].

To address these limitations, the World Health Organization developed a comprehensive airborne infection risk framework integrating particle size distribution, environmental conditions (temperature, humidity), particle dynamics (emission, transport, deposition, decay), occupant characteristics, and behavioral factors including masking [127]. This standardized model enables systematic evaluation of mitigation strategies under realistic conditions. Despite widespread adoption of ventilation and air cleaning measures, a fundamental question remains regarding how much clean air delivery can realistically reduce airborne infection risk in operational commercial buildings, relative to other mitigation measures.

In summary, the effectiveness of HVAC-based clean air delivery for airborne infection risk mitigation remains insufficiently understood, with several gaps in the existing literature:

- **Over-reliance on simplified models**: the majority of existing studies used Wells-Riley-based models where ventilation was the only controllable variable, with critical factors such as particle size distribution, respiratory deposition, and short-range transmission either oversimplified or ignored entirely. These simplifications may systematically overestimate the relative impact of clean air delivery.
- **Lack of comprehensive, multi-factor evaluation**: COVID-19 transmission is inherently multifactorial, yet most studies fix behavioral and biological variables while varying only ventilation, preventing a fair assessment of its marginal contribution relative to other mitigation strategies such as masking and social distancing.
- **No rigorous validation with state-of-the-art models**: the WHO has developed a comprehensive airborne infection risk framework that integrates particle dynamics, environmental conditions, and behavioral factors. However, this framework has not been systematically applied to evaluate the practical controllability of infection risk through HVAC interventions in real building environments.

These gaps motivate an exploratory assessment in this dissertation, using the WHO infection risk model with real measured indoor environment data, to determine the conditional role of HVAC-based clean air delivery in airborne infection risk mitigation — specifically, whether ventilation serves as the primary available defense when infector presence is unknown, or a supportive complement to behavioral interventions when infector presence is identified.

2.5 Dissertation Contributions

Table 3 summarizes existing studies across the key dimensions addressed by this dissertation: physics-informed machine learning modeling, advanced control comparison, multi-agent scalability, model-based learning, and real-building implementation. Several patterns emerge from this comparison. Studies employing PIML models have been limited to single-zone or small-scale applications (six zones or fewer) and have not extended to multi-agent settings. Conversely, studies that adopt multi-agent formulations operate exclusively in simulation without leveraging physics-informed dynamics models. No prior work has systematically compared multiple advanced control strategies (MPC, RL, and DPC) in the same real building, and the largest multi-agent deployment reported in the literature involves 15 agents in a simulation environment. This dissertation is the first to unify all six dimensions — PIML modeling, control comparison, multi-agent scalability to 18 zones, model-based learning, and real-building implementation — within a single integrated research program.

Table 3. Comparison of existing studies on PIML modeling and advanced control for building HVAC systems.

Study	PIML	Control Comparison	No. of Agents	No. of Zones	Model-Based Learning	Implementation
[91]			3	5		
[93]			5	5		
[83]	✓		1	1		
[94]			3	1		✓
[92]			1	7	✓	
[105]			15	15		

[44]		✓	3	3		
[31]		✓	1	1		
[55]	✓		1	3		
[25]	✓		1	6		
[107]			3	1		
[62]	✓		1	1		✓
[61]	✓		1	6		✓
[114]			2	10		
[115]			2	6		
This Study	✓	✓	**18**	**18**	✓	✓

Based on the research gaps identified, this dissertation makes the contributions as shown in Figure 2, which are described as follows:

1. **A control-oriented PIML architecture for single-zone building dynamics**. A physics-consistent input convex neural network (PCICNN) is proposed, embedding temporal causality through partial connectivity inspired by the lower-triangular structure of multi-step state-space models, input convexity through non-negative weight constraints, and physical consistency through both architecture design and loss function augmentation. The PCICNN provides accurate multi-step predictions without recurrence, making it inherently suited for predictive control and RL agent training.
2. **The first long-term real-building comparison of MPC, RL, and DPC**. Three advanced control strategies: hierarchical data-driven predictive control (HDDPC), differentiable predictive control (DPC), and reinforcement learning (RL), are implemented in the same real building for more than five months using the same PCICNN dynamic model. This provides the first experiment-based, head-to-head comparison under consistent conditions,

revealing the practical trade-offs between optimality, robustness, computational cost, and deployment complexity that simulation-based comparisons cannot capture.

3. **The first dual-phase benchmarking framework for PIML models**. A systematic evaluation framework is developed that jointly assesses PIML models in both open-loop prediction (accuracy, physical consistency, training efficiency) and closed-loop real-building control (energy savings, comfort maintenance, equipment impact). Three representative PIML architectures: PCNN, PCICNN, and ModNN, are benchmarked under this framework, establishing PCICNN as the most control-oriented architecture and providing model selection guidelines for practitioners.
4. **A scalable PIML architecture for multi-zone building dynamics**. A physics-consistent graph neuron network (PCGNN) is developed that replaces manual adjacency-matrix neuron connections with group-shared multi-scale causal convolutions and heat diffusion graph mechanisms. The causal convolutional structure is mathematically equivalent to the lower-triangular Toeplitz matrix of multi-step state-space models but is significantly more parameter-efficient and scalable, enabling multi-zone temperature prediction beyond the 6-zone limit of prior PIML models.
5. **A scalable model-based multi-agent RL framework with neighborhood-truncated critics**. A κ-neighborhood truncated critic design is introduced for multi-agent soft actor-critic algorithm, where each agent's critic receives only the states and actions within κ-hop neighborhood rather than the full building state. This design exploits the exponential decay of thermal coupling with distance for multi-zone temperature dynamics, reducing the critic input dimension from scaling with the total number of zones to scaling with the local neighborhood size. Combined with model-based training using PCGNN, the framework is

validated at two scales: a 6-zone simulation study and an 18-zone real-building deployment achieving 35-70% energy savings, representing one of the largest real-building MARL deployments for HVAC control in literature.

6. **Exploratory assessment of HVAC's role in airborne infection risk mitigation**. As a complementary exploration, WHO's comprehensive airborne infection risk model is applied to real measured indoor environment data, a rigorous feasibility assessment is conducted through Monte Carlo simulation, generalized linear modeling, and machine learning-based sensitivity analysis. The study demonstrates that when the presence of an infector is known, behavioral interventions such as masking outperform HVAC-based clean air delivery by factors of 3–10. However, when infector presence is unknown, ventilation is the only continuously available mitigation measure, serving as the primary line of defense under conditions of uncertainty.

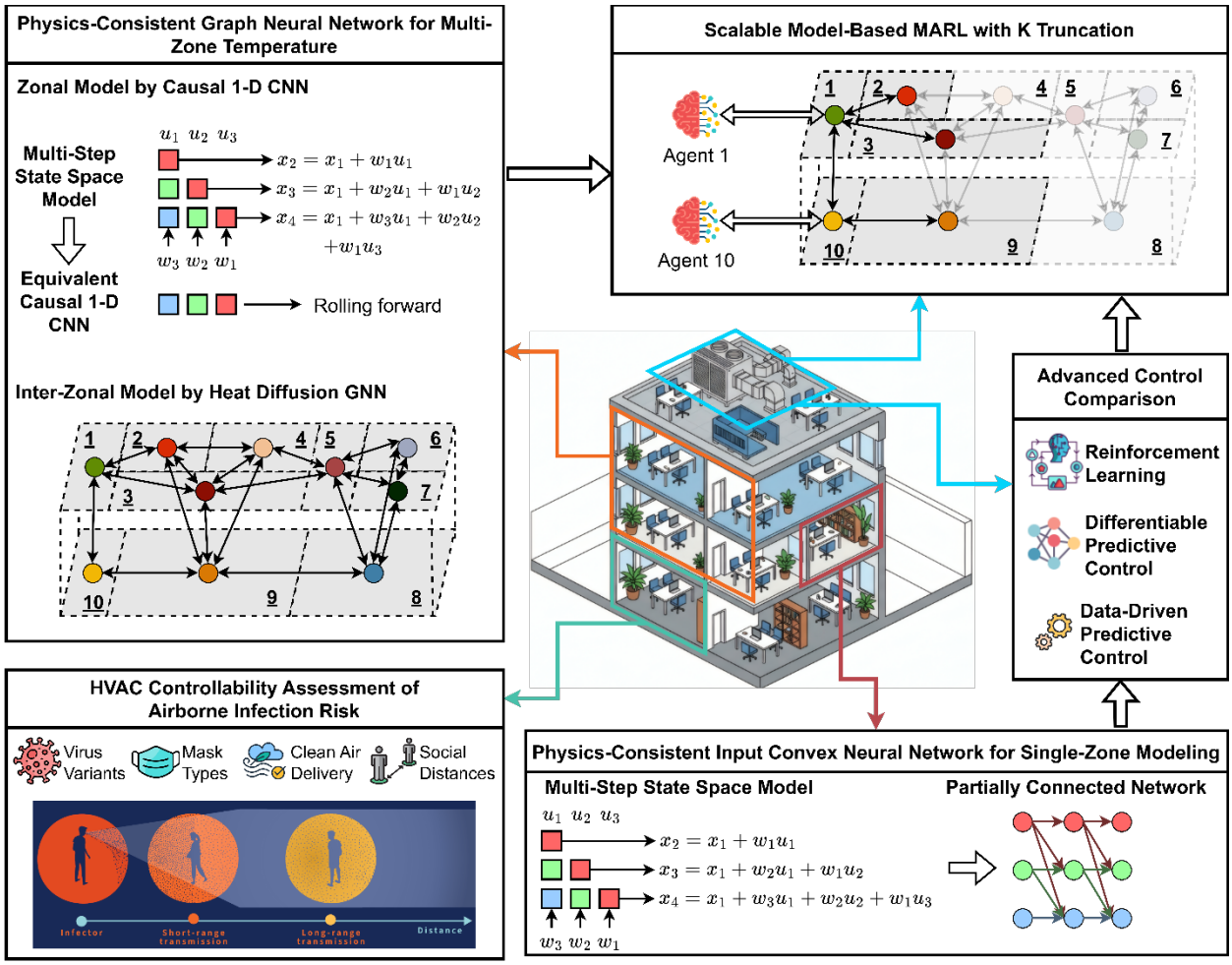


Figure 2. Overview of dissertation methods and contributions.

Chapter 3. Methodology I: PCICNN and Advanced Control for Single Zone

This chapter presents the methodology for single-zone building dynamic modeling and advanced control. Section 3.1 introduces the physics-consistent input convex neural network (PCICNN), a control-oriented dynamic model that embeds temporal causality and input convexity through architectural modifications inspired by multi-step state-space models. Sections 3.2 through 3.4 describe the three advanced control strategies built upon the PCICNN: hierarchical data-driven predictive control (HDDPC), differentiable predictive control (DPC), and reinforcement learning (RL), along with the weather and occupancy forecasting models that support them. Section 3.5 describes the simulation study designed to validate the PCICNN's prediction accuracy and physical consistency. Section 3.6 details the real-building experimental setup for the long-term comparison of the three advanced controls, including the testbed, baseline control logic, implementation workflow, and evaluation metrics. Section 3.7 presents the dual-phase evaluation framework for benchmarking the PCICNN against two representative PIML architectures in both prediction and closed-loop control settings.

3.1 Physics Consistent Input Convex Neural Network

As the physics-based model requires expertise to develop and vanilla black-box models were not designed for control purposes, we propose a control-oriented data-driven model inspired by the structure of the physics-based model.

A general form of the linear one-step state-space model is $x(t+1) = Ax(t) + Bu(t) + EW(t)$, where x, u, and W are state, controllable, and disturbance variables, respectively; A, B, and E are coefficient matrices. For predictive control purposes, the one-step state-space model is usually reformed as a multi-step state-space model:

$$x(t+1{:}\,t+h) = A_h x(t) + B_h u(t{:}\,t+h-1) + E_h W(t{:}\,t+h-1) \tag{1}$$

Where h is the prediction horizon; A_h, B_h, E_h and are obtained based on coefficient matrices of the one-step model and expressed as:

$$A_h = [A \quad A^2 \quad \cdots \quad A^h]^T \tag{2}$$

$$B_h = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{h-1}B & A^{h-2}B & \cdots & B \end{bmatrix} \tag{3}$$

$$E_h = \begin{bmatrix} E & 0 & \cdots & 0 \\ AE & E & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{h-1}E & A^{h-2}E & \cdots & E \end{bmatrix} \tag{4}$$

When solving the optimization problem in each control step, the multi-step state-space model, rather than the one-step state-space model, plays the key role of predicting indoor environmental dynamics. Therefore, this research seeks to use a black-box model as a surrogate of the multi-step state-space model.

Classical MLP models were not aimed for control purposes and lacked the physical meaning of model parameters. To address these drawbacks, modifications are needed. In multi-step RC model (equation (1)), the time correlation between inputs and outputs are guaranteed by the lower triangular structure of coefficient matrices (equations (3) and (4)). Furthermore, the elements along the main and parallel diagonals are the same. Exemplifying by temperature prediction, given a sequence of HVAC load $[u(t) \quad \cdots \quad u(t+h-1)]^T$, the prediction of temperature at any time steps only depends on the historical HVAC load sequence within the prediction horizon, e.g., for the prediction of temperature at $t+1$ time step, the effect of $u(t+1:t+h-1)$ is cancelled out by zeros in the first row of the matrix in equation (3); only $u(t)$ takes effect on the prediction of temperature at time step $t+1$. Moreover, given any prediction time steps, the coefficient corresponding to the impact of HVAC load at the previous, for example, one-time step, on the prediction of indoor temperature, which is the same as long as the step difference between input and output is the same. Because the coefficient represents temperature increase per unit load, which relates to indoor air properties and is assumed as time-invariant constant. Therefore, two constraints should be considered for neural network development. Firstly, the weight matrix of each layer should follow a lower triangular structure to guarantee the time correlation between inputs and outputs. Secondly, the control-input- and disturbance-related weight matrices should share the same elements along the main and parallel diagonals to represent air and envelope

properties. Such lower-triangular and shared weights result in a lower-triangular block Toeplitz matrix form. The modified neural network can be then formulated as follows:

For the input layer,

$$\begin{bmatrix} z_1(t) \\ z_1(t+1) \\ \vdots \\ z_1(t+h-1) \end{bmatrix} = g_1\left(w_1 \begin{bmatrix} x(t) \\ u(t) \\ \vdots \\ u(t+h-1) \end{bmatrix} \right) \tag{5}$$

$$w_1 = \begin{bmatrix} w_{1,t,x(t)} & w_{1,t,u(t)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ w_{1,t+h-1,x(t)} & w_{1,t+h-1,u(t)} & \cdots & w_{1,t+h-1,u(t+h-1)} \end{bmatrix} \tag{6}$$

Where $z_1(i)$ ($i \in [t, t+h-1]$) is the output of the first layer that contains the information up to the ith time step; $w_{1,i,u(j)}(i, j \in [t, t+h-1])$ is the weight related to the input at time step j for the output of $z_1(i)$; g_1 is the activation of the input layer.

For the same non-negative $i-j$, the weight should be the same, meaning for a fixed time interval, the unit effect from inputs should be the same on the prediction, e.g., $w_{1,t,u(t)} = w_{1,t+1,u(t+1)} = w_{1,t+2,u(t+2)} = \cdots$ and $w_{1,t+1,u(t)} = w_{1,t+2,u(t+1)} = w_{1,t+3,u(t+2)} = \cdots$.

For negative $i-j$, the weight should be zero, meaning the information of more than time step i will not be included for the prediction of $i+1$ step, e.g., $w_{1,t,u(t+1)} = w_{1,t,u(t+2)} = \cdots = 0$.

Compared to the coefficient matrices of multistep RC model, the weight matrix w_1 is similar to the concatenation of matrices (2) and (3). Note that the disturbance variables were not included in the formulation because they can be treated in the same way as control variables.

For hidden layers,

$$\begin{bmatrix} z_k(t) \\ z_k(t+1) \\ \vdots \\ z_k(t+h-1) \end{bmatrix} = g_k\left(w_k \begin{bmatrix} z_{k-1}(t) \\ z_{k-1}(t+1) \\ \vdots \\ z_{k-1}(t+h-1) \end{bmatrix} \right) \tag{7}$$

$$w_k = \begin{bmatrix} w_{k,t,z_{k-1}(t)} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ w_{k,t+h-1,z_{k-1}(t)} & \cdots & w_{k,t+h-1,z_{k-1}(t+h-1)} \end{bmatrix} \tag{8}$$

Where $z_k(i)$ is the output of the kth hidden layer that contains the information up to the ith time step. $w_{k,i,z_{k-1}(j)}$ is the weight related to the output from previous hidden layer; g_k is the activation function of the kth hidden layer. Note that the shared weights of the input layer also applies to the weights of hidden layers.

Similarly, for the output layer,

$$\begin{bmatrix} x(t+1) \\ x(t+2) \\ \vdots \\ x(t+h) \end{bmatrix} = g_o\left(w_o \begin{bmatrix} z_k(t) \\ z_k(t+1) \\ \vdots \\ z_k(t+h-1) \end{bmatrix} \right) \tag{9}$$

$$w_o = \begin{bmatrix} w_{o,t,z_k(t)} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ w_{o,t+h-1,z_k(t)} & \cdots & w_{o,t+h-1,z_k(t+h-1)} \end{bmatrix} \tag{10}$$

Where $x(i+1)$ is the prediction of state variable at the $i+1$th time step. $w_{o,i,z_k(j)}$ is the weight related to the output from the last hidden layer; g_o is the activation function of the output layer.

The final structure of the PCICNNs is shown in Figure 3.

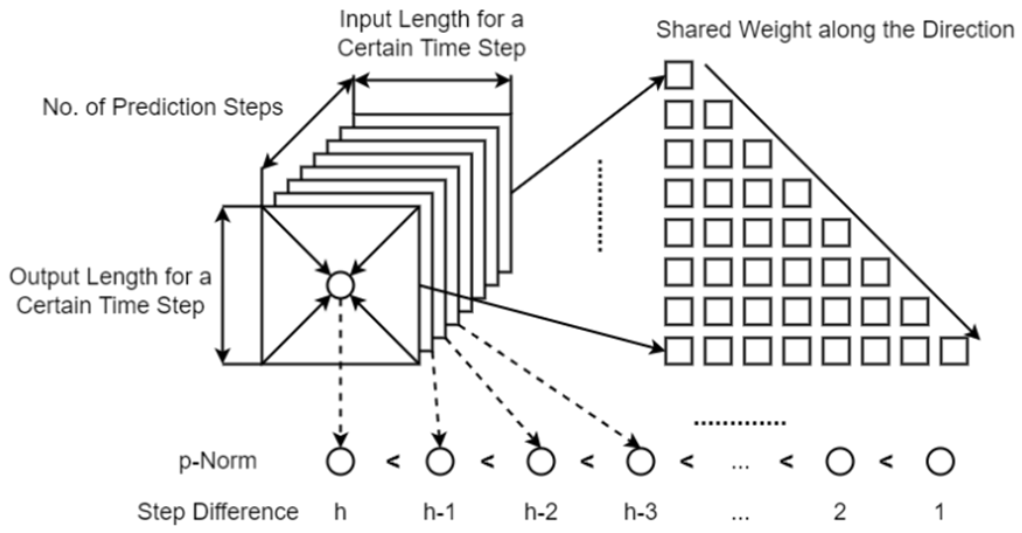


Figure 3. PCICNN model. The lower triangular matrix provides input-output time correlation. Shared weight forces PCICNN to learn physical properties. Norm penalty reinforces near-step attention.

In a compact form, the mathematical formulation of the PCICNN (physics-consistent input convex neural network) is as follows. For the input layer, the weight matrix is:

$$\boldsymbol{w}_1 = [\boldsymbol{w}_x \quad \boldsymbol{w}_u] \tag{11}$$

where $\boldsymbol{w}_x$ and $\boldsymbol{w}_u$ are weights related to state variable and input variables. For indoor environment dynamics, the state variables could be indoor temperature and CO2 concentration, and the input variables could be HVAC load and ventilation rate. More specifically, the input-variable related weights are expressed as:

$$\boldsymbol{w}_u = \begin{bmatrix} \boldsymbol{w}_1 & 0 & \cdots & 0 \\ \boldsymbol{w}_2 & \boldsymbol{w}_1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \boldsymbol{w}_{h-1} & \boldsymbol{w}_{h-2} & \cdots & \boldsymbol{w}_1 \end{bmatrix} \tag{12}$$

in which the shared weight is applied along the main and parallel diagonals. The state-variable related weights only have one step, so the $\boldsymbol{w}_x$ can be constructed similar to the first column of $\boldsymbol{w}_u$.

For the hidden layers, the weight matrices are in the same form as equation (12) but maybe with different dimensions.

Besides the structural modifications, the training loss function was modified as:

$$\ell = \ell_{MSE} + \rho_1 \ell_{norm} + \rho_2 \ell_{bias} \tag{13}$$

Where ℓ_{MSE} refers to mean squared error loss and ℓ_{norm} and ℓ_{bias} are formulated as:

$$\ell_{norm} = \sum_{i}^{h-2} \max(|\boldsymbol{w}_{i+1}|_p - |\boldsymbol{w}_i|_p, 0) \tag{14}$$

$$\ell_{bias} = \sum b \tag{15}$$

ℓ_{norm} forces the neural network to focus on the effect from the inputs at closer steps to the output. ℓ_{bias} penalizes large biases to avoid overfitting from vanishing weights. ρ_1, and ρ_2 are coefficients for balancing different loss terms.

In [24], there are three principles of physics-informed learning: 1) observational biases; 2) inductive biases; and 3) learning biases. The architecture modification and introduction of new loss terms belong to inductive biases and learning biases, respectively, which are two principles of physics-informed learning. Therefore, our proposed network belongs to a hybrid approach of physics-informed learning. As mentioned in [25], loss function augmentation and neural network architecture alteration are both variants of physics-informed networks, which reinforces the statement in [24] that different physics-inform learning principles towards physically consistent solutions are not exclusive. Besides, we further constrain the weights of the network to be non-negative, which is true if the parameters represent the building thermal property. With convex activation functions such as ReLU, the neural network is convex with respect to its input features. Therefore, the proposed network is a variant of the physics-informed learning model and is referred to as a physics-consistent input convex neural network.

3.2 Hierarchical Data-Driven Predictive Control

In MPC, heating or cooling load is typically optimized with indoor temperature maintained within a predefined range. When indoor air quality (IAQ) is taken into consideration, the coil-side and fan energy consumption should be optimized instead. From the supervisory level, i.e., the load itself is the optimal variable, the optimization can be quadratic and therefore convex if the indoor environmental model is convex. However, when the optimal variables are airflow rates and supply temperature, bilinear terms result in nonconvexity. Here, we formulate the problem as two levels of optimization problems. One level (room level) plans the best HVAC load trajectory, and another level (system level) follows the trajectory cost-effectively. Based on it, optimization of the room level was formulated as follows.

$$\min_{\mathcal{P}_{seq}} \sum_{i=1}^{h-1} \mathcal{P}_{seq}^2(i) \tag{16}$$

$$s.t. \underline{\mathcal{P}_{seq}} \leq \mathcal{P}_{seq} \leq \overline{\mathcal{P}_{seq}} \tag{17}$$

$$T_{seq} = \mathcal{N}_T(T_0, \mathcal{P}_{seq}) \tag{18}$$

$$\underline{T} \leq T_{seq} \leq \bar{T} \tag{19}$$

Where $\mathcal{P}_{seq}$ is the sequence of HVAC load and the control variable on the room level, T_{seq} is the trajectory of indoor temperature, $\mathcal{N}_T$ is PCICNN for indoor temperature dynamics, and lower and upper bars denote lower and upper bounds.

The optimization of the system level was formulated as below.

$$\min_{u} \sum_{i=1}^{h-1} \mathcal{P}_{sys}^2(i) + \sum_{i=2}^{h-1} \rho_3(u_{seq}(i) - u_{seq}(i-1)) \tag{20}$$

$$s.t.\ \underline{u} \leq u_{seq} \leq \bar{u} \tag{21}$$

$$c_{seq} = \mathcal{N}_c(c_0, \mathcal{Q}_{o_{seq}}) \tag{22}$$

$$\mathcal{P}_{sys} = f_1\left(T_{seq}, \mathcal{Q}_{o\,seq}, \mathcal{P}_{seq}\right) = c_p \rho \mathcal{Q}_{o\,seq}(T_o - T_{seq}) - \mathcal{P}_{seq} \tag{23}$$

$$\mathcal{P}_{seq} = f_3\left(\mathcal{Q}_{s_{seq}}, T_{s_{seq}}, T_{seq}\right) = c_p \rho \mathcal{Q}_{s_{seq}}\left(T_{s_{seq}} - T_{seq}\right) \tag{24}$$

$$\underline{c_{seq}} \leq c_{seq} \leq \overline{c_{seq}} \tag{25}$$

$$u = [\mathcal{Q}_s, \mathcal{Q}_o, T_s] \tag{26}$$

Where $\mathcal{P}_{sys}$ is system energy usage, u is optimal variable, c is IAQ index, $\mathcal{N}_c$ is PCICNN for IAQ, $\mathcal{Q}_s$ and $\mathcal{Q}_o$ are supply and fresh airflow rates, f_1 and f_3 refer to coil and HVAC load calculated based on specific heat c_p, air density ρ, supply and outdoor airflow rates, and temperature differences of room temperature from outdoor temperature T_o and supply air temperature T_s. Note that $\mathcal{P}_{seq}$ is the output from room-level optimization. In equation (20), the first term is to optimize the load on the system level, and the second term is to make the control sequence as smooth as possible. The coefficient ρ_3 is used to balance the importance of load and smooth operation. The control variables on the system level are supply airflow rates $\mathcal{Q}_s$ and outdoor airflow rates $\mathcal{Q}_o$. The supply air temperature T_s is implicitly affected by $\mathcal{Q}_s$ through equation (24).

The hierarchical formulation turns the original nonconvex problem into two quasi-convex problems. As the hierarchical formulation is used in this study, the term HDDPC is used hereafter rather than the general term DDPC.

3.3 Differentiable Predictive Control

One of the applications of neural networks is solving optimization. For an unconstrained problem, the objective function can be used as the loss function to train an agent-like neural network.

Whereas for constrained problems, relaxation can be used as additional terms in the loss function. For predictive control, the loss function can be formulated based on the formulation of HDDPC.

$$\ell = \rho_{obj}\ell_{obj} + \rho_{comf}\ell_{comf} + \rho_{op}\ell_{op} \tag{27}$$

Where ℓ_{obj} is the loss from objective and refers to equation (20), ℓ_{comf} is the loss for discomfort and refers to equations (19) and (25), ℓ_{op} is the loss for dangerous operations and refers to equations (17) and (21), and ρ_{obj}, ρ_{comf}, and ρ_{op} are coefficients balancing different loss terms. Both ℓ_{comf} and ℓ_{op} are quantified by bound violations. The training of the control law network is illustrated in Figure 4. The outputs of the control law network are supply airflow rate, OA flow rate, and supply air temperature. The control decision is made based on current state measurements and disturbance predictions such as weather and occupancy forecasts. The control decision is implemented into PCICNN for indoor environmental responses. Then the three loss terms can be calculated accordingly. Note that PCICNN is only used to train the control law network. Upon implementation, only the part highlighted by the red dashed box in Figure 4 is used to make control decisions.

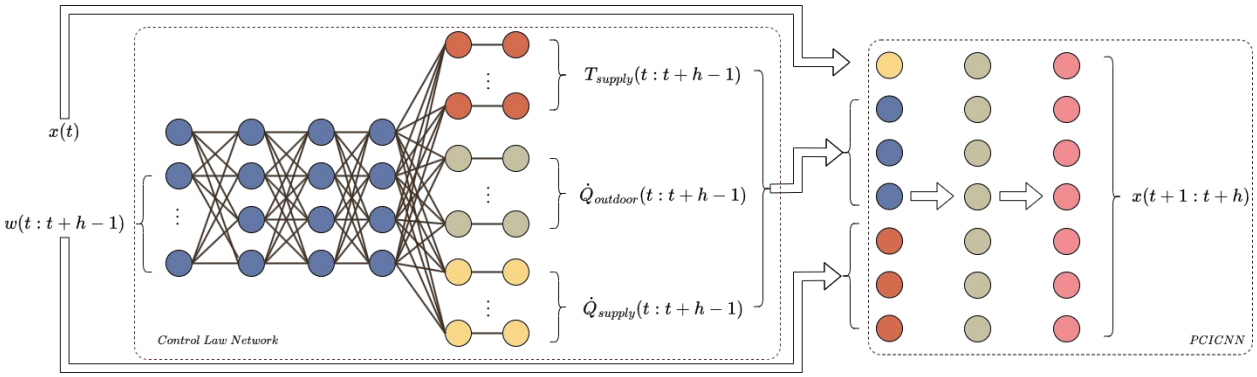


Figure 4. Differentiable predictive control with PCICNN. The control law neural network provides control sequences, and the PCICNN provides environmental responses.

3.4 Reinforcement Learning

Ideally, RL agent training does not require any models in the environment, as the agent can interact directly with the real environment, i.e., the HVAC system and indoor space. However, the exploration nature of the agent will potentially lead to destructive behaviors. Therefore, most agent training is in a simulation environment where the PCICNN can be involved.

The RL formulation is based on the Markov decision process, which is quite different from HDDPC and DPC and needs step-by-step interactions. The PCICNN therefore should be single-step-in and single-step-out, so the original sequence-to-sequence PCICNN should be changed. Thanks to our special structure of PCICNN grouping weights to represent different time steps, we only need to extract $\boldsymbol{w}_1$ in equation (12), $\boldsymbol{w}_x$, and 1-step related weights in hidden and output layers. A new PCICNN can then be assembled to provide single-step prediction. Such a reassemble is illustrated in Figure 5 where the numbered blocks represent weights in equation (12).

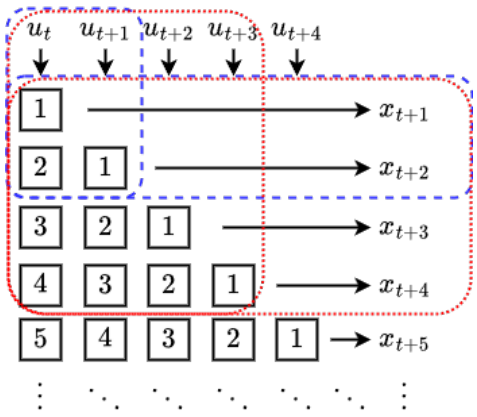


Figure 5. Reassemble weights for different prediction horizons.

The action space in our RL formulation is the same as DPC and HDDPC, which is continuous. Therefore, an actor-critic-based approach, specifically, soft actor critic (SAC), is used. The reason for using SAC is that it balances exploration and exploitation well. Because firstly it is an off-policy approach with replay buffer to record historical experience, which improves data efficiency.

Secondly, it uses the entropy-regularized Bellman target instead of gradually shrinking the exploration rate. The entropy-regularized Bellman equations are shown below:

$$V_\pi(s) = \mathbb{E}_\pi[Q_\pi(s,a)] + \alpha H\big(\pi(\cdot\,|s)\big) \tag{28}$$

$$Q_\pi(s,a) = \mathbb{E}_\pi[R(s,a,s') + \gamma V_\pi(s)] \tag{29}$$

$$H\big(\pi(\cdot\,|s)\big) = E_\pi[-\log \pi(\cdot\,|s)] \tag{30}$$

Where V and Q are state and action values, s and a represent state and action, R is the reward, π is action policy, γ is the discount factor, α is the temperature parameter, and H is entropy. The reward function is designed as follows.

$$R(s, a, s') = r_1 s_{vio} + r_2 a_{vio} + r_3 \mathcal{P}_{sys} + r_4 s_{comf} + r_5 a_{smooth} \tag{31}$$

Where s_{vio} and a_{vio} are state and action violations. They are 0 if the states and actions are within the bounds. Otherwise, they are calculated by the magnitude of bound violations. s_{comf} indicates the status of comfortable indoor environment quality (IEQ), which is 1 when both temperature and CO2 are within the bounds and without action violations, and 0 otherwise. a_{smooth} is the smoothness of the actions calculated by taking the difference between current actions and the previous ones. r_1, r_2, r_3, r_4, and r_5 are balancing weights. Note that each term in equation (31) is calculated in the same way as DPC.

The agent training algorithm is summarized in **Algorithm 1**, where duplicated state and action value neural networks are used for improving learning stability.

Algorithm 1. SAC agent training algorithm.

Initialize environment, replay buffer *D*

Initialize 5 neural networks: $\boldsymbol{\pi_\theta}$, $\boldsymbol{v_{\theta_v}}$, $\boldsymbol{\hat{v}_{\theta_{\hat{v}}}}$, $\boldsymbol{q^1_{\theta_{q_1}}}$, $\boldsymbol{q^2_{\theta_{q_2}}}$

Repeat:

Observe state s and select action $a \sim \pi(\cdot \mid s)$

Execute action a and observe new state s', reward r, and done.

Store $(s, a, r, s', done)$ in replay buffer D

If stored experience is enough for batch training:

Randomly sample batch $B = \{(s, a, r, s', done)\}$ from D

Sample action $\tilde{a}$ based on the latest π_θ

Evaluate $q^1_{\theta_{q1}}(s, \tilde{a})$, $q^2_{\theta_{q2}}(s, \tilde{a})$, $q^1_{\theta_{q1}}(s, a)$, $q^2_{\theta_{q2}}(s, a)$, $\hat{v}_{\theta_{\hat{v}}}(s')$

Bellman label for state value y_v: $\min\left(q^1_{\theta_{q1}}(s, \tilde{a}), q^2_{\theta_{q2}}(s, \tilde{a})\right) - \log(\pi(\tilde{a}|s))$

Bellman label for action value y_q: $r + \hat{v}_{\theta_{\hat{v}}}(s')$

Bellman objective for policy J_θ: $\frac{1}{B}\sum^B \left[\min\left(q^1_{\theta_{q1}}(s, \tilde{a}), q^2_{\theta_{q2}}(s, \tilde{a})\right) - \log(\pi_\theta(\tilde{a}|s))\right]$

Update parameters based on: $\min_{\theta_v} MSE(v_{\theta_v}(s), y_v)$, $\min_{\theta_{q1}} MSE(q^1_{\theta_{q1}}(s, a), y_q)$,

$\min_{\theta_{q2}} MSE(q^2_{\theta_{q2}}(s, a), y_q)$, $\max_\theta J_\theta$ for π_θ, v_{θ_v}, $q^1_{\theta_{q1}}$, $q^2_{\theta_{q2}}$

Update $\hat{v}_{\theta_{\hat{v}}}$ by weighted average of $\hat{v}_{\theta_{\hat{v}}}$ and v_{θ_v} parameters

3.5 Weather and Occupancy Forecasting

For the implementation of advanced control strategies, especially HDDPC and DPC, forecast models are needed for disturbances such as weather and occupancy. The occupancy forecast model

was developed in [133] to predict occupancy numbers in an autoregressive way using LSTM model.

For the weather forecast, we use two LSTM models for future 6 hours of outdoor temperature and solar radiation with 15-minute interval. The input sequence lengths were historical 6 hours for the temperature forecast model, and historical 3 days for the solar radiation forecast model. Both models had two LSTM layers with 50 neurons. The loss function was MSE loss for the outdoor temperature LSTM model. For the solar radiation LSTM model, the loss function had an additional term to penalize negative predictions:

$$Loss = MSE + \sum\left(-min(\hat{y}, 0)\right) \tag{32}$$

The weather data was acquired from Syracuse University Weather Station. Two-year data for 2019 and 2020 was used to train models.

3.6 Case Study 1: Simulation Study for PCICNN Performance

We first test the PCICNN's performance using real-building data. The indoor environment and system data was acquired from the student office shown in Figure 6. The HVAC system of the office was a dedicated air handler unit with four variable air volume terminal devices. The measuring of four VAVs was aggregated for model development and test.

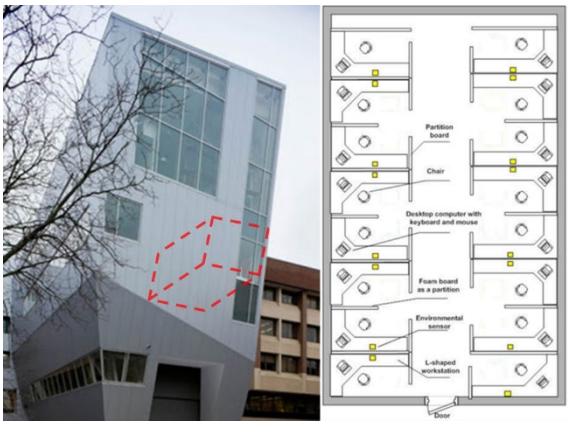


Figure 6. Outside view and floor plan of testing site.

Even though the neural networks were formed to be physics-informed and convex with respect to inputs, they still need to be tested to guarantee the control performance. The model sanity check includes three parts:

a) Effect of accumulative errors: in real implementation and model training, the prediction of next-h-hour indoor temperature and CO2 was based on the current measurement, along with control inputs and disturbances. However, in a simulation environment for RL training, such measurement was not included. Instead, the model prediction was treated as the response from the real environment and used recurrently in the next step, which would cause accumulative errors. Therefore, the model performance under such a recurrent process needs to be tested.

b) Responses to control inputs: we need to check whether the neural networks have reasonable responses to different control inputs. For example, the indoor air temperature and CO2 prediction should be higher (lower) than the measurement if less (more) cooling load (fresh air) was provided. On the other hand, the effects of future inputs on the current time step prediction need to be tested by different scenarios.

c) Response to disturbances: Lastly, we also need to check the response of neural networks with different disturbance inputs. For the indoor air temperature neural network, the prediction should respond physics-consistently with disturbances.

Therefore, in this study, the model sanity was checked by following cases with different scenarios:

1. All input features are the same as measurement, except that the input indoor state values were from previous time step.
2. The control inputs were increased or decreased by a certain amount. Different scenarios were tested in each case. In the first scenario, all control inputs were changed. In the second scenario, the first control input was the same as the measurement, but the remaining control inputs were changed, and so on. In the last scenario, the control inputs were the same as the measurement except for the last one. Different scenarios were to test if future inputs have effects on the prediction for the current time step.
3. It is essentially the same as the second model sanity check, but instead of control inputs, disturbance inputs were changed.

3.7 Case Study 2: Comparison of Advanced Control in Real-Building Implementation

Testbed Description

To address the research gaps of inconsistent comparison results and limited number of real-building implementations of advanced control. We deploy all three advanced controls in the real building and compare their performances. The testbed in this study is a student office (Figure 7) at the Syracuse Center of Excellence in Environmental Energy Systems. The indoor environment is controlled by a dedicated AHU-VAV system. The space is partitioned into 12 workstations. The system has two modes. One is for personalized environmental control, which has each workstation as a thermal zone and a dedicated VAV supplying air under the desk. Another is for typical office environmental control, which has mixing ventilation through a VAV on the ceiling. The whole system has 13 VAVs in total.



Figure 7. Outside and inside view of the testbed.

Baseline Control Logic

In this study, mixing ventilation through the VAV on the ceiling is used. The VAVs under the desks are turned off. As a baseline control case, the whole system operates based on a fixed schedule. From 6:00 AM to 6:00 PM, the system will be on with daytime temperature setpoint. During nighttime, the system will be off with temperature setback. During the daytime, the ventilation rate from the system is controlled by floor area and occupancy number based on ASHRAE Guideline 36 [137].

$$OA = (R_p \times P_{z,max} + R_a \times A_z)/E_z \tag{33}$$

where R_p is the outdoor airflow rate required per person, $P_{z,max}$ is the maximum population of the room, R_a is outdoor airflow rate required per unit floor area, A_z is zone floor area, and E_z is ventilation efficiency of the air distribution system.

Workflow of Advanced Controls

Loosely speaking, HDDPC and DPC are both MPC, but HDDPC is implicit and DPC is explicit. Therefore, the workflow of HDDPC and DPC are similar, and we only show the one for HDDPC in Figure 8.

The control is executed on a daily routine as follows. At midnight, a wipe-off function is executed to release all overriding in the system. Another program then checks whether all models required for the control are available and decides whether the model training function needs to be executed. Once all required models are ready, at each scheduled control step, weather data, indoor environment measurement, and occupancy data are acquired in parallel and sent to corresponding forecast models. The room-level controller, based on the weather and occupancy forecasts and indoor temperature prediction, decides and predicts the required HVAC load for the testbed space. The system-level controller then decides the outdoor airflow rate, supply airflow rate, and supply air temperature based on forecasts, CO2 prediction, and required HVAC load. The outputs from the system-level controller are overridden into the Building Management System (BMS). The control process loops through the whole day until midnight.

The workflow of DPC is similar to the HDDPC. The differences are 1) the IEQ prediction is not needed because the IEQ prediction from PCICNNs is used for control law training but not for real-time decision making as shown in Figure 4, and 2) two levels of the controller are replaced by the control law neural network. The model training process for HDDPC and DPC does not require large computational resources, they are trained every day at midnight with the most updated data.

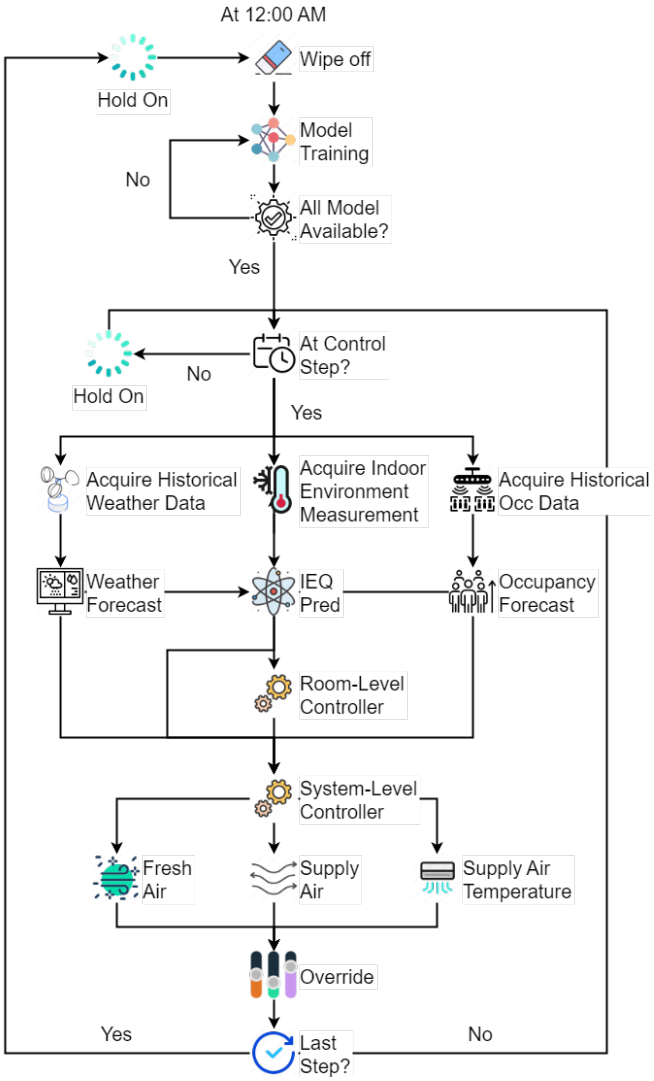


Figure 8. Workflow of HDDPC.

The workflow of RL is a bit different from DPC and HDDPC as shown in Figure 9. RL is notorious for being difficult to train. Online agent training is not feasible, so we separate the agent training from the control logic. The agent is trained on the GPU cluster. To improve training efficiency, we propose a hybrid method. On the right of the top part in Figure 9, we first construct the (state, action, reward) tuples based on the historical measurement from the BMS. These tuples are then sent to and stored in the replay buffer. On the left, we have the PCICNN as a virtual environment

of the testbed to receive the actions from the agent and give the response. The sample tuples are constructed based on the simulation results from the virtual environment and sent to the replay buffer as well. Therefore, the replay buffer is filled with both measured data from BMS readings and simulated data from interactions between the agent and PCICNNs. At the beginning of the training, the measurement data takes more portions and is more frequently used, which makes the agent learn from all the control strategies that have been implemented in the system. As the training proceeds, the simulated data will be pushed into the replay buffer after each interaction with PCICNNs, so the simulated data will be gradually increased. If the simulated days are greater than the days of the measurement data, two kinds of data will be balanced. Since then, the simulated data will take more and more portions as training proceeds. The simulated data has more exploration actions to further improve the agent. Once the training is done, the agent will be uploaded to the cloud. The computer running the online control will download the agent for the execution.

The online control part of RL (bottom part in Figure 9) in essence is similar to DPC and HDDPC but does not require forecast and prediction models because the agent only relies on the current measurements of indoor environment, weather, and occupancy numbers for decision making.

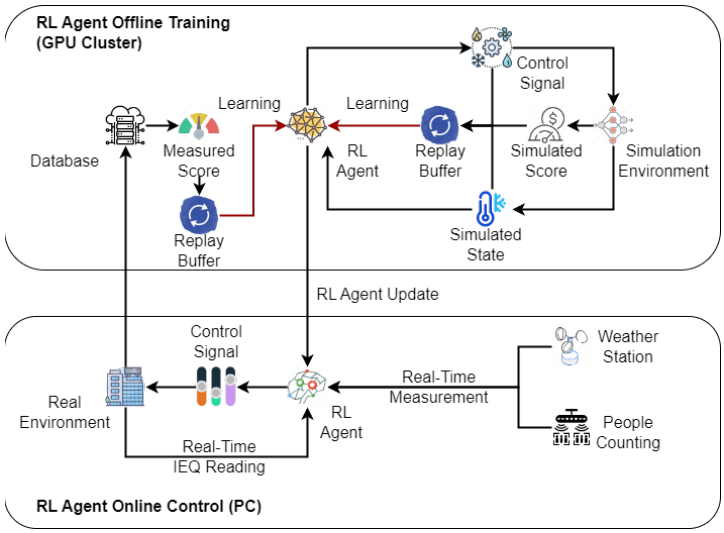


Figure 9. Workflow of RL.

Control Settings for Temperature and Ventilation

The control settings for different controls are shown in Table 4. The indoor and supply temperature and supply airflow bounds of baseline, HDDPC, DPC, and RL are determined based on the default settings in the AHU-VAV system. In the AHU-VAV system, even though there is an option for the CO2 setpoint on the VAV interface, the actual control on the AHU side is not programmed. Instead, the ventilation rate is controlled based on the user-defined setpoints. Therefore, in the baseline control, to follow the default control logic, we do not define the CO2 setpoint, and we use equation (33) to calculate the fixed outdoor airflow rate, while in the advanced controls, OA is dynamically optimized.

Table 4. System settings for different controls.

Control	Baseline	HDDPC	DPC	RL
Temperature setpoint	21.7~24℃, 6:00AM to 6:00PM			

	18.3~26.7℃, otherwise	
CO2 setpoint	None	600 PPM, 6:00AM to 6:00PM 1000 PPM, otherwise
Supply airflow	0.09~0.28 m^3/s, 6:00AM to 6:00PM 0~0.28 m^3/s, otherwise	
Outdoor airflow	Guideline 36	≤ Actual supply airflow
Supply temperature	>12.8℃	

During the implementation, the control step is 15 minutes for HDDPC, DPC, and RL. The prediction horizon of HDDPC and DPC is 6 hours. The PCICNNs have 3 layers with 5 neurons for each prediction step. The control law network shown in Figure 4 is a multi-head neural network. It has a backbone part which branches into three head parts for supply airflow rate, outdoor airflow rate, and supply air temperature. The control law network has 5 layers with 40 hidden neurons. The networks (value networks and policy networks) in RL are all 5-layers with 256 hidden neurons. The key hyperparameters are listed in Table 5.

Table 5. Hyperparameters for HDDPC, DPC, and RL.

Hyperparameters	Settings
ρ_3	1
ρ_{obj}	3
ρ_{op}	30
ρ_{comf}	20 (6:00 AM ~ 6:00 PM) 10 (6:00 PM ~ 6:00 AM)
r_1	-0.1
r_2	-0.5
r_3	-0.05
r_4	0.04
r_5	-0.001
γ	0.9
Episode length	1-day, 2-day
Agent learning rate	1e-4
Agent training epochs	3e4

All controls are programmed and realized through Python with modularized functions for each element shown in the workflows on the laptop equipped with Intel i7-1185G7 CPU and integrated graphics. The GPU cluster for RL agent training is equipped with AMD Ryzen 9 3950 X 16-Core Processor, 2 NVIDIA Quatro RTX 5000 GPUs, and 117 GB memory.

Control Signal Implementation

To implement the control action accurately, we compared the readings from BMS with our calibrated sensors. The supply airflow rate is measured by a TSI air capture hood. The outdoor airflow rate and supply air temperature are measured by a TSI multi-function ventilation meter. The measurements of supply airflow rate and air temperature are almost the same as BMS readings. The outdoor airflow rate readings are different as shown in Figure 10.

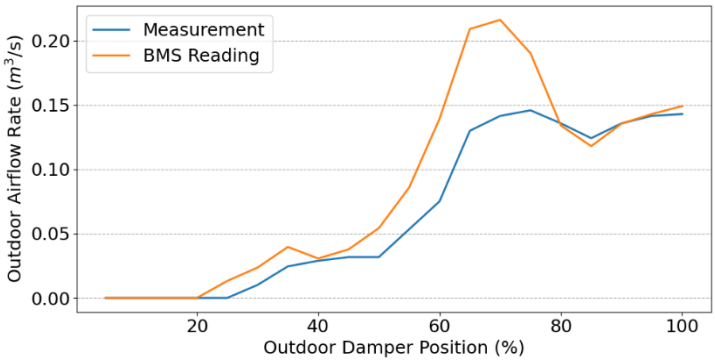


Figure 10. A snapshot of comparison of outdoor airflow measurement and system reading.

To address such inaccuracy, we measured the outdoor airflow rate in different outdoor damper positions and fan speed and then developed a performance curve for outdoor airflow rate with fan speed and outdoor air damper position as inputs. The fan speeds are set at 0%, 25%, 50%, 75%, and 100%. The outdoor air damper is changed from 0 to 100% with 5% as the interval. The fitted curve is in the form:

$$f(x,y) = a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 + a_6x^3 + a_7x^2y + a_8xy^2 + a_9y^3 + a_{10}x^4 + a_{11}x^3y + a_{12}x^2y^2 + a_{13}xy^3 + a_{14}y^4 \quad (34)$$

Where x is the fan speed and y is the outdoor air damper position. The curve-fitting result is shown in Figure 11. The overall mean absolute error is 0.013 m^3/s.

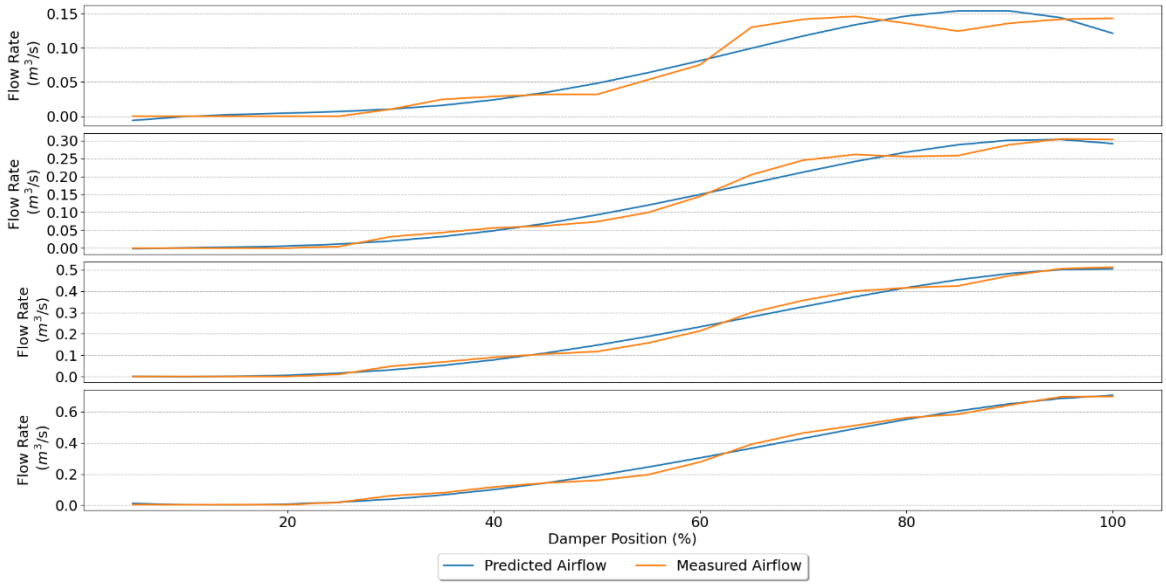


Figure 11. Fitted curve of outdoor airflow rate as a function of damper position. From top plot to bottom plot, the fan speeds are 25%, 50%, 75%, and 100%, respectively.

Therefore, given the fan speed signal and outdoor airflow requirement, we can use Eq. (23) to calculate the required damper position and override the corresponding value in the system. As for the other two control signals, supply airflow rate and supply air temperature, they can be implemented by overriding corresponding setpoints.

All overriding is achieved through the BACnet.

Control Performance Evaluation

The objective of the advanced controls is to maintain IEQ, i.e., indoor temperature and indoor CO2 concentration, cost-effectively. Besides, the computation time for decision-making is critical for the hardware requirement of implementation. Therefore, the control performance is quantified from three perspectives:

1. *Daily average IEQ bound violation*. Based on the control settings in Table 4 and the measurement data, the bound violation can be calculated by $\max(0, T - \bar{T}) + \max(0, \underline{T} - T)$ for indoor temperatures and $\max(0, C - \bar{C}) + \max(0, \underline{C} - C)$ for indoor CO2 for each control step. The daily average IEQ violation can be calculated by averaging the daily total IEQ violation over the daily total number of violated control steps.
2. *Energy usage.* The energy consumption is monitored by energy meters installed on the cooling, heating, pre-heating, and reheating coils. The energy usage is analyzed based on the meter readings.
3. *Computation time.* In the training and implementation program of the advanced control, the time elapsed to train the agents and get the control decision is recorded, which is used to quantify the computation time of the advanced controls.

Experiment Schedule

The whole experiment ran from September 19th, 2023, to February 16th, 2024. Baseline control, HDDPC, and DPC were implemented first. 2-day and 5-day pilot experiments were conducted for HDDPC and DPC, respectively. RL was implemented starting November, with 12-day pilot experiments. To balance the number of experiment days for each control, we first altered each

control day by day as they were implemented and then scheduled more days for those implemented later than others. The full experiment schedule is shown in Figure 12.

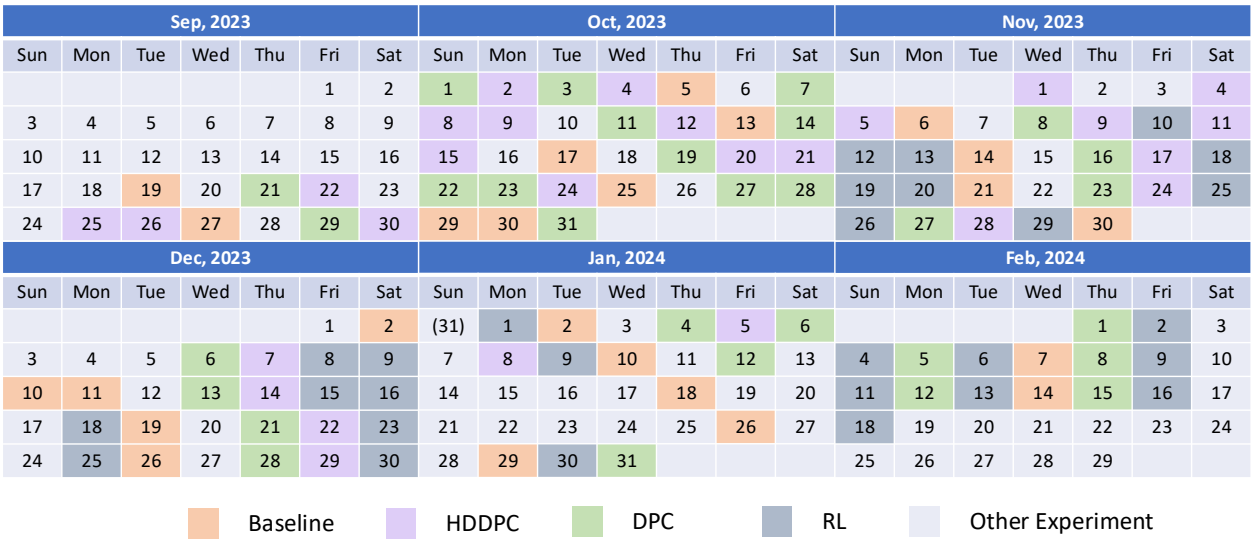


Figure 12. Experiment Schedule.

3.8 Case Study 3: PIML Model Benchmarking

Among all the publications on PIML models for indoor temperature dynamics, only three are open-sourced [23][27][59], each representing a different PIML approach. Models with physics-informed loss functions follow the PINN approach, which aims to solve differential equations faster, particularly partial differential equations and nonlinear ordinary differential systems. However, in [23], the linear grey-box model, which can be easily solved by either integration or discretization, is incorporated as physics-informed loss terms. The development of the grey-box model constitutes the largest portion of the total development effort. Consequently, the PIML model that utilizes the grey-box model in its loss functions does not reduce but rather increases the development effort due to the involvement of duplicated models. Therefore, in this study, the model from [23] is not used for the evaluation. Instead, the other two models, the physics-consistent neural network

(PCNN) [27] and the modularized neural network incorporating physical priors (ModNN) [59], along with the physics-consistent input convex neural network (PCICNN), will be used.

Comparative Evaluation Design

PCNN, ModNN, and PCICNN were originally developed based on different buildings ranging from living labs to office rooms. Comparing the three models on either dataset of their development will introduce bias in their performance. Moreover, real buildings have to be maintained under limited operation ranges, making the testing of model performance in extreme scenarios difficult. Therefore, to address the identified research gaps and enable systematic comparison of PIML models, we design a comprehensive evaluation framework that combines controlled comparative analysis with real-world validation.

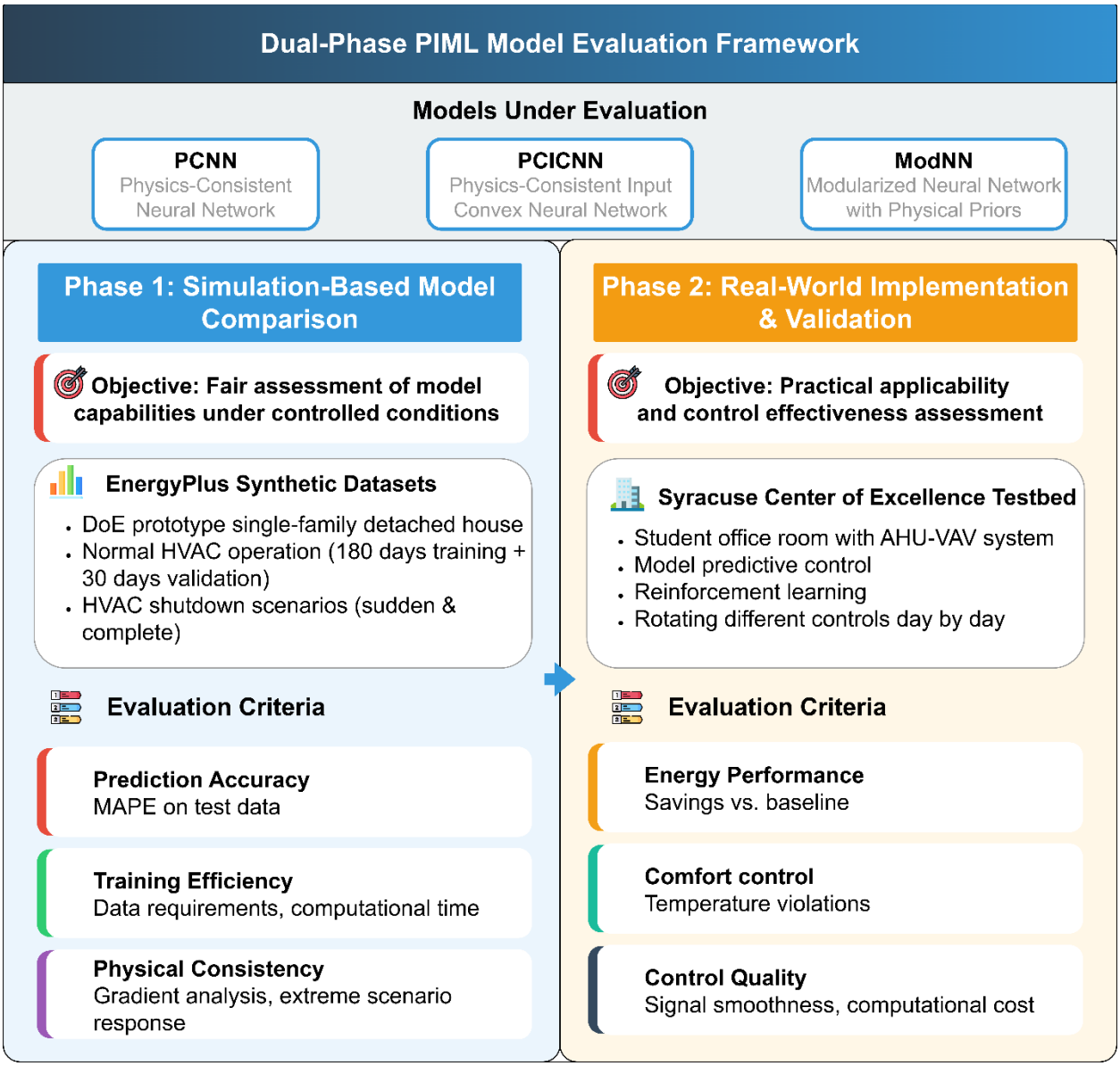


Figure 13. Dual-phase PIML model evaluation framework.

The dual-phase evaluation framework is shown in Figure 13. In the first phase, we employ standardized EnergyPlus simulation data to provide a neutral testing ground where all models can be evaluated under identical conditions. This approach eliminates confounding factors related to building-specific characteristics and training data advantages, enabling fair assessment of each model's fundamental capabilities and architectural strengths. Then in the second phase, we

implement the three models in an actual building environment using both model predictive control and reinforcement learning strategies. While this phase inherently reflects realistic deployment constraints and favors the building-specific model (PCICNN), it provides essential insights into practical performance and control effectiveness under real operational conditions.

The rationale for this dual approach stems from the need to separate model capability assessment from deployment validation. The simulation-based evaluation serves as an unbiased comparative baseline, while the real-world implementation validates practical applicability and reveals how model differences translate to control performance.

Phase 1: Simulation-Based Model Comparison

Three synthetic datasets will be used to evaluate the performance, in terms of prediction accuracy, physical consistency, and computational cost, of the three PIML models.

The datasets are generated based on DoE EnergyPlus prototype model of single-family detached house, which is different from the original building types that the three models were developed on. The first three different datasets represent different operations of HVAC systems: 1) normal HVAC operation based on setpoints; 2) suddenly turning-off the HVAC system starting from August; and 3) no HVAC system for open-loop indoor temperature dynamics.

Different datasets are used for different evaluations, representing different case studies:

1) To evaluate the prediction accuracy, training efficiency, and physical consistency, the dataset with normal HVAC operation is used.

2) To evaluate the model performance under changed cooling/heating load, the two synthetic datasets with abnormal operations are used. Specifically, the models will be trained using the normal operation data and tested using the data during abnormal operation periods.

For fair comparison, hyperparameters for all models were tuned using the same approach. The hyperparameters include both standard machine learning hyperparameters (learning rate, number of layers, and hidden layer size) and physics-specific hyperparameters unique to each model architecture. Since exhaustive grid search across all hyperparameters is computationally prohibitive, we employed a systematic approach: first identifying the most performance-critical hyperparameters through preliminary sensitivity analysis, then conducting grid search within predefined ranges for these selected parameters. All models were trained using the first 180 days of data and validated on the data for the next 30 days, with hyperparameter selection based on validation performance. Table 6 shows the hyperparameters tuned for each model. Note that for PCNN, there are two learning rates. One is for the black-box module (unforced dynamics), and another is for physical module (energy emulator). Additionally, we identified a critical division factor that balances the outputs of these two modules—while not explicitly discussed in the original study, our sensitivity analysis revealed this parameter significantly impacts model performance and was therefore included in the tuning process.

Table 6. Hyperparameters for each model.

Hyperparameter Category	Hyperparameter	PCNN	ModNN	PCICNN
Standard ML Hyperparameters	**Learning Rate**	✓ (unforced dynamics)	✓	✓
	No. of layers	✓	N/A	✓
	Hidden layer size	✓	✓ (external and internal	N/A

			modules)	
	Neurons per time step	N/A	N/A	✓
	Enable/disable bias	N/A	N/A	✓
Physics-Specific Hyperparameters	**Learning rate (physics module)**	✓ (energy accumulator)	N/A	N/A
	Division factor	✓	N/A	N/A
	Coefficient of weight-related loss	N/A	N/A	✓
	Coefficient of bias-related loss	N/A	N/A	✓

Models	**PCNN**	**ModNN**	**PCICNN**
Standard ML Hyperparameters	Learning rate – unforced dynamics	Hidden layer size – external module	No. of layers
	No. of layers	Hidden layer size – internal module	Enable/disable bias
	Hidden layer size	N/A	N/A
Physics-specific Hyperparameters	Learning rate – energy emulator	N/A	Neurons per time step
	Division factor	N/A	Coefficient of weight-related loss
	N/A	N/A	Coefficient of bias-related loss

Phase 2: Real-World Implementation

In Phase 2, we implement the HDDPC and RL with three models in the testbed shown in Figure 7. The implementation starts from April 28th, 2025, with one week of the pilot test and ends on June 15th, 2025. The full implementation schedule is shown in Figure 14. Because the implementation period is during summer break, the office is unoccupied. To mimic occupancy heat gains during the summer break when the office was unoccupied, we deployed 8 heat lumps throughout the testbed, one at each workstation. The heat lumps operated on predefined schedules to simulate typical occupancy patterns: most (4 heat lumps) operated from 9:00 AM to 6:00 PM,

with some variations to represent different work schedules (e.g., 1 heat lump operates from 9:45 AM to 5:45 PM, 1 heat lump operates from 10:00 AM to 5:00 PM, and 2 heat lumps operate from 9:00 AM to 5:00 PM) to create a more realistic and diverse internal heat gain profile.

The system settings of different controls are shown in Table 4. Baseline control follows the inherent control logic of the system and uses ASHRAE Guideline 36 for outdoor airflow rate. For most settings, advanced controls are the same as the baseline except that the outdoor airflow rate of advanced controls is no more than the total supply airflow rate.

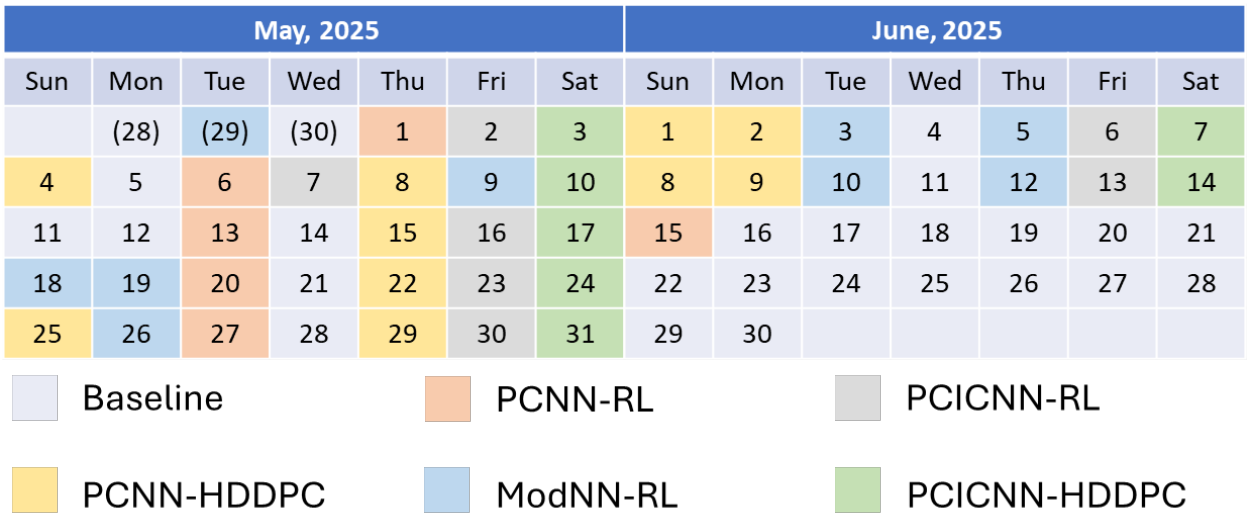


Figure 14. Implementation Schedule.

Evaluation Criterion for Simulation Studies

For a comprehensive comparison, the performance of PCNN, PCICNN, and ModNN will be evaluated in terms of accuracy, training efficiency, physical consistency, and scalability as shown in Table 7. The details of each evaluation are as follows.

Table 7. Evaluation Criterion of the PIML models performance.

Evaluation Type	Quantitative		Qualitative
Accuracy	MAE	MAPE	n/a
Training efficiency	Least data requirement	Computation time	Hyperparameter tuning
Physical consistency	Accuracy under changed dynamics		Gradient

The training efficiency will be evaluated from three aspects, both quantitatively and qualitatively: 1) The minimum data requirement will be quantified based on the shortest dataset length needed for good performance; 2) Computing time will be assessed by the number of updates and the computing time for each update during model training; 3) the hyperparameter tuning will be qualitatively evaluated based on the number of general machine learning hyperparameters and physics-prior related hyperparameters, as well as the methods used to tune them.

The evaluation of accuracy is straightforward, utilizing mean absolute error (MAE) and mean absolute percentage error (MAPE).

The physical consistency will be evaluated based on the gradient and prediction accuracy under changed dynamics.

For each model, the gradient of sequential predictions with respect to the sequential inputs will be calculated. The purposes of gradient evaluation come from twofold: 1) verifying positive correlation between input and outputs; 2) verifying the thermal effect of inputs.

Besides the gradient evaluation, two scenarios will be tested for the prediction accuracy of changed dynamics:

1) *Suddenly turning off the HVAC system*: the models will be trained on normal operation data but tested on the data when the HVAC system was suddenly turned off.

2) *Open-loop indoor dynamics*: the models will be trained on normal operation data but tested on the data when the HVAC system was always off.

Evaluation Criterion for Real Building Implementation

For the real-building implementation, the performance of the control for each model will be evaluated based on energy usage and temperature violation.

The energy usage is recorded every 15 minutes from the BAS, which measures the thermal energy of coils. Because the implementation is short term that only convers 1.5 months, the energy reading will be normalized based on the degree day method. As the testbed is a LEED platinum building, a lower benchmark temperature, 55 Fahrenheit (12.8 Celsius), is used. The aggregated normalized energy usage of advanced controls based on different models will be compared with baseline to quantify the energy saving.

The temperature violation of different controls will be quantified from two aspects: 1) daily violation degree-hours and 2) total number of steps of violations. The daily violation degree-hours are calculated as:

$$T_{vio}^{daily} = \frac{1}{D}\sum_{d=0}^{D}\sum_{i=0}^{96}[\max(0, T - \bar{T}) + \max(0, \underline{T} - T)]/4 \quad (35)$$

Where D is the total number of days for a specific control and i is the control steps that is 15 minutes in this study.

Chapter 4. Results I: PCICNN and Advanced Control for Single Zone

This chapter presents the results corresponding to the methodologies introduced in Chapter 3, organized by the three case studies. Section 4.1 reports the simulation results for the PCICNN, including prediction accuracy under accumulative errors and model sanity checks with perturbed control inputs and disturbances, establishing that the PCICNN produces physically consistent predictions suitable for use in advanced control. Section 4.2 presents the results of the long-term real-building comparison of HDDPC, DPC, and RL, covering indoor environmental quality, energy savings, experimental failure analysis, the impact of RL training episode length, and a detailed comparison between agent-based and non-agent-based control strategies. Section 4.3 reports the PIML model benchmarking results across both simulation and real-building phases, comparing prediction accuracy, physical consistency, training efficiency, and the impact of model selection on closed-loop control performance.

4.1 Case Study 1: Simulation Study for PCICNN Performance

We used data measured from a student office (Figure 6) to evaluate the performance of proposed neural networks and DDPC strategy. The measured data was collected every 5 minutes and resampled as 15 minutes and included indoor environmental information, weather information, and HVAC operational information.

According to [28], the input convex neural network (ICNN) performance improved significantly with more than four-week training data. In this study, with the physics-informed structure, we found that starting from three-week training data could guarantee prediction errors less than sensor errors, which was 0.5 ℃ (0.9 ℉) for temperature sensors and 50 PPM for CO2 sensors. Since the indoor air quality data was available after June 23rd, 2021, we started our testing on July 14th, 2021. To mimic a real implementation, we increased the amount of training data by days as the testing date moved. Because students went on vacation after the week of August 13th, the simulation study stopped on August 13th. The input features of the indoor temperature model included the current step indoor temperature and 6 hours, with a 15-minute resolution, of cooling loads, solar radiation, occupant numbers, and ambient temperature. The input features of the indoor CO2 model were the current sensor reading and 6 hours of fresh airflow rate and occupant number with the same time interval. The outputs of the two models were the next 6-hour indoor temperature and CO2 concentration, respectively. Both models were constructed with 3 hidden layers, with 40 and neurons in each block, respectively for indoor temperature and CO2 models. Table 8 shows parameters for model development.

Table 8. Parameters of indoor temperature and CO2 neural networks.

Model parameters	Indoor temperature model	Indoor CO2 model
Number of layers	3	3
Step to predict	24	24
Neurons for each step	40	20
Epochs	10e4	2e4
Initial training dataset size	2016 points * 97 inputs	2016 points * 49 inputs
Training data set increment	96 points *97 inputs	96 points * 49 inputs
Testing dataset size	96 points *97 inputs	96 points * 49 inputs

In the model training and testing, the input state values were from the measurement. However, in the simulation, we treated the response from indoor dynamic models as from the real indoor environment. Therefore, we first presented the results of the model sanity check case, in which the prediction was used as next-step input to evaluate accumulative model errors during each testing day.

Model Sanity Checks for Actual Control Inputs

As shown in dark green lines in Figure 15 and Figure 16, indoor temperature and CO2 concentration predictions on July 23rd followed the same trend as the measurement over different steps if the actual control input was used. Even without correction from measured data at each time step, the MAPE was around 1% for the temperature prediction and 4.2% for the CO2 prediction. The absolute error of both kinds of predictions was within the sensor errors at most time steps. Both models exhibited such performance on each testing day, and the overall model performance is listed in Table 9 and Table 10. The average indoor temperature prediction MAPE during the one-month test was around 1.1% for all prediction steps. The average MAE of indoor temperature prediction was around 0.44 ℃ (0.8 ℉) for all prediction steps, which was less than the temperature sensor error (0.5 ℃ / 0.9 ℉). On the other hand, the average indoor CO2 prediction MAPE and MAE were around 4.7% and 21 PPM, respectively, which was also less than the sensor error (50 PPM).

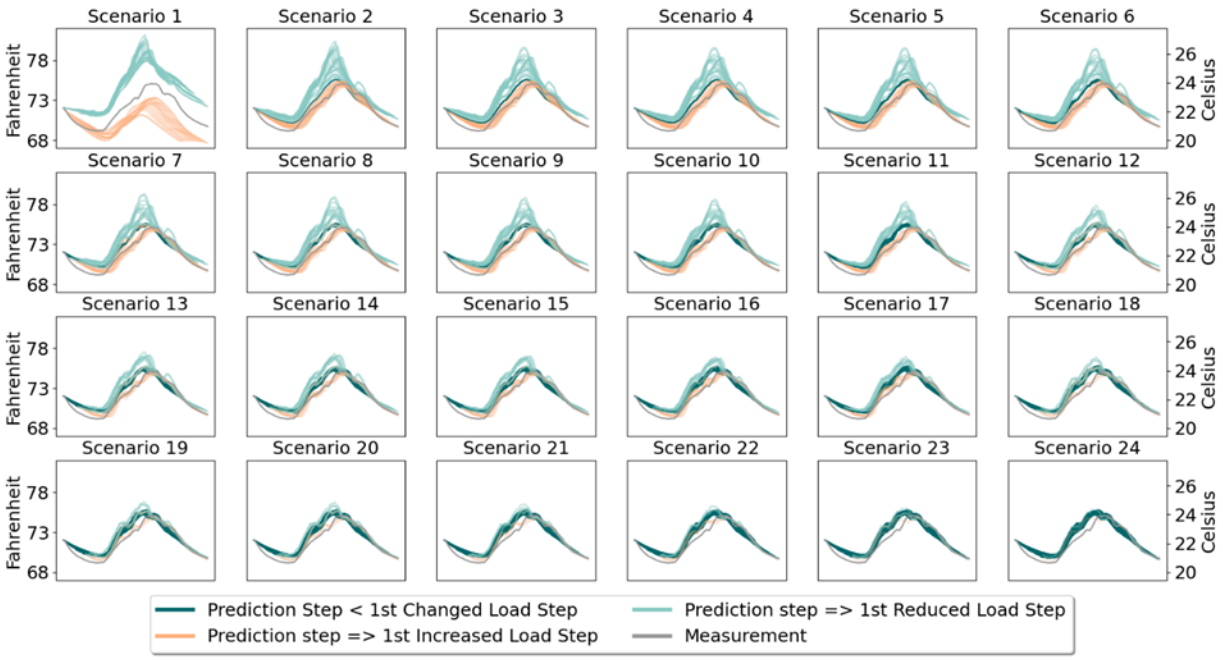


Figure 15. Temperature model sanity checks with different cooling loads on July 23rd. Light green lines were results with reduced cooling load. Orange lines indicate results with increased cooling load. Dark green lines indicate results with measured cooling loads. Grey lines were measurements.

Table 9. Average MAPE (%) and MAE (℃) of indoor temperature model of each step during the testing period.

Steps	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
MAPE	1.09	1.10	1.10	1.10	1.09	1.09	1.09	1.09	1.10	1.10	1.10	1.10
MAE	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44
Steps	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
MAPE	1.10	1.10	1.10	1.11	1.12	1.12	1.11	1.10	1.09	1.10	1.09	1.09
MAE	0.44	0.44	0.44	0.45	0.45	0.45	0.45	0.45	0.44	0.44	0.44	0.44

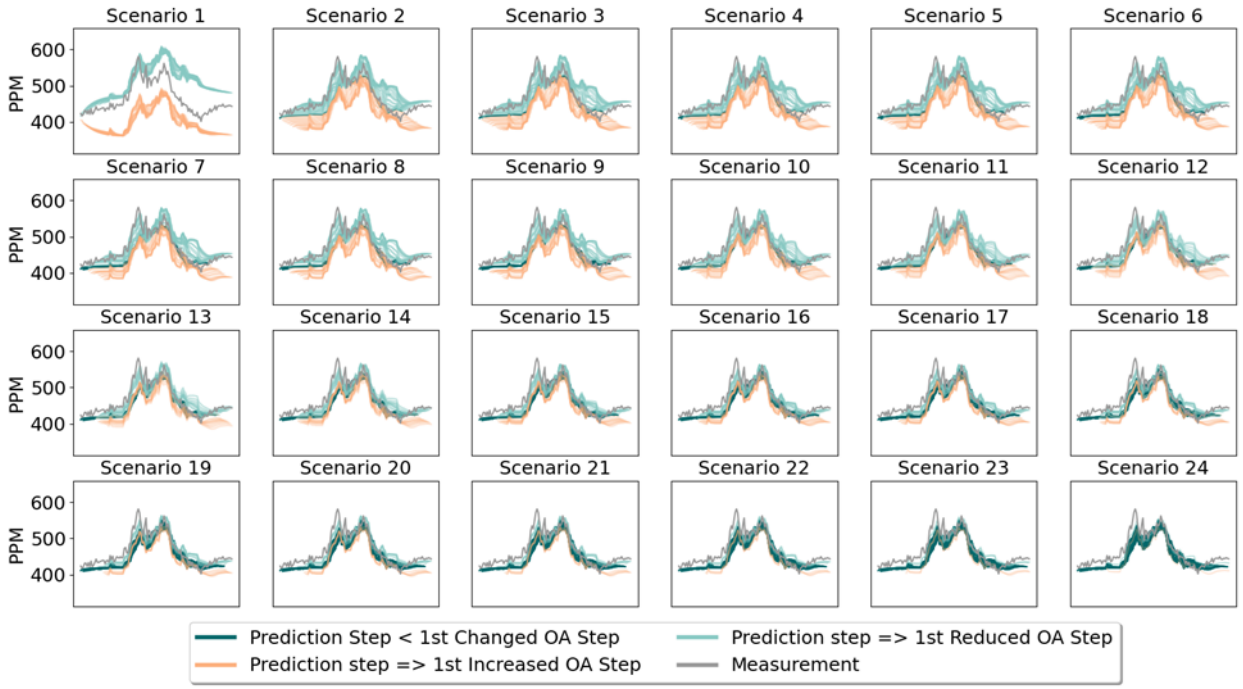


Figure 16. CO2 model sanity checks with different outdoor airflows on July 23rd. Light green lines indicate results with reduced outdoor airflow. Orange lines indicate results with increased outdoor airflow. Dark green lines were results with original outdoor airflow. Grey lines were measurements.

Table 10. Average MAPE (%) and MAE (PPM) of indoor CO2 model of each step during the testing period.

Steps	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
MAPE	4.64	4.67	4.7	4.70	4.71	4.70	4.73	4.75	4.76	4.77	4.79	4.79
MAE	21.8	21.9	22.0	22.1	22.1	22.0	22.2	22.3	22.4	22.4	22.5	22.5
Steps	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
MAPE	4.79	4.84	4.84	4.75	4.77	4.72	4.69	4.64	4.61	4.61	4.58	4.61
MAE	22.5	22.7	22.7	22.3	22.3	22.1	21.1	21.7	21.5	21.5	21.4	21.5

Model Sanity Checks for Perturbed Control Inputs

For the model sanity check with more and fewer control inputs, we only present the results of July 23rd to illustrate, as shown in light green and orange lines in Figure 15 and Figure 16.

To check indoor temperature model responses, we gave 50% less and 100% more cooling loads in the aforementioned two cases, respectively. In each case, 24 scenarios were tested, in which different steps of cooling load were changed.

Case 1 for temperature model: Model response with less cooling loads. In the first plot of Figure 3, in which all cooling loads were changed, indoor temperature deviated from measurement. The indoor temperature prediction increased significantly in the case without any cooling load. The maximum outdoor temperature was 26.7 ℃ (80 ℉), and the maximum solar radiation was 1020W/m2 on July 23rd. Also considering internal heat gain from occupants and devices, such an increase was physically interpretable.

Case 2 for temperature model: Model response with more cooling loads. When 100% more cooling load was consistently given, the indoor air temperature prediction decreased but not incredibly. For the scenarios in which the cooling load was not changed in the first certain steps, the corresponding temperature predictions converged the results with the measured cooling load.

For the indoor CO2 model, 2 cases were tested with 24 different scenarios.

Case 1 for CO_2 model: Model response with less fresh airflow rate. For the case with more outdoor airflow rate, doubled fresh air was given. For the other case, 30 CFM of fresh air was given. Similarly, the indoor CO2 prediction increased (decreased) with less (more) outdoor airflow rate. Even with the maximum outdoor airflow rate, the indoor CO2 predictions were still around 400 PPM, which was similar to the outside CO2 concentration.

Case 2 for CO_2 model: Model response with more fresh airflow rate. On the other hand, when less outdoor air was given, the indoor CO2 concentration increased during the day and reached a maximum of more than 600 PPM in the afternoon when there were 6 students in the

office. When the fresh air flow rate was kept the same as the measurement in the first certain steps, the corresponding CO2 prediction was neither affected by the future fresh airflow rate, as shown in scenarios 2 to 24. Such behavior in both models indicated that future control inputs didn't affect the prediction for the current steps, which validated our physics-informed model structure.

Model Sanity Checks for Disturbances

For the model sanity check with different disturbances, similarly, we present the temperature model response with increased (decreased) outdoor temperature and solar radiation, respectively, in Figure 17 and Figure 18. For the case with different outdoor air temperatures, we increased (decreased) measured values by 5.5 ℃ (10 °F), shown as light green (orange) lines in Figure 17. Similarly, for the case with increased solar radiation, we increased measured values by 375 watts per square meter and set the radiation to 0 in another case with decreased solar radiation, shown as light green and orange lines in Figure 18, respectively. As indicated in the two figures, the indoor temperature model had physics-consistent predictions with different disturbances. Moreover, scenarios 2 to 24 indicated that the future outdoor temperature and solar radiation had no effect on the prediction of the current time step, which validated our model structure.

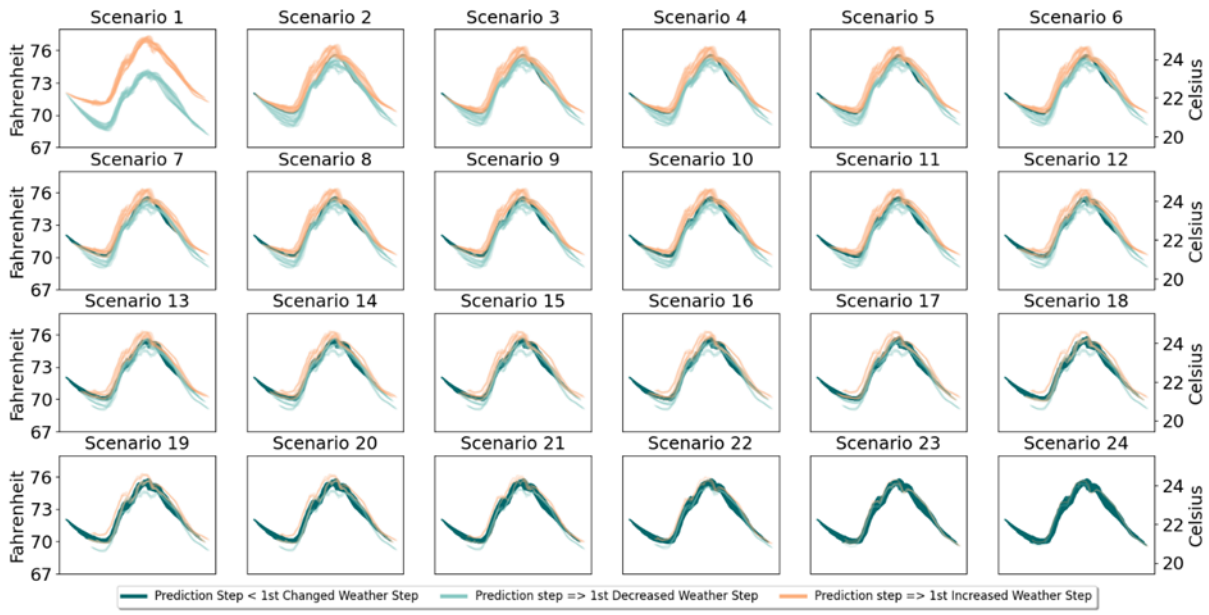


Figure 17. Temperature model sanity checks with different outdoor air temperatures on July 23rd. Light green lines were results with decreased outdoor air temperature. Orange lines indicate results with increased outdoor air temperature. Dark green lines indicate results with measured outdoor air temperature.

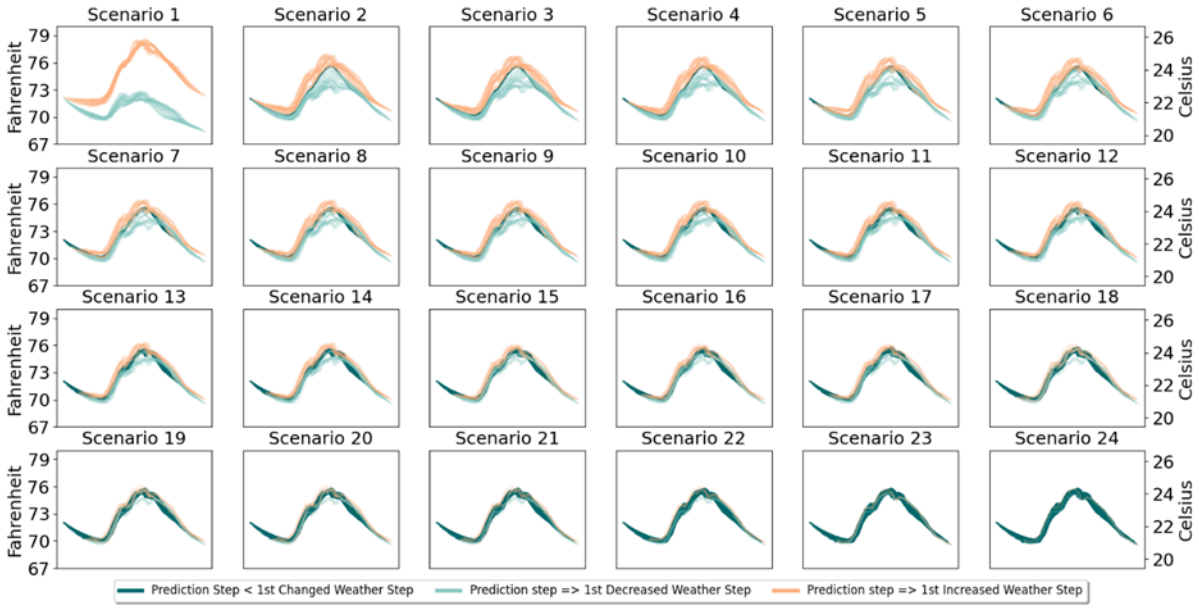


Figure 18. Temperature model sanity checks with different solar radiation on July 23rd. Light green lines were results with decreased solar radiation. Orange lines indicate results with increased solar radiation. Dark green lines indicate results with measured solar radiation.

In summary, the model prediction errors were within sensor errors for different prediction steps. Moreover, with different control inputs and disturbances at different prediction steps, the models had physically consistent responses, which benefited from physics-informed model structures.

4.2 Case Study 2: Comparison of Advanced Control in Real-Building Implementation

Training Results of DPC and RL

As the focus of this study is on the comparison of different control strategies, we will not have a dedicated subsection to present the results of PCICNN. In fact, the performance of PCICNN can be indirectly reflected in the training results of the control law neural network in DPC and the agent in RL.

PCICNN, DPC control law, and RL agent were trained using data after August 4th, 2023. During the experiment, PCICNN was trained every day at midnight, and the DPC control law was trained at midnight every time before it is implemented. RL agent was trained once a week until December 18th, 2023, shifting to monthly training thereafter. These training sessions consistently utilized the latest available data. The snapshots of the training results of the DPC control law network and RL agent are shown in Figure 19 and Figure 20. Figure 19 demonstrated that the DPC control law network effectively maintained IEQ within desired limits. Meanwhile, Figure 20 reveals the RL agent initially struggled, indicated by a total score below -100. However, as training progressed, there was a notable improvement in the score, quickly stabilizing at a positive value, signifying the agent's ability to sustain good IEQ in a cost-efficient manner.



Figure 19. DPC control law network training results.

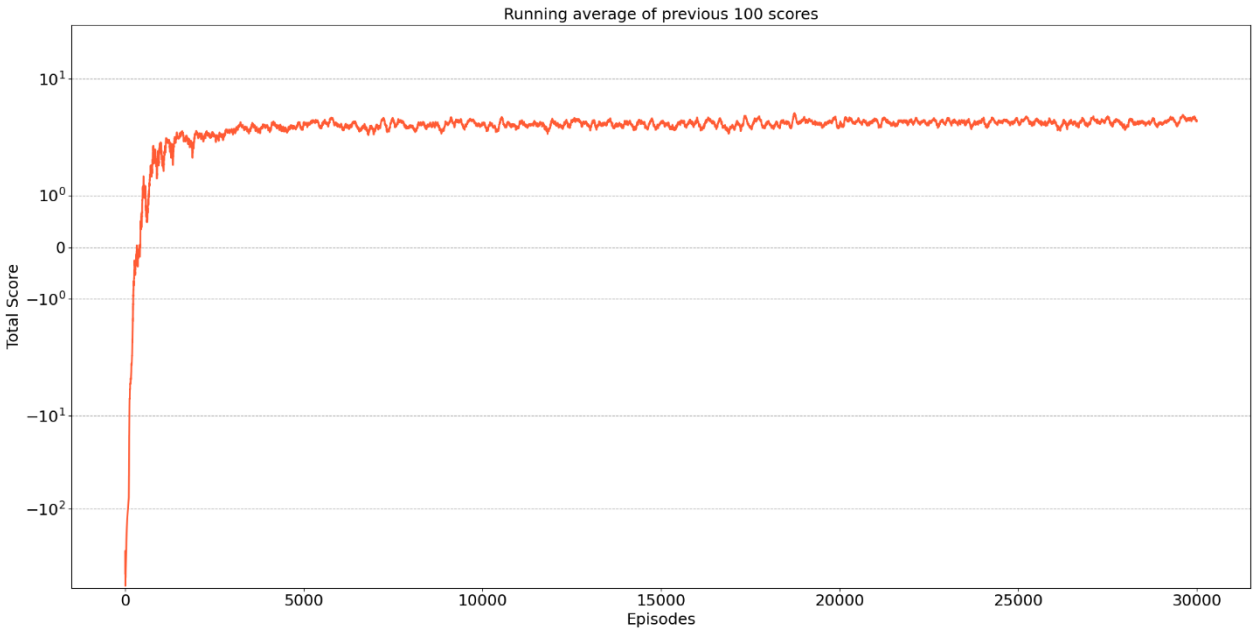


Figure 20. RL agent training results.

One-Day Experimental Results of HDDPC, DPC, and RL

The whole experiment spanned more than half a year. For a clarity, we showcase the one-day detailed results of HDDPC, DPC, and RL, depicted in Figure 21, Figure 22, and Figure 23, and the one-day control signal results in Figure 24.

Figure 21 and Figure 22 show results of both model prediction and measurement. The results of indoor temperatures and CO2 concentrations are predictions from PCICNN and readings from the BMS, while results of outdoor temperatures, solar radiation, and occupancy numbers are from the specific forecast models and sensor readings. Because the DPC does not rely on the IEQ predictions for decision-making, there are no predictions for indoor temperature and CO2 in Figure 22. In line with standard predictive control practices, we apply and present only the initial step from the control sequence for comparison against actual measurements. The upper and lower bounds for indoor temperature and CO2 represent the requirements in Table 4. The bounds for

occupancy prediction are with ± 1 occupant counting error of the measurement as acceptable errors. Figure 21 illustrate that both the indoor temperature and CO2 were controlled as anticipated, closely adhering to the upper bound in pursuit of energy efficiency. This not only underscores the successful execution of our approach but also validates the precision of our PCICNNs. In Figure 22, without the guidance of IEQ predictions, the control law network still controlled the IEQ within the comfort range. The outdoor temperature and solar radiation forecasts were also very accurate with MAPE less than 5%. As for the occupancy prediction, it was within the bounds most of the time. We also observed an underestimation in Figure 22, which caused the bound violation of the indoor CO2 concentration, but such a violation was not significant.

In Figure 23, only actual measurements are presented for the RL outcomes, reflecting the model-free nature of the SAC approach used. In this implementation, the RL agent operates independently of predictive models for its control decisions. Despite the absence of predictive model guidance, the RL agent effectively kept indoor temperature and CO2 levels within our specified bounds, which demonstrated the capabilities of the model-free RL.

Figure 24 compares the control commands sent from advanced controllers with the actual feedback from the system. Overall, the control commands were realized by the system, as two lines overlap most of time.

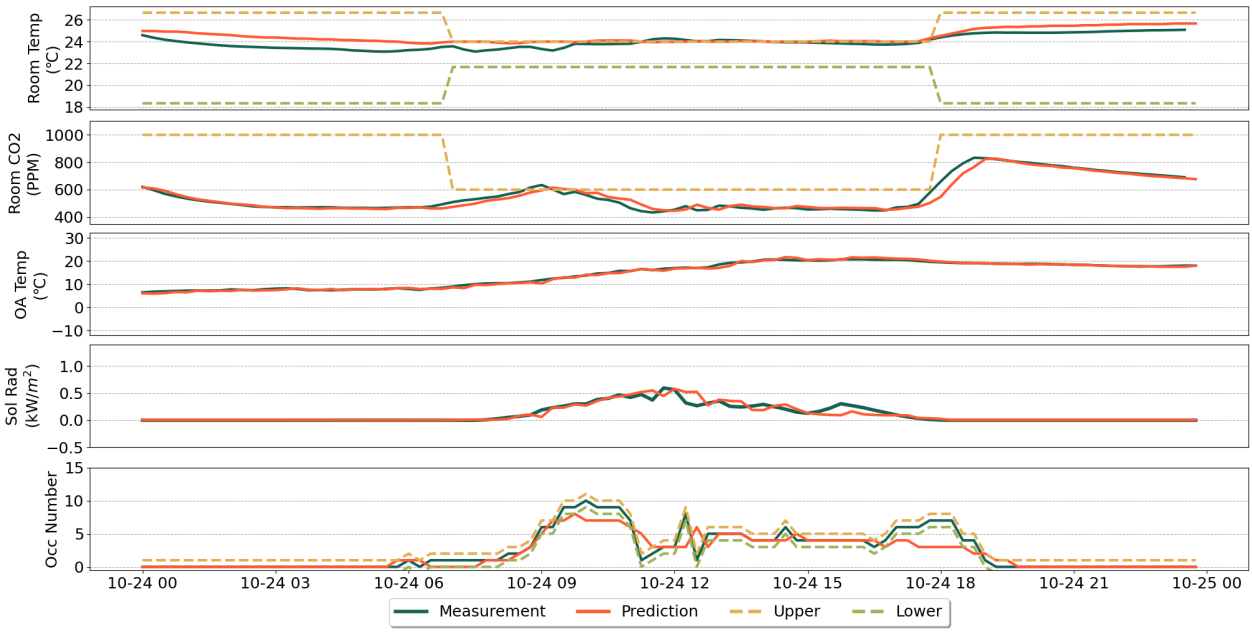


Figure 21. One-day results of HDDPC.

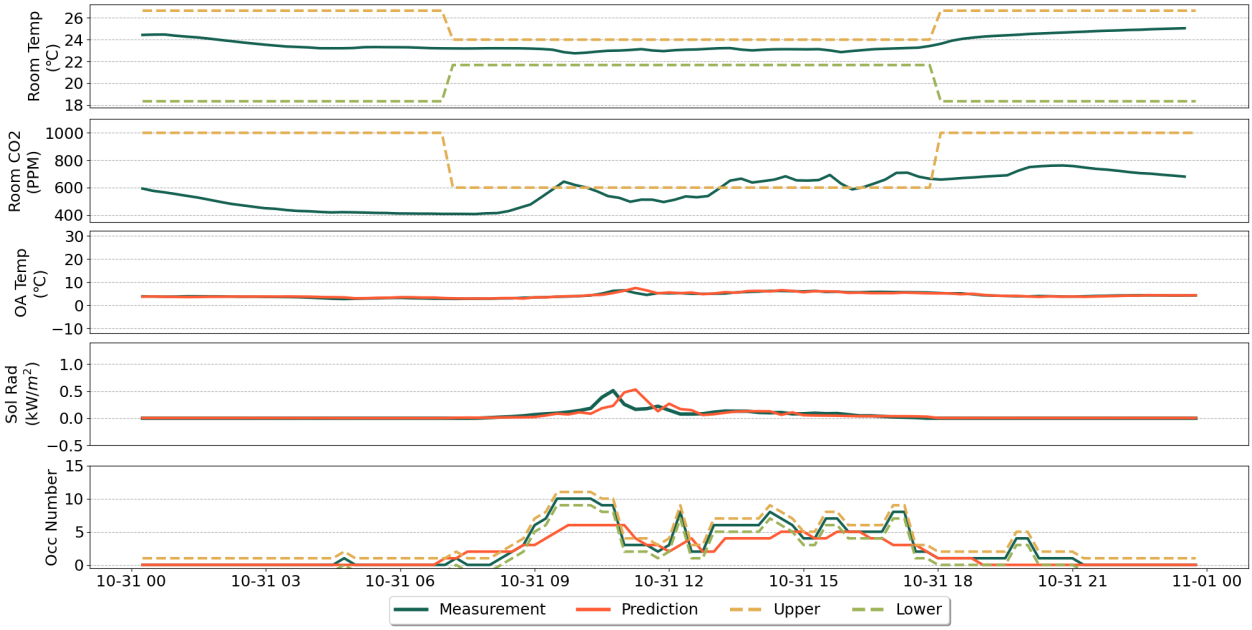


Figure 22. One-day results of DPC.

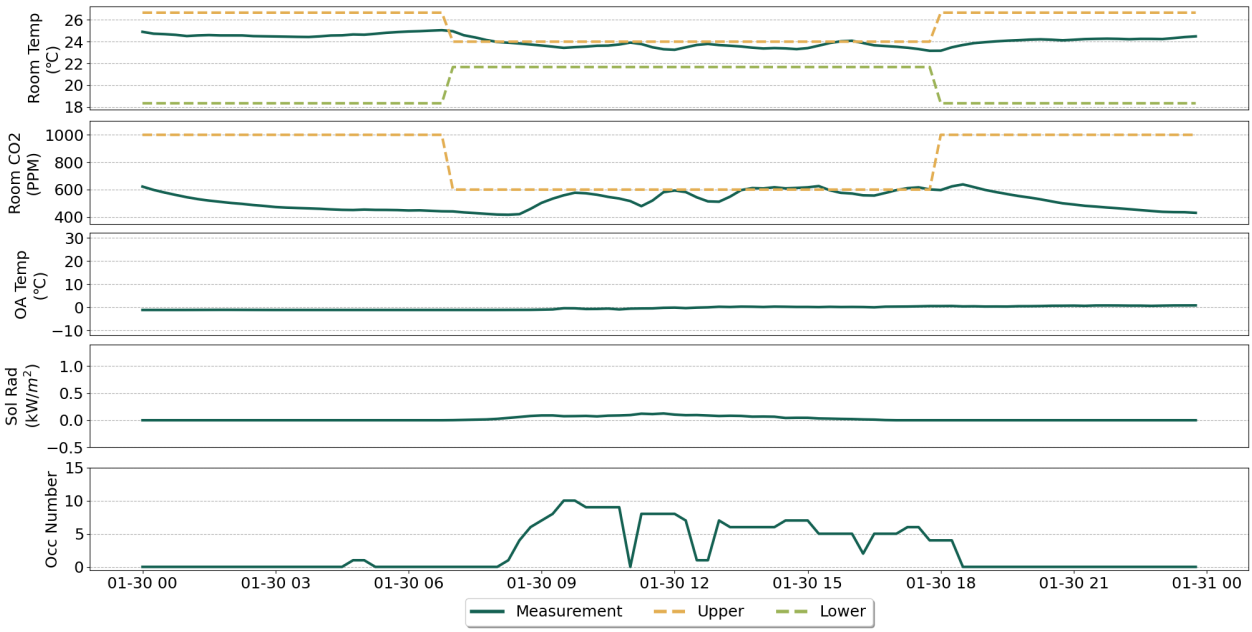


Figure 23. One-day results of RL.

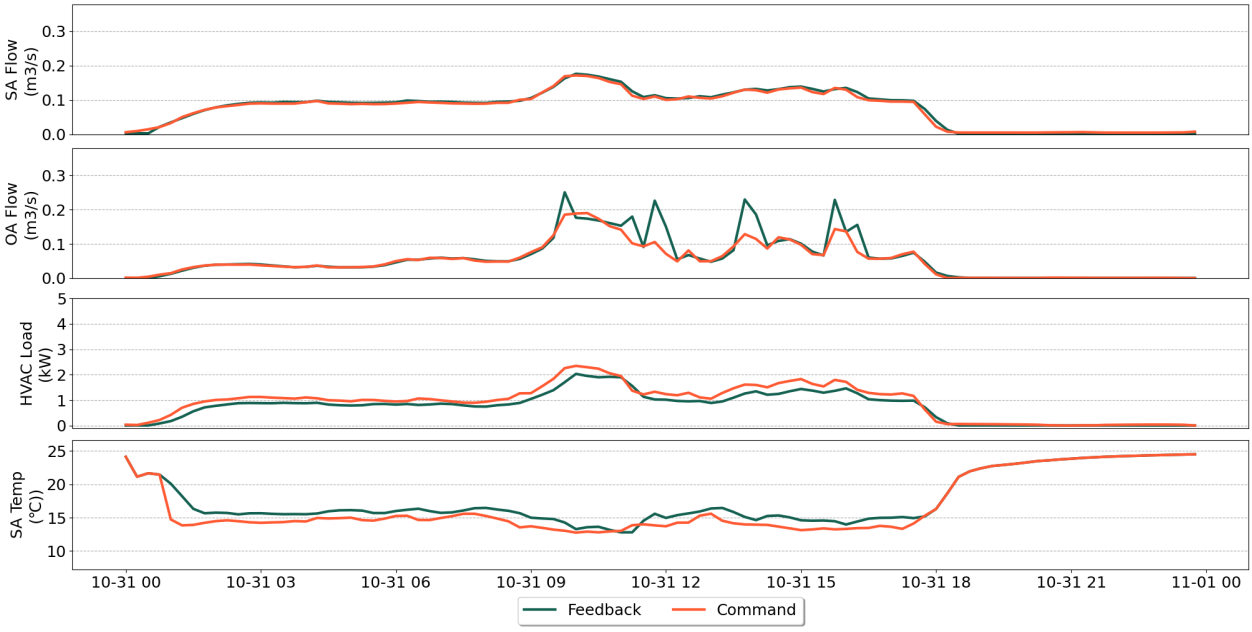


Figure 24. One-day results of control signals.

Overall Performance Evaluation of Different Controls

The overall IEQ performance is shown in Figure 25. For all controls, indoor air temperature is well maintained within our requirements. Temperature bound is only violated by DPC and RL by a small amount at noon. It is reasonable because during the training of the control law and agent, different loss terms are balanced based on the weight coefficient. From the optimization viewpoint, this results in a soft-constrained problem, so the temperature will not be maintained as good as hard-constrained optimization like HDDPC. Indoor CO2 concentrations are all well maintained within our requirements, whereas the baseline control had high CO2 concentrations, which indicates that we cannot meet 600 PPM CO2 concentration requirements based on ASHRAE guideline 36.

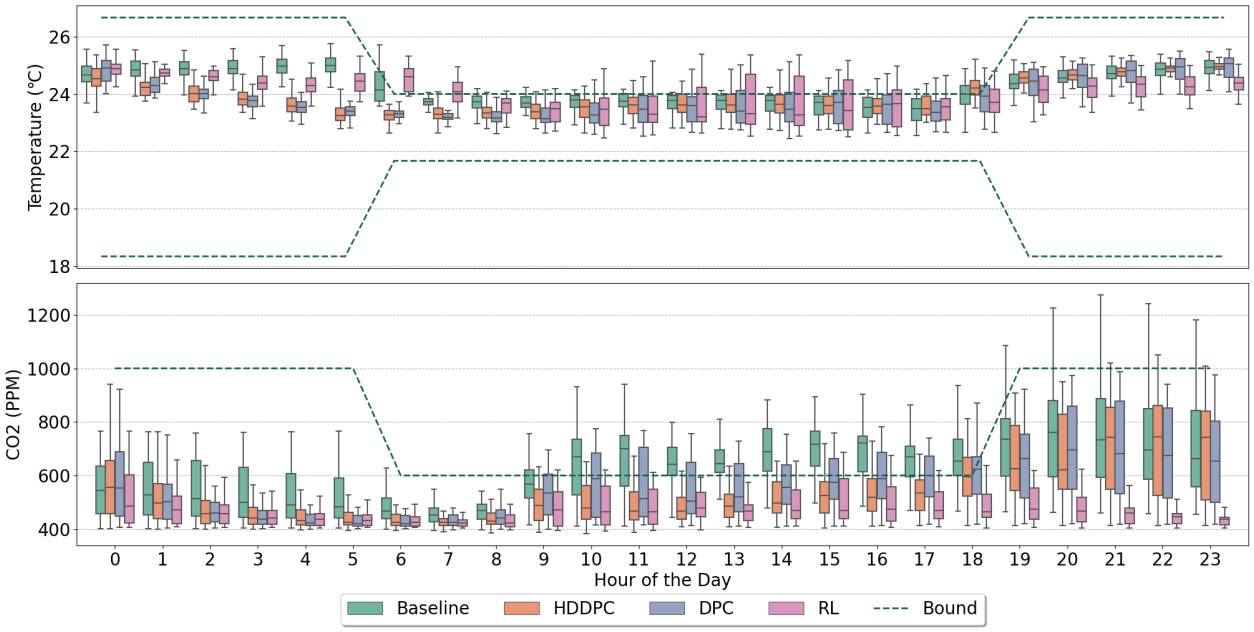


Figure 25. Overall IEQ performance of baseline control, HDDPC, DPC, and RL.

A more detailed comparison of IEQ requirement violations is shown in Figure 26, where we present the daily mean temperature violation against the daily mean CO2 violation. The daily violations were normalized based on their maximum and minimum values for a fair comparison. This figure highlights the confidence regions for each control strategy as shaded areas, with distance from the bottom left corner indicative of the extent of violations. For example, the green area is farther away than the orange area from the bottom left along the y-axis. It means that baseline control resulted in more CO2 violation than the HDDPC. Another example is that the pink area is farther away than the orange area from the bottom left corner along the x-axis. It indicates that the RL results in more temperature violation than the HDDPC.

HDDPC demonstrates superior performance in maintaining IEQ, as evidenced by the orange area's proximity to the bottom left corner compared to the green area, which represents baseline control. Both RL and DPC outperform baseline control regarding CO2 violation, while their temperature regulation is similar to the baseline performance, occasionally worse than the baseline. DPC's performance varies the most as the blue area is the largest among the four. It may be due to the stability of the training. Despite the requirement violation, during the experiment, we did not receive any complaints about the IEQ from the occupants.

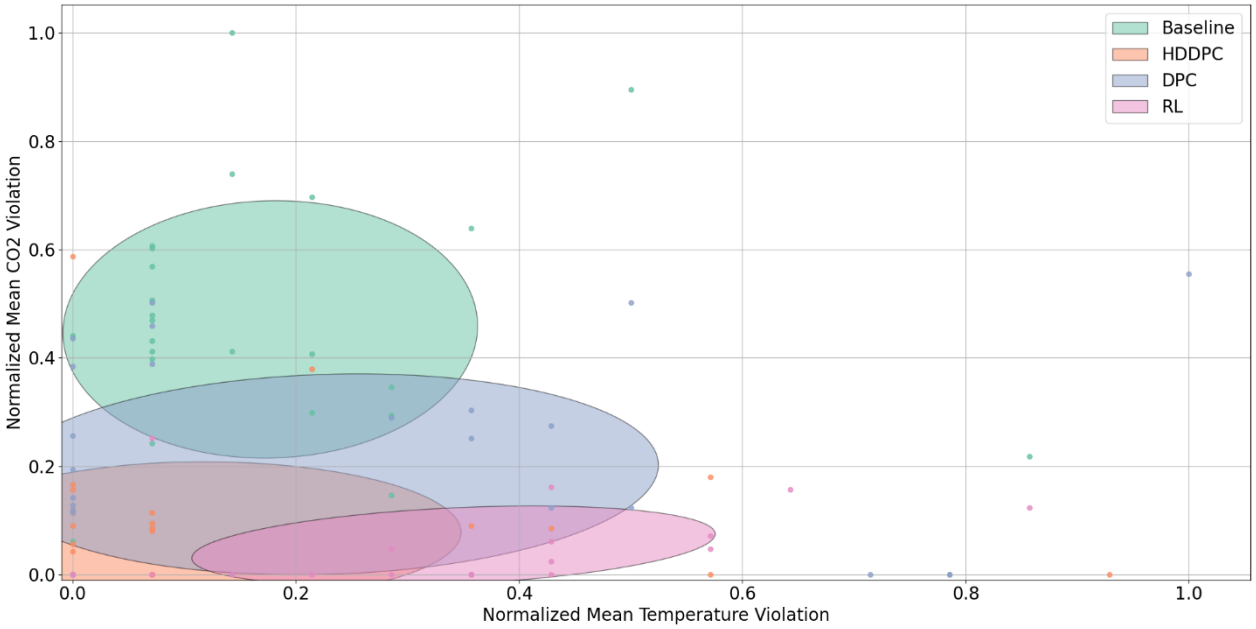


Figure 26. Temperature violation vs. CO2 violation of different controls. The further away the area is from the bottom left corner, the more violation it represents.

Figure 27 depicts the HVAC load dynamics across various control strategies at different times of the day over the whole testing period. The top plot indicates that all advanced controls exhibited pre-action (heating or cooling) behavior, whereas the baseline control only operated during the scheduled time. Such pre-action resulted in smoother load profiles for the advanced controls, avoiding the sharp spikes characteristic of the baseline's start-up moments. The lower graph further illustrates that throughout the day, all advanced controls reduced coil loads compared to the baseline. Comparing three advanced controls, DPC had a 5-hour pre-action with relatively higher coil load whereas RL and HDDPC had a 3-hour pre-action. During night hours, both HDDPC and DPC successfully shut down the system, while RL maintained minimal HVAC and coil loads rather than turning the system off. This difference stems from the deterministic nature of HDDPC and DPC's control decisions versus the stochastic control outcomes associated with RL. The stochastic sampling nature of RL agents makes it infeasible to turn off the system.

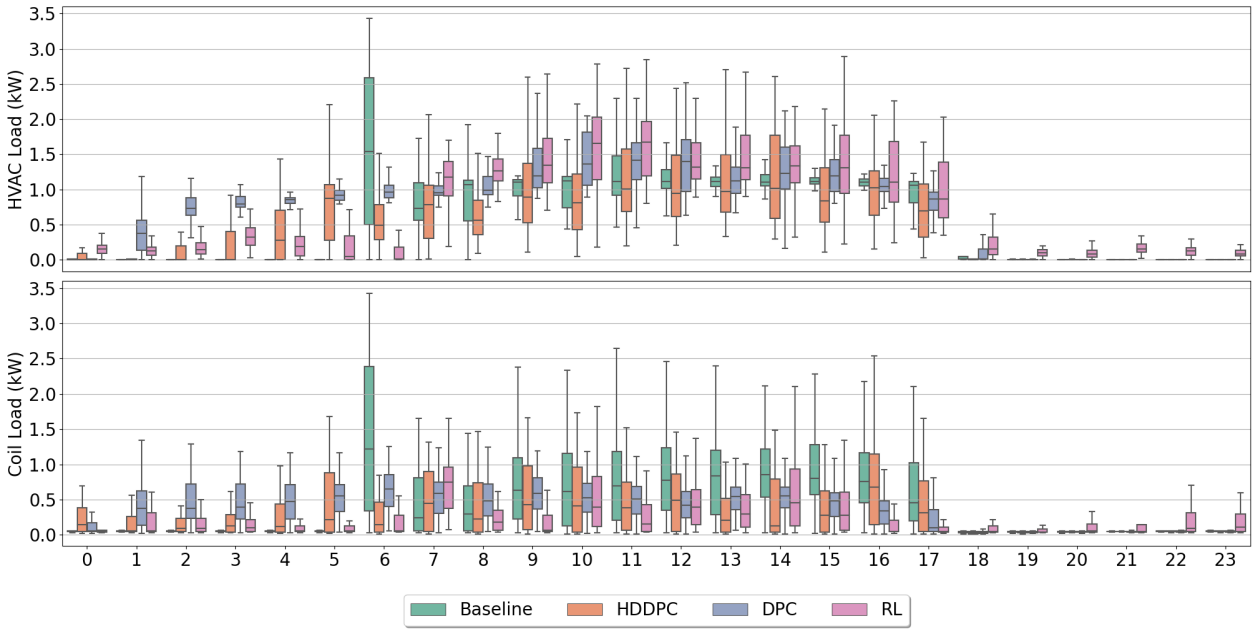


Figure 27. Overall load performance of baseline control, HDDPC, DPC, and RL.

To quantitively compare different controls, we categorized the daily results based on the daily average outdoor air temperature throughout the 108-day experimental period. After excluding data from a preliminary pilot study (19 days) and 20 days' experimental failures, we categorized the 69 days' results into intervals of every 5°C increase in mean outdoor air temperature, as detailed in Table 11. This approach was chosen over the degree-day method to mitigate biases in comparison.

Table 11. Number of experiments by daily mean outdoor air temperature (°C).

Median interval temperature (°C)	-5	0	5	10	15	20
Days. of experiments	1	36	11	13	7	1

The experiments in the intervals with median temperature -5 and 20 °C are not used in our analysis because they only contain one day of experiments respectively. Energy consumption data from experiments conducted at median outdoor temperatures of 0, 5, 10, and 15 °C were analyzed to evaluate the energy efficiency of advanced control systems compared to the baseline. Daily energy

consumption was measured using the energy meters installed on the coils. Table 12 presents the average energy consumption and its reduction (compared with the baseline) for each temperature interval across different controls. It is worth noting that the testbed space is an internal zone, so it requires the cooling load most of the time. This explains why the baseline coil energy increases with outdoor temperature. The results reveal that across all temperature intervals, advanced controls achieved significant energy savings up to 56.98%. Specifically, HDDPC led to the greatest reduction in energy use by 56.98% at maximum, followed by RL by 48.39% and then DPC by 30.6%. This result of energy savings is consistent with the results of a recent simulation study [31], where black-box MPC outperformed other methods, with RL coming in second, followed by the approximated MPC.

Table 12. Averaged coil energy (kWh) in each outdoor temperature interval (°C). The values in the parathesis are the reduction compared to the baseline.

Median interval temp. (°C)	0	5	10	15
Baseline	9.57	11.17	13.04	14.95
HDDPC	9.53 (0.42%)	9.77 (12.53%)	5.61 (56.98%)	6.83 (54.31%)
DPC	7.67 (19.85%)	10.61 (5.01%)	10.9 (16.41%)	10.38 (30.6%)
RL	6.68 (30.2%)	7.03 (37.06%)	6.73 (48.39%)	None

Experiment Failure Analysis

The study experienced 20 days of failed experiments. These failures were determined based on whether the control program operated throughout the entire day. The left plot on Figure 28 shows the total and successful number of experiments for each control strategy, while the right plot presents the causes and frequencies of failures.

The outer circle of the left plot in Figure 28 shows the total days of implementation for each control strategy. The inner circle presents the total days of successful implementation. The light-shaded area of the inner circle presents the faction of failed implementation for each control strategy. Among the four controls, DPC had the biggest fraction of failures, followed by HDDPC. RL and baseline control have the least number of failures.

The right plot shows that there are 20 failures in total and their causes. DPC and HDDPC encountered the highest incidence of failures, with five instances attributed to DPC due to issues with the plotting function after online training. This function, activated to assess training efficacy, led to increased memory usage over time as training data accumulated, resulting in program crashes. Beyond program malfunctions, the other cause of experiment failure was related to issues with the application programming interfaces (API) of the weather station and occupancy sensor. Specifically, the failures in communication of the control program with the weather station API caused three implementation failures, and the occupancy sensor's cloud service experienced five outages which also caused communication issues. These API-related issues were most prevalent during DPC and HDDPC implementations, accounting for their higher failure rates. Therefore, stable cloud services of occupancy sensors and weather stations are vital for the success of the advanced controls. In addition to those failures, there have been 5 other failures including 3 equipment failures and 2 failures due to time changes. The equipment failure is from the chiller, which resulted in insufficient capacities. The incidents of failure observed with the advanced controls underscore the recommendations made in [19]. It is advised to deploy an alarm system capable of sending out notification emails upon detecting a control failure. Additionally, the implementation of a backup controller is recommended to mitigate risks of discomfort due to

control failures, ensuring a fail-safe mechanism in place to maintain comfortable indoor environments.

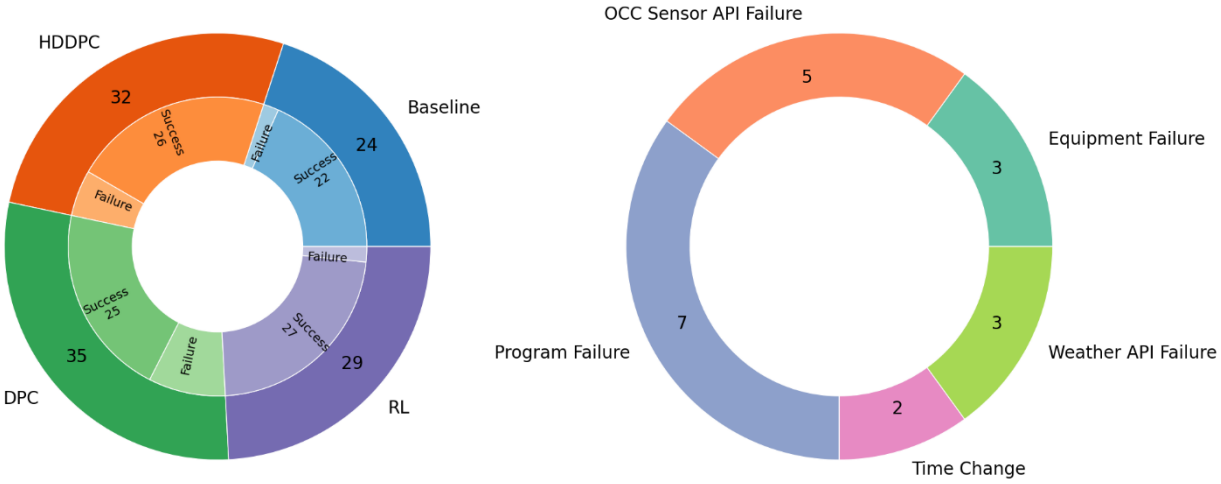


Figure 28. Number of failed experiments for each control and failure reasons.

The Impact of Training Episode on Real-Time RL Performance

The training of the RL agent was conducted with two different episode lengths. Initially, the RL agent was trained over 1-day episode lengths and evaluated within a simulation framework during the pilot testing period. Although simulation results were promising, real-world implementation revealed that RL consumed more energy than anticipated. To mitigate this issue, the training was adjusted to extend the episode length to 2 days. The measured performance of the two agents trained by 1-day and 2-day episodes is shown in Figure 29.

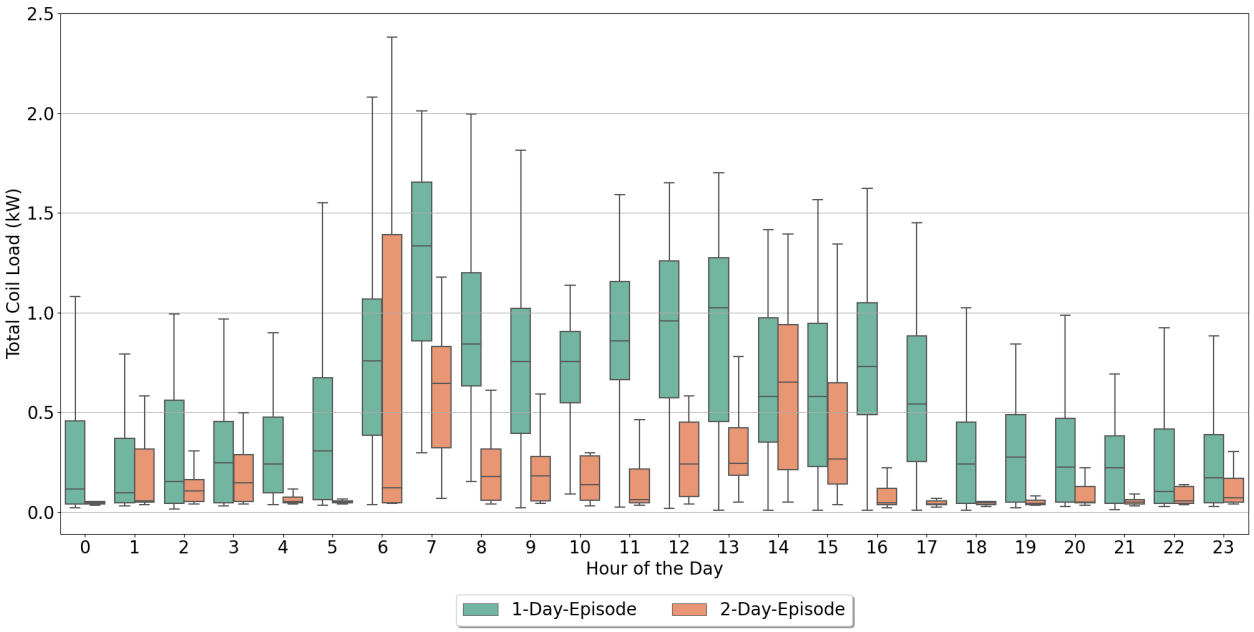


Figure 29. RL agent's performance by different episode length.

Adjusting the episode length significantly altered the RL agent's performance. The agent trained on one-day episodes exhibited higher energy consumption during the day and struggled to minimize the load after 6:00 PM. Figure 30 provide insights into these behaviors. The green represents the agent trained by the one-day episode, while the orange represents the agent trained using the 2-day episode. The white edge indicates the agent's behavior during the day before 6:00 PM, and the black edge indicates the behavior at night. During the day, the temperature difference between mixed air and supply air was higher under the control of the agent trained with one-day episodes. Furthermore, for the same temperature difference, the supply airflow rate was substantially higher. This combination of a greater temperature difference and increased airflow rate led to elevated energy consumption during daytime hours for the agent trained using the 1-day episode.

At night, the agent trained with two-day episodes maintained the supply air temperature closer to room temperature and had a smaller temperature difference between supply and mixed air, which resulted in a fully recirculated mode. It also managed to maintain the supply airflow rate to nearly zero. These adjustments enabled the two-day episode trained agent to consume less energy during night-time hours, illustrating the impact of extended episode length on energy efficiency.

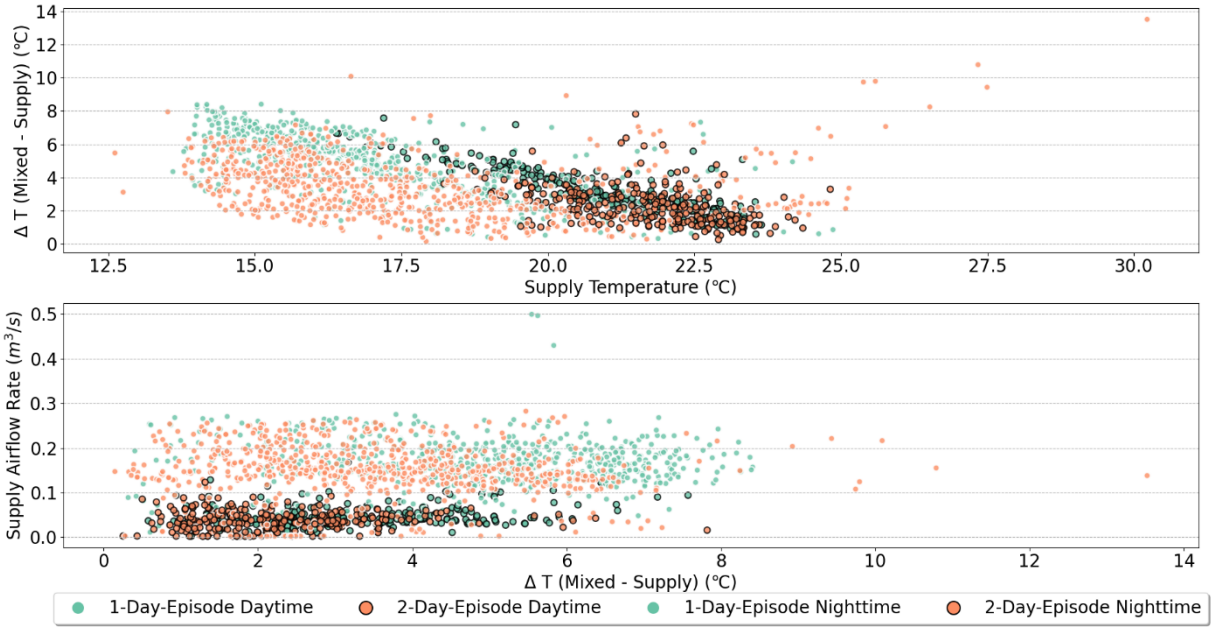


Figure 30. The agents' behavior during the day and at night.

The differences of such behavior can be further explained theoretically. In SAC, the state and action values are calculated based on the current and next states and actions by approximating the following calculation:

$$v(s) = E_{\tau}\{R_{t+1} + \gamma \cdot R_{t+2} + \gamma^2 \cdot R_{t+3} + \gamma^3 \cdot R_{t+4} \dots | S_t = s\} \tag{36}$$

Figure 31 shows the plot with $R_i = 1$. It shows that with a one-day episode, the information is truncated at the 96th step. This phenomenon will reduce the actual state and action values as the

time step approaches to the end of the day, which makes the agent take the states and actions after 6:00 PM less important. Therefore, implementing RL in a real building requires careful selection on the length of the training episode.

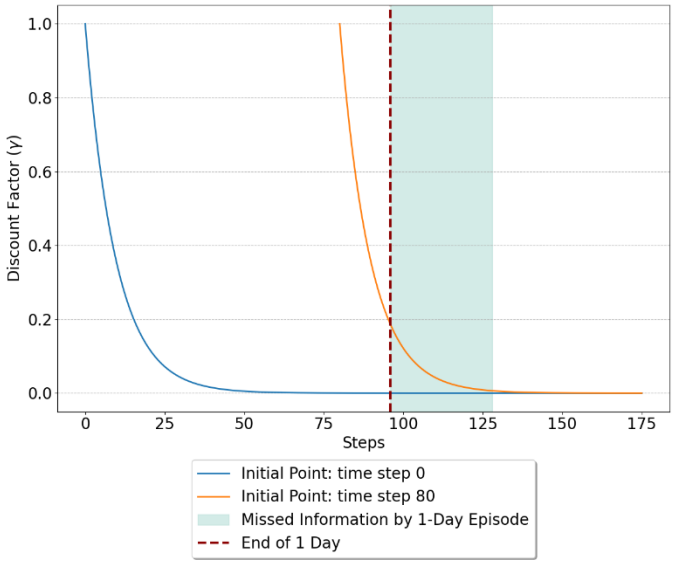


Figure 31. Information difference of different training episodes.

Agent-Based Control vs. Non-Agent-Based Control

In this section, we compare the agent-based and non-agent-based controls in terms of HDDPC, DPC, and RL. Specifically, the non-agent-based control is HDDPC because the control decisions are made by online optimizations. The agent-based controls are DPC and RL because the control decisions are made by neural networks.

Information Gap in HDDPC

Table 12 shows the energy reduction performance varies a lot not only for different controls but also for the same control under different outdoor temperature intervals. In Table 12, HDDPC provides the most energy saving when the outdoor temperature is more than 10 ℃. However, as the outdoor temperature decreases, the energy saving decreases significantly. At 0 ℃, HDDPC performed almost the same as the baseline, whereas DPC and RL still provide energy savings. Such a phenomenon can be explained by the formulation of different control strategies.

We formulated the optimization of HDDPC as a hierarchical control to compromise for the convexity as follows equations of room-level (equation (36)) and system-level controls (equation (37)):

$$\mathcal{P}_{seq} = \mathcal{P}_{HVAC} = \dot{Q}_{sup} c_p \rho (T_s - T_{room}) \tag{37}$$

$$\begin{aligned} \mathcal{P}_{sys} = \mathcal{P}_{coil} &= \dot{Q}_s c_p \rho \left(\frac{\dot{Q}_o}{\dot{Q}_s} T_{out} + \frac{\dot{Q}_s - \dot{Q}_o}{Q_s} T_{room} - T_s \right) \\ &= \mathcal{P}_{HVAC} + \dot{Q}_o c_p \rho (T_{room} - T_o) \end{aligned} \tag{38}$$

where c_p is the specific heat of air, ρ is the air density, and T_{room} and T_{out} are indoor and outdoor air temperature.

As shown in equation (36) and equation (37), when solving room-level optimization, the second term on the right-hand side of equation (37) is not considered. This is not an issue if the HVAC load is not close to zero as the second term of equation (37) will have more space to be optimized. However, when not required; load is required, the minimization of equation (36) does not necessarily lead to the minimization of equation (37). One example is that when room temperature and CO2 are close to or above the upper bound in the winter, the room-level controller will require

no HVAC load because the temperature setpoint is met, whereas the system level control will provide maximum outdoor airflow rate to meet IAQ requirement and lift the supply air temperature to be the same as the room temperature so that the room-level HVAC load is calculated to be 0. It will result in energy consumption at the system level. However, the system can provide free cooling to the room, which is more cost-effective. Such an information gap between the two-level controllers reduced the HDDPC performance in terms of energy saving.

If we do not care about the convexity, such two-levels controllers are integrated into one and treated the same formulation as in the DPC and RL. Such integration eliminates the information gap, which made the DPC and RL outperform HDDPC when outdoor temperatures are cold. The performance difference between non-agent-based and agent-based controls due to the information gaps was also found for drone control [138].

Impacts from non-optimal control decisions

Besides the information gap, non-optimal control decisions will also affect the performance of HDDPC and cause a difference in different controls. To facilitate real-time implementation, 2000 is set as the maximum iteration for two-level optimization. If the solver failed to find optimal solution within the maximum iterations, non-optimal control decisions will be made. As shown in Figure 32, when the non-optimal decisions were made, HDDPC consumed more energy than DPC and RL because the supply airflow rate and temperature difference between mixed and supply air were higher. Figure 33 shows the correlation between the number of non-optimal decisions and disturbances, i.e., outdoor air temperature and occupant number. The ticks on the perimeter represent the hour of the day. The ticks on the radius represent the occurrence of non-optimal decisions and the value of disturbances. For better readability, we normalized the outdoor

temperature to be on a scale of 0 to 25. From the left plot, the number of non-optimal decisions on the room-level increases as the normalized outdoor temperature is above 15. The increases in room level non-optimal decisions are 3-to-4 hours ahead of the increase in outdoor temperature. In the right plot, the peak of system-level non-optimal decisions is 1 hour ahead of the occupancy peak. Such behavior is related to PCICNNs, in which 4-hour outdoor temperatures affect the indoor temperature prediction the most, while it is 1 hour for the effect of occupants on the prediction of indoor CO2.

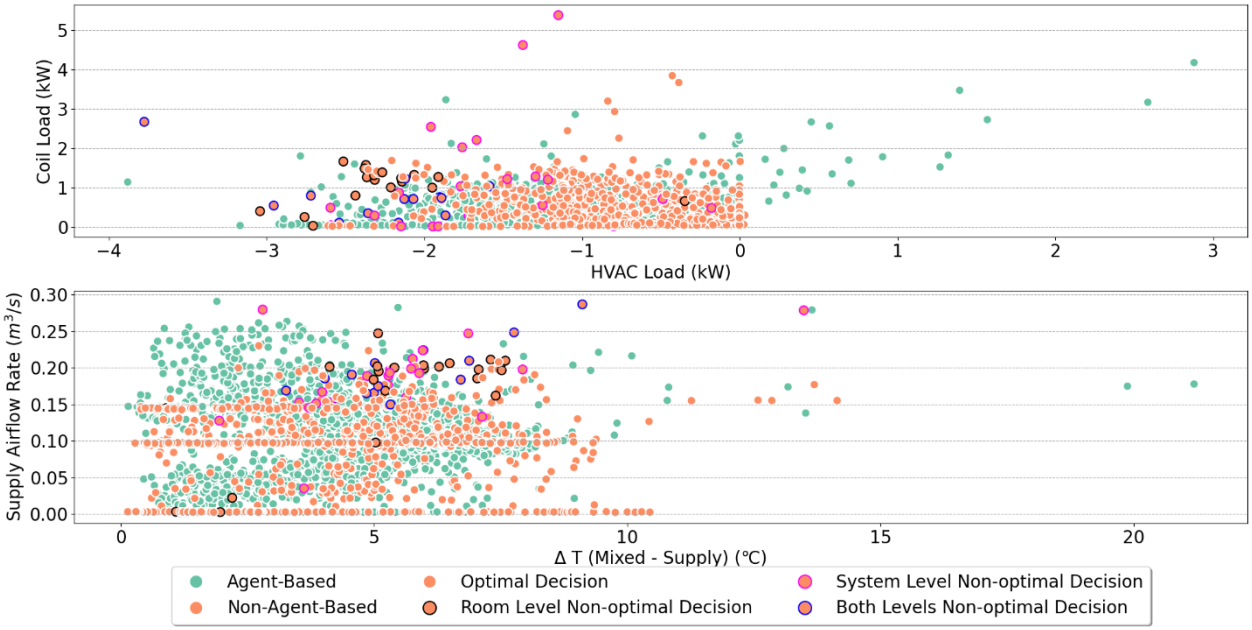


Figure 32. Comparison of control decisions of different controls under different scenarios.

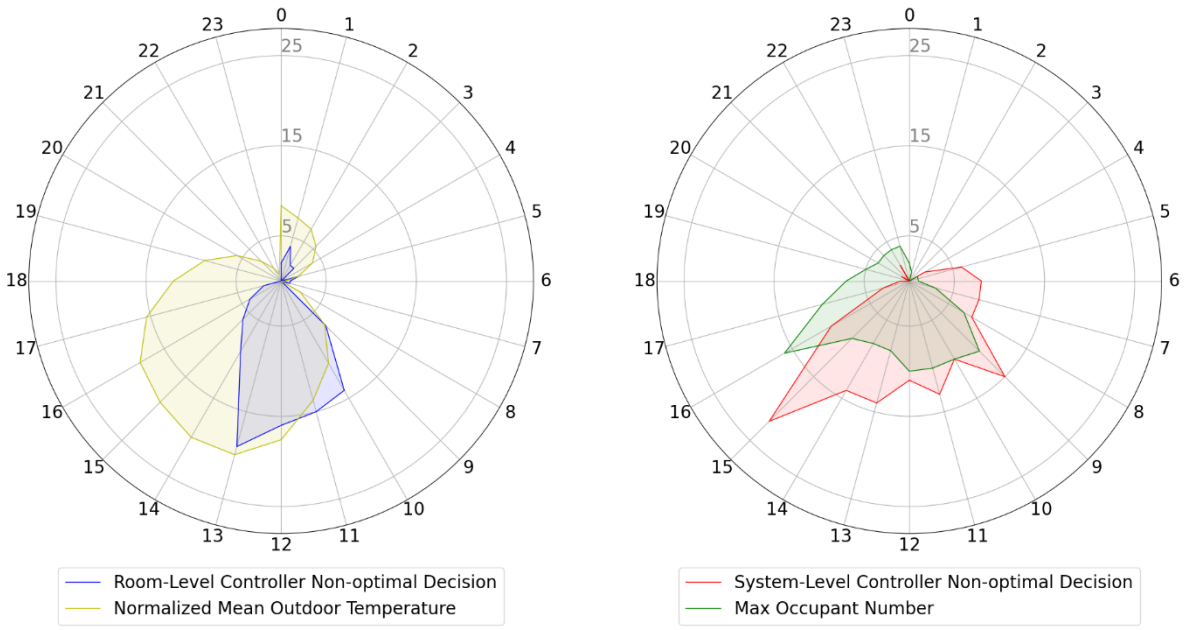


Figure 33. Correlation between non-optimal decisions and disturbances.

Computational Costs

Except for the performance on energy saving, another important aspect is the computational cost. Table 13 shows the time needed to make a step of control decision and the time to train the control law neural network and agent. From an online computing viewpoint, HDDPC requires more computational resources as it takes the longest to make control decisions. DPC and RL only need milliseconds. Therefore, the implementation of HDDPC has higher hardware requirements than the other two. The training of the DPC control law is directly based on the gradient of loss functions, so it is much faster than RL. Even with the measured data and replay buffer to increase the data efficiency and GPU cluster to speed up, the RL agent still needs more than 2 hours to train, but its energy performance is better than the DPC by 28%.

Table 13. The computational cost of different controls

Controls	HDDPC	DPC	RL
Control decision time (Seconds)	$10^{1}\sim10^{2}$	$10^{-3}\sim10^{-2}$	$10^{-3}\sim10^{-2}$
Training time	None	~20 min	~ 120 min

In summary, under different outdoor temperature scenarios, the control performance is different. If the information gap is small and optimal decision is made, methods like HDDPC which directly solving the optimization, lead to a better performance in terms of energy reduction. RL agent, despite saving a bit less than HDDPC, performs consistently for energy savings under different outdoor temperature intervals. From a computational cost perspective, HDDPC has the highest hardware requirements for implementation. DPC and RL both have lower online computing resource requirements. In addition, RL requires the most offline computing resources.

In this case study, we implemented HDDPC, DPC, and RL in a real building testbed over more than 5 months. The testbed is modeled by PCICNNs for online optimization of HDDPC and offline training of DPC and RL. The three advanced controls are compared with the baseline control which follows the default control logic of the system and ASHRAE Guideline 36 for ventilation requirement. The experimental results indicate that all advanced controls could maintain the IEQ within the requirements, with a few cases where DPC and RL performed slightly worse than the baseline. In terms of energy savings, all three advanced controls outperformed the baseline control. HDDPC provided more than 50% energy reduction at the maximum, followed by RL with 48%, and DPC with 30.6%. The experimental results reinforce the results from state-of-the-art simulation in literature. The failure analysis revealed that the stability of the communications between control program and occupancy sensor and weather station APIs is the most relevant to

the success of real-time implementation. We also found that the length of the training episode plays an important role for RL performance. Short episode length will lead to missing information in the value calculation and therefore degrade the RL performance. By comparing agent-based and non-agent-based control, we found that even though the hierarchical formulation of HDDPC could guarantee convexity and provide the most significant energy-saving most time, the information gap between the two-level controllers leads to performance degradation. Such an issue does not happen to agent-based controls, i.e., DPC and RL because we do not need to make compromises for the convexity. In addition, if not finding the solution within the maximum iteration, the non-optimal control decisions will degrade the HDDPC's performance significantly. In terms of the computational resource requirements, HDDPC requires the most online computing resources, which will raise the hardware requirement for its implementation. DPC and RL require fewer online computational resources as the burden is shifted to the offline. DPC outperformed the RL in the training time, but RL outperformed in IEQ control and energy saving. The findings of this research shed light on a comprehensive understanding and future research direction of advanced controls in buildings.

4.3 Case Study 3: PIML Model Benchmarking

Beyond the control comparison of advanced controls based on PCICNN, we evaluate PCICNN's performance with another two state-of-the-art PIML models. Synthetic datasets are used for a fair comparison of the model's performance. The dataset with normal HVAC operation is used to evaluate training efficiency of the models. The dataset consists of 240 days of data, with 180 days of data for training, 30 days of data for validation, and 30 days of data for testing. For the hyperparameter tuning, we first performed an informal sensitivity analysis to identify which hyperparameters had the greatest influence on model performance. The grid search was then

restricted to those influential parameters, while non-sensitive parameters were fixed to default values from the original implementations. For each combination of hyperparameters, the best model is saved based on the smallest loss on validation dataset. Early stopping was not standardized across models; instead, each model used the stopping rule provided in its original source code. The final value of each hyperparameter is selected based on the smallest loss on the testing dataset. Table 14 shows the identified hyperparameters, their ranges, and final value of each hyperparameter for each model.

Table 14. Results of hyperparameter tuning of the models.

Model	Hyperparameter	Search Range	Final Value
PCNN	Learning rate – unforced dynamics	[5e-2, 5e-3, 5e-4]	5e-4
	No. of layers	[2, 3]	3
	Hidden layer size	[32, 64, 128]	128
	Learning rate multiplier – energy emulator	[1, 10, 30]	10
	Division factor	[10, 30]	30
ModNN	Hidden layer size – external module	[4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28]	16
	Hidden layer size – internal module	[4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28]	12
PCICNN	No. of layers	[3, 4, 5]	4
	Enable/disable bias	[False, True]	True
	Neurons per time step	[5, 10, 15, 20]	15
	Coefficient of weight-related loss	[1, 5, 10, 20]	10
	Coefficient of bias-related loss	[1, 5, 10, 20]	1

Prediction Accuracy

The overall testing performance of the three models and the error distribution for each testing day is shown in Figure 34. All models capture a similar trend to the ground truth. The average MAPE

for the three models are 3.35%, 3.02%, and 5.03%, respectively, for PCICNN, ModNN, and PCNN. From the error distribution, the daily MAPE of PCICNN and ModNN are less than 5% except 1 day for PCICNN and 3 days for ModNN, indicating a stable performance on the testing dataset. For PCNN, even though it predicts a similar trend to the ground truth, its performance is not as stable as the other two models from its error distribution since almost half of the testing performance has more than 5% MAPE.

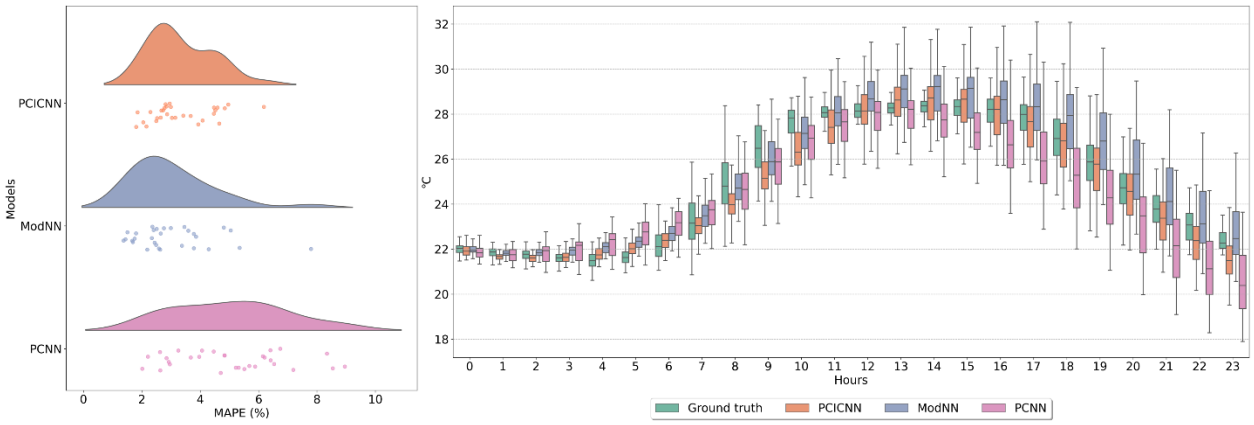


Figure 34. Testing Performance of the models

Physical Consistency Analysis

To evaluate the physical consistency of the models, two extreme cases are tested. The first case simulates the temperature dynamics when the HVAC system is suddenly off. The second case simulates open-loop temperature dynamics. Figure 35 shows the results of the first case for the three models. When the HVAC is suddenly turned off, all models have physically consistent predictions as the temperature predictions increase. However, the accuracy differs between models. PCICNN and ModNN not only have physically consistent trends but also have accurate predictions, whereas PCNN only exhibits an upward trend with relatively low accuracy.

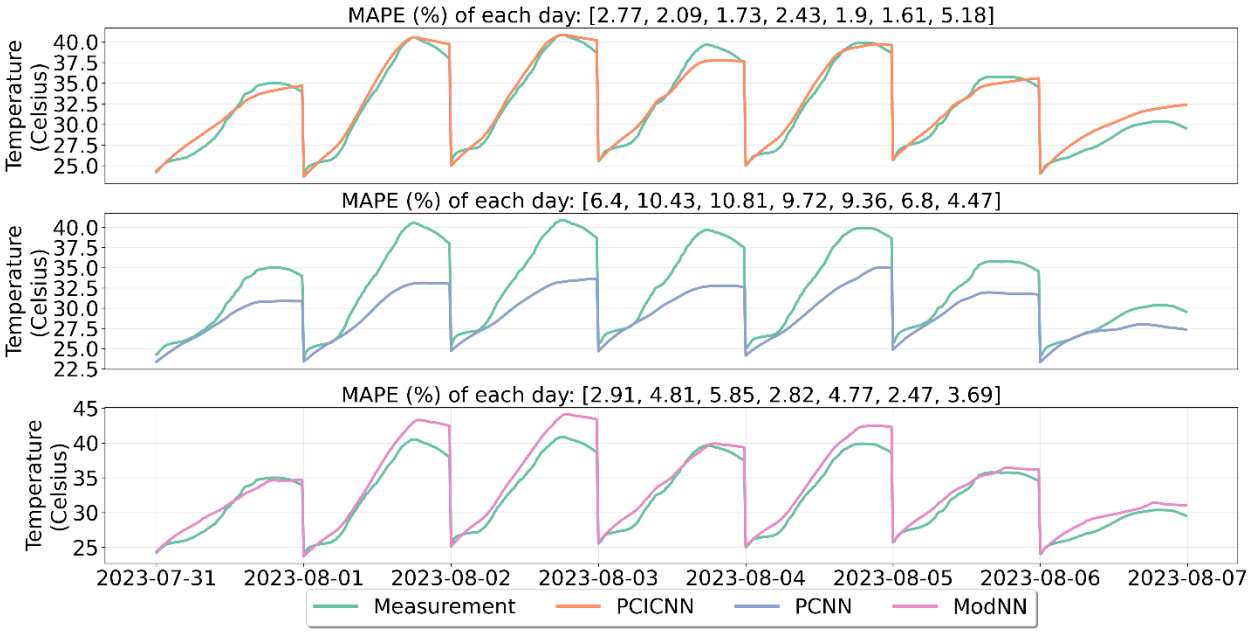


Figure 35. Performance of the case when HVAC is suddenly off.

Figure 36 shows the models' performance for the open-loop case that HVAC is always off. Daily reset is used, i.e., the room temperature is reset to the ground truth every night. Similarly to the first case, PCICNN and ModNN predict the temperature dynamics accurately. However, PCNN failed to provide accurate predictions. The predicted temperature from PCNN monotonically decreases instead of decreasing at night and increasing during the day.

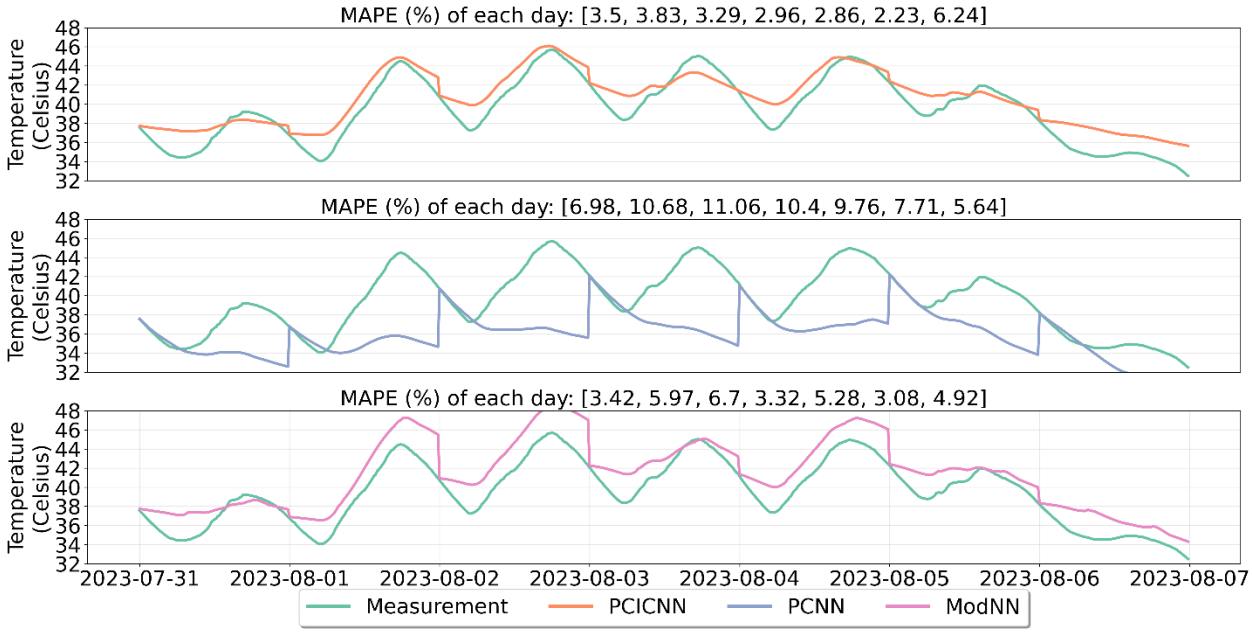


Figure 36. Performance of the case when HVAC is always off.

Besides the quantitative comparison of the prediction accuracy under extreme cases, Figure 37 shows the gradient of the models with respect to its common inputs. The three rows represent PCICNN, ModNN, and PCNN, respectively. The five columns represent the gradient with respect to current indoor temperature and the sequence of HVAC load, outdoor temperature, solar radiation, and occupancy, respectively. The gradients are normalized between -0.1 and 1 for better visualization. The color bar from bottom to top indicates the magnitude of the gradient. The gradient is calculated for 6 hours and averaged for the entire testing dataset. The y-axis indicates the time steps 1 to 24 of the input from top to bottom. The x-axis indicates the time steps 1 to 24 of the temperature prediction from left to right.

The three models exhibit similar gradients of the temperature prediction with respect to the current temperature. As the prediction time step leaves the current time step, the impact of current indoor temperature monotonically decreases. Similar behavior can also be observed from the second

column of plots for the HVAC load. The HVAC load that is one step before the prediction time step takes the most impact, and the impact decreases as the step difference between output and input increases. For outdoor and solar radiation, which are shown in the third and fourth columns of the plots, PCICNN and ModNN have similar behavior. Instead of monotonically decreasing with the increase of step difference, the gradient first increases and then decreases. The peak gradient is when outdoor temperature and solar radiation is around 1.5 hours before the prediction step. This is physically meaningful because it indicates that PCICNN and ModNN learned thermal lagging effect of the building.

However, for the PCNN, monotonically decreasing is still observed for outdoor temperature gradient, and solar radiation has no effect on the indoor temperature prediction as the gradients are zero. The zero gradient with respect to solar radiation for PCNN also explains why PCNN predicts indoor dynamics accurate before sunrise but poorly during the day in Figure 35 and Figure 36. The gradient with respect to occupancy, which is in the fifth column of the plots, is also worth noting. The three models learned differently for the occupancy. The gradient of PCICNN makes no sense because the impact increases with the increase of the step difference, which is counterintuitive. Interestingly, for PCNN, the lagging effect is learned to be associated with occupancy. Moreover, negative gradients are observed for PCNN, which is also counterintuitive and does not make any physical sense. Lastly, ModNN learns the most plausible gradient for the occupancy.

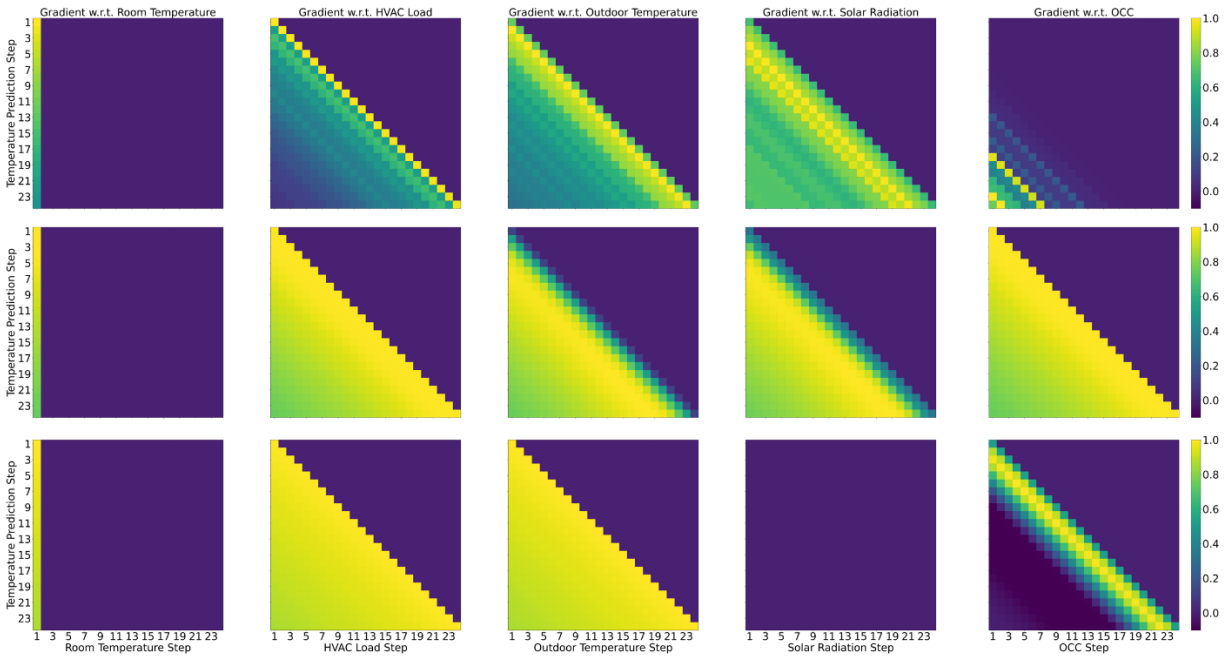


Figure 37. Gradient plot of the three models. The three rows represent PCICNN, ModNN, and PCNN, respectively. The five columns represent the gradient with respect to indoor temperature, HVAC load, outdoor temperature, solar radiation, and occupancy, respectively.

Training Efficiency

Besides the model accuracy and physical consistency, the training efficiency is also evaluated. The dataset with normal HVAC operation is used for training. The random seed is fixed as 42, which is the same as hyperparameter tuning. Figure 38 shows the prediction accuracy versus the training dataset length. The models are trained starting with 5 days of data with 7 days of data increasing. The maximum training dataset length is 180 days. With 5 days of data, PCNN's MAPE is only around 6%, but with the increase of the data, its performance does not improve much. ModNN's performance is more stable with respect to the training dataset length. The performance keeps improving until there is more than 75 days of data. For PCICNN, although the overall trend improves with more data, it still fluctuates, indicating an instability of the training performance.

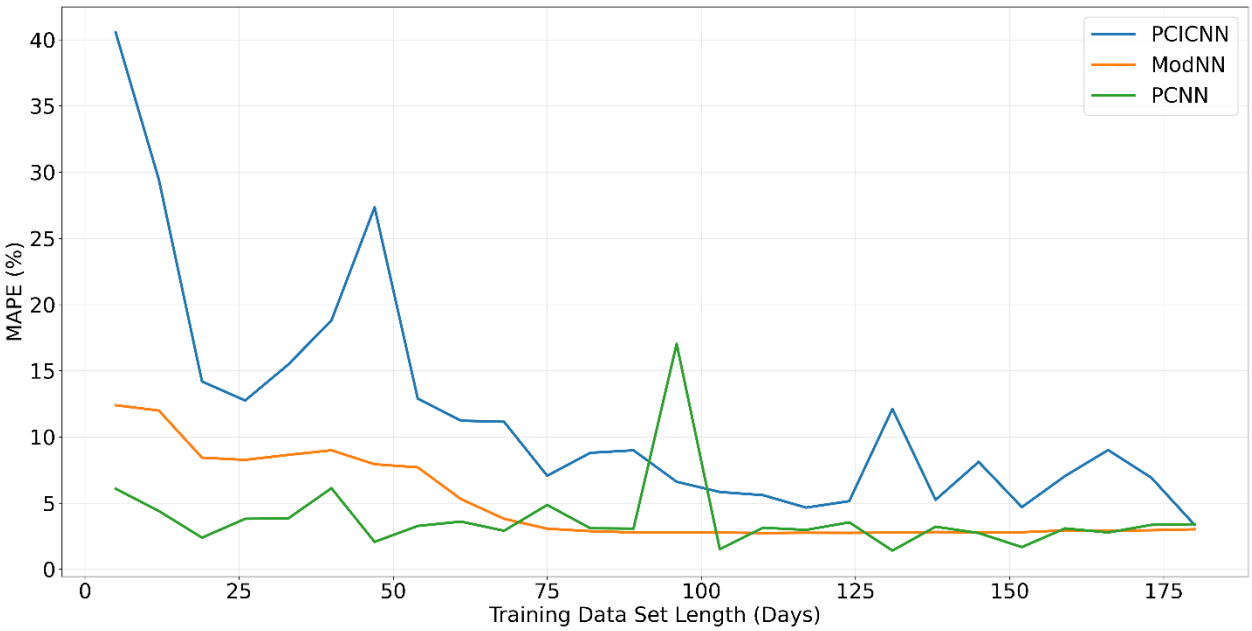


Figure 38. Model accuracy vs. training dataset length.

Table 15 shows the computational cost for model training. The computational cost is evaluated based on the number of updates, total training time, and time per update to get the best model. Overall, ModNN is the most computational effective, as it requires the least updates and training time. Although PCICNN requires more updates, its total training time is not significantly increased compared to ModNN. Lastly, PCNN is the most computational expensive. It requires the most updates and more than 1 hour to finish training.

Table 15. Computational cost of the three models.

Model	Number of updates	Total time (min)	Time per update (s)
PCICNN	2935	13.25	0.27
ModNN	1377	5.76	0.25
PCNN	3441	104.77	1.83

Real-World Implementation

For the three models, we implemented RL and HDDPC to our testbed. The experiment lasts for 49 days. The first 7 days were pilot tests. During the formal testing, there were 6 days with failure due to various reasons such bad cloud communications and equipment failures. Figure 39 shows the indoor temperature results during the implementation. Note that for ModNN, we only implemented RL as the coupling of the input features makes it unable to implement model predictive control. Overall, the temperature was controlled within the bonds. For RL with different models, slight temperature upper bond violations were observed. RL training can be treated as solving multi-objective optimization, with soft constraints in the objective. The trade-offs between constraints and energy usage are inevitable.

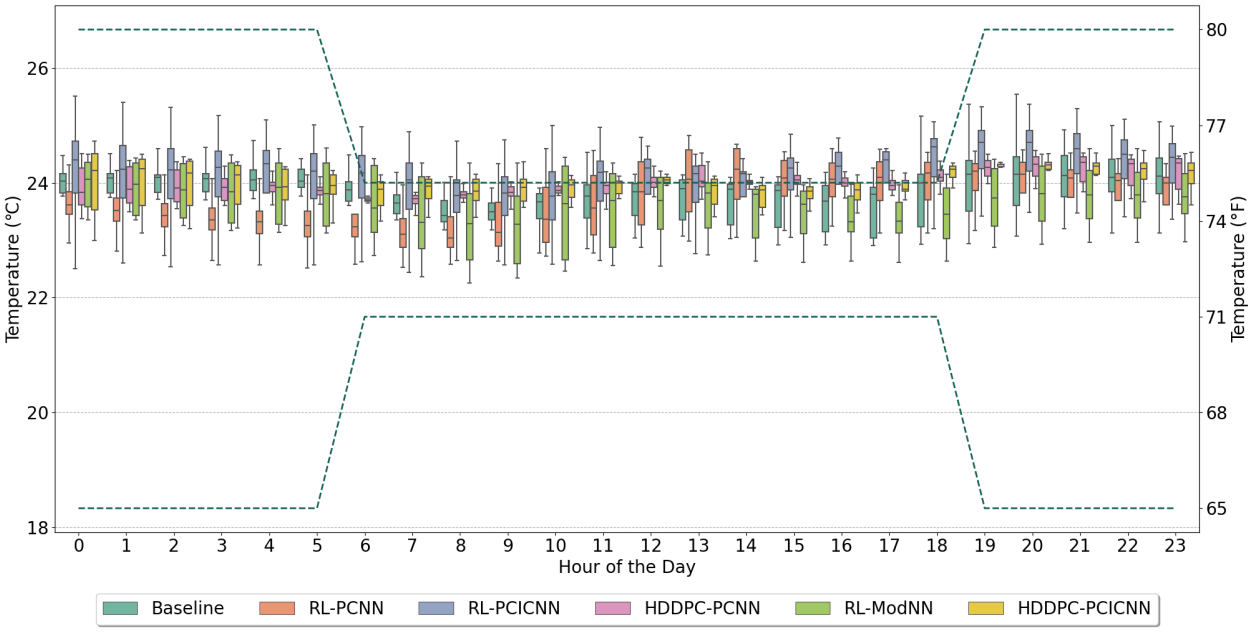


Figure 39. Indoor temperature results of different controls during implementation.

The quantitative evaluation of the control performance is shown in Table 16. The experiment above benchmark temperature is used. Because different controls have different numbers of

implementation days and outdoor weather conditions, we normalize the energy usage based on the cooling degree day and the number of implementation days, shown as energy intensity in the table. The saving is calculated by comparing the energy intensity with the baseline control. For different controls, energy saving ranges from 35.9% to 88.73%. The highest energy saving comes from RL based on PCNN. RL based on ModNN achieves the least energy saving. As for the temperature violation, it is quantified based on the average degree hours of violation. RL with all three models exhibit higher violation than baseline. Again, the violation of RL is inevitable due to the trade-off between different objectives and discrepancies between simulated and real environments. The HDDPC with both PCICNN and PCNN improves indoor comfort by reducing daily degree-hour violations compared to the baseline.

Table 16. Energy and temperature violation results of different controls during implementation.

Control strategy	No. of days	Avg. degree-days (°C·day)	Total energy (kWh)	Energy intensity (kWh/°C·day)	Energy saving (%)	Comfort violation (steps)	Comfort violation (°C·h/day)
Baseline	6	7.2	82.6	11.5	–	59	1.26
RL-PCNN	3	5.3	7.1	1.3	88.7	67	4.3
RL-PCICNN	5	6.0	32.4	5.4	53.0	144	5.0
HDDPC-PCICNN	4	5.5	28.9	5.3	53.5	63	0.7
HDDPC-PCNN	5	6.0	33.6	5.7	50.8	87	0.9
RL-ModNN	4	6.5	47.4	7.4	35.9	77	2.4

As for the computational speed, RL based on the three models requires less than a second to make decisions, which is consistent with our previous study. For the HDDPC-PCICNN, it needs 10 to 100 seconds to solve the two levels of optimization problems. For the HDDPC-PCNN, it needs 1

to 10 seconds to solve the problems. The difference between the computational time of the two HDDPC is due to nonlinear and linear optimization for PCICNN and PCNN, respectively.

Impact of Model Performance on Control

From Table 16, HDDPC-PCICNN and HDDPC-PCNN have a similar performance in terms of energy saving. The two controls are nonlinear and linear programming, respectively. Existing studies [50][139] suggested that linear models are competitive with non-linear models. However, from the HVAC system viewpoint, a different conclusion can be made. Figure 40 to Figure 43 show the results of indoor temperature and control signals for HDDPC-PCNN and HDDPC-PCICNN for June 2nd and 7th, respectively. For the two days of implementation, indoor temperature aligns with the upper bound, indicating it is controlled in a cost-effective manner. However, the cycling of control signals is observed for HDDPC-PCNN. Note that the formulation of the two controls is the same except for predictive models of indoor temperature. Action smoothness is also considered in the objective function. The control signal difference can be explained by the prediction difference of indoor temperature. Figure 40 shows that PCNN either overestimates or underestimates the temperature when it is close to the upper bound, whereas Figure 42 indicates that PCICNN has accurate predictions. The misestimation of PCNN on indoor temperatures makes the control decision cycle like on/off control. For HDDPC-PCICNN, the accurate prediction of indoor temperatures makes the control signal smooth. Although HDDPC-PCICNN and HDDPC-PCNN perform similarly in terms of temperature violation and energy savings, HDDPC-PCNN can make harmful decisions for the HVAC system: repeated cycling increases actuator wear, shortens equipment lifetime, and may reduce occupant comfort due to fluctuating airflow. While short-term energy savings may appear favorable, unstable or oscillatory control signals have important practical consequences. Rapid cycling of airflow rates or supply air temperature

increases actuator wear, shortens damper and valve lifetimes, and can trigger unnecessary alarms or overrides in BAS supervisory logic. In occupied spaces, such cycling can also lead to noticeable drafts or temperature fluctuations, negatively affecting comfort perception. These real-world consequences highlight why physical consistency and smooth control trajectories are critical metrics – not just energy savings – when evaluating PIML models for building deployment. In contrast, the smooth control behavior of HDDPC-PCICNN places less mechanical stress on the equipment and is more suitable for long-term deployment. The discrepancy between predictions of PCNN and measurements may result from the model expressiveness and underestimation of solar radiation. Considering both energy and system reliability, HDDPC-PCICNN therefore outperforms HDDPC-PCNN, highlighting the advantage of physically consistent nonlinear models in predictive control applications.

The results of this study align partially with existing findings in the literature. Prior works have shown that linear models can perform competitively with nonlinear models in predictive control when the prediction horizon is short or disturbances are mild [50][139], which is consistent with our observation that HDDPC-PCICNN and HDDPC-PCNN achieves similar energy savings. However, our results diverge from previous studies in terms of control quality: we show that physical inconsistency of PCNN model produces oscillatory actuation patterns that are not visible when performance is evaluated only through energy metrics. This highlights a limitation in earlier work, where evaluation criteria were dominated by energy savings and did not explicitly account for actuator stress, comfort stability, or physical consistency. By contrast, the dual-phase evaluation used in this study reveals that models with stronger embedded physics (e.g., ModNN and PCICNN) yield smoother, more reliable control trajectories, even when energy savings appear similar.

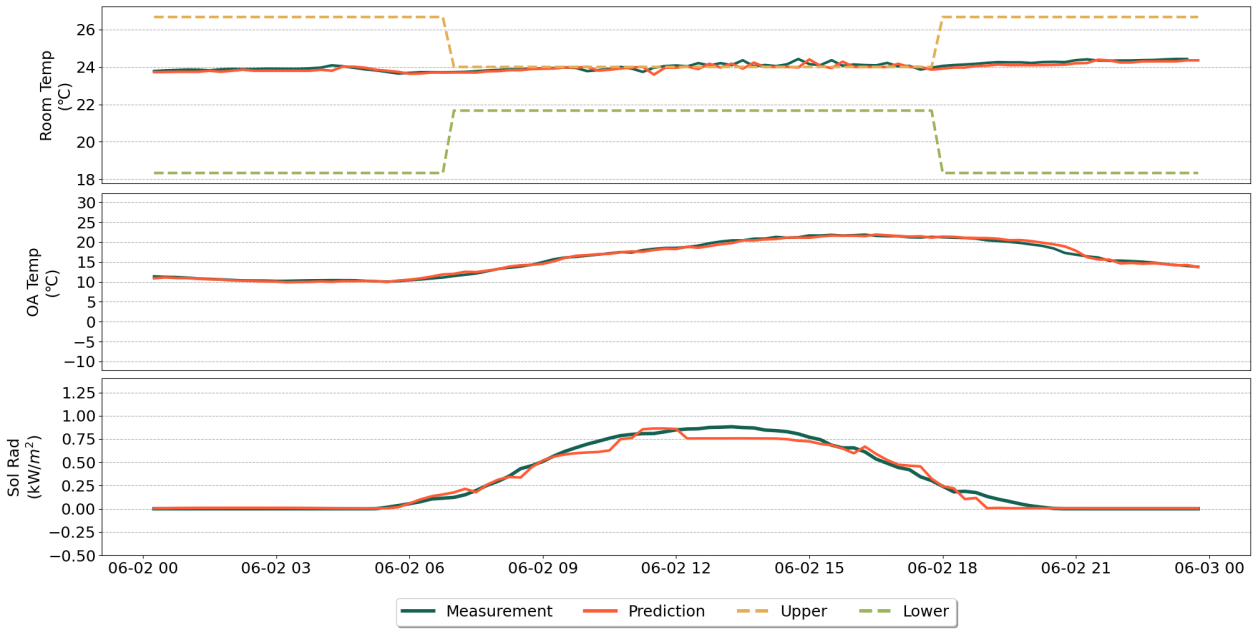


Figure 40. Indoor temperature results of HDDPC-PCNN on June 2nd.

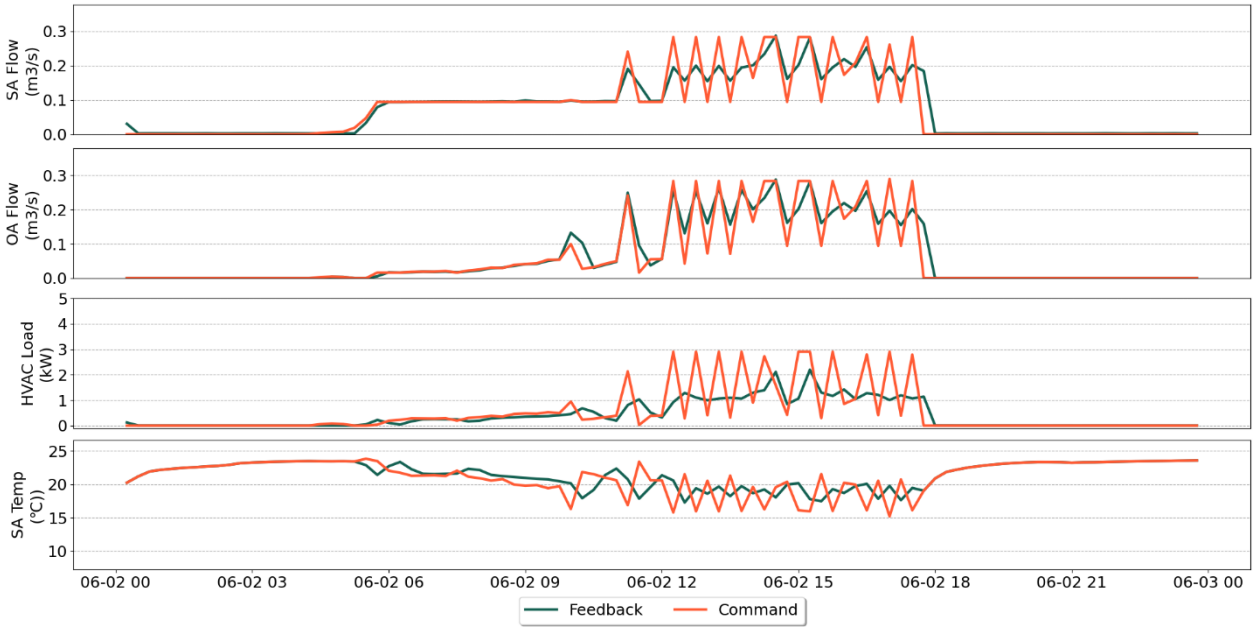


Figure 41. Control signal of HDDPC-PCNN on June 2nd.

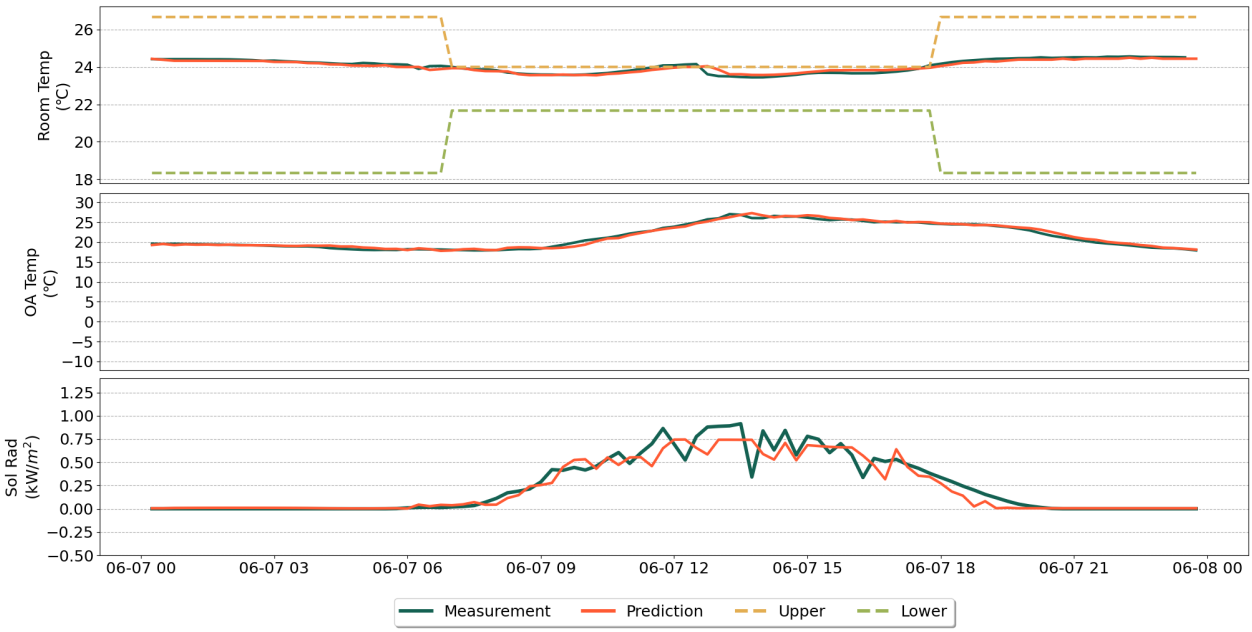


Figure 42. Indoor temperature results of HDDPC-PICCNN on June 7th.

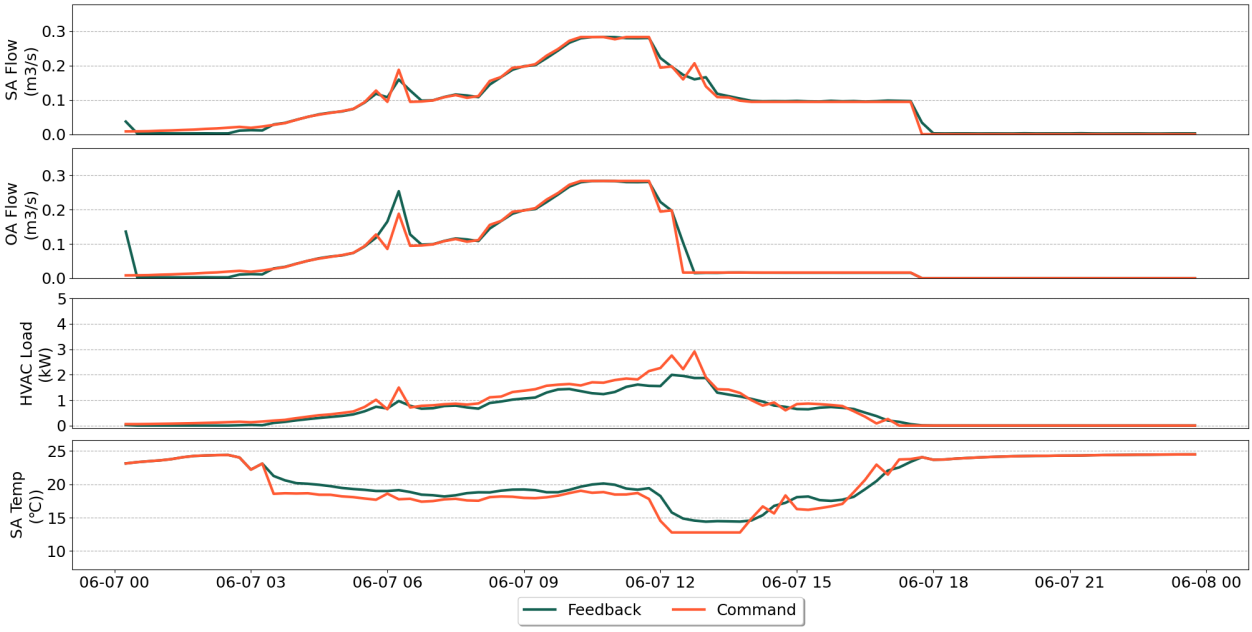


Figure 43. Control signal of HDDPC-PCICNN on June 7th.

For the RL implementation, RL-PCNN provides the highest energy saving. It may be due to experimental uncertainty. The total days of experiment and degree-day of RL-PCNN is the lowest,

indicating most RL-PCNN is implemented during mild days. Moreover, two days of implementation had cloudy weather conditions, which may alleviate the impact from the underestimation of solar radiation of PCNN. It is expected that the performance RL-PCNN will be degraded with hotter weather in a long period of implementation. It can be supported by the RL training results shown in Figure 44. The hyperparameters are the same for different agent training. The only difference is the predictive model used in the environment. From the RL training results, it is expected that RL-PCICNN and RL-ModNN would achieve much better performance than RL-PCNN in long-term implementations.

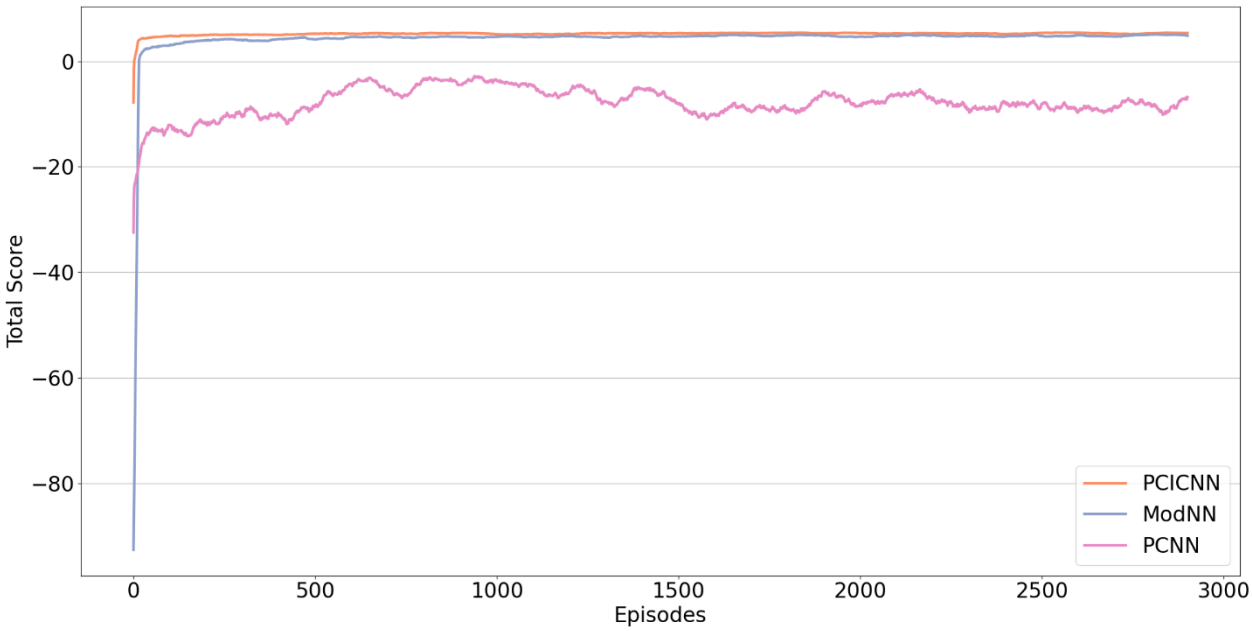


Figure 44. Total score of RL agent training based on the three models.

In summary, the results presented in this chapter establish three key findings for single-zone building dynamic modeling and advanced control. First, the PCICNN demonstrates prediction accuracy within sensor error bounds (MAPE of approximately 1% for temperature and 5% for CO2) while maintaining physically consistent responses to perturbed control inputs and

disturbances, validating its suitability as a control-oriented dynamic model. Second, the long-term real-building comparison reveals that all three advanced controls outperform the baseline, with HDDPC achieving the highest peak energy savings (over 50%) but suffering from information gaps and non-optimal decisions under certain conditions, RL providing the most robust and consistent savings (up to 48%) across varying outdoor temperatures, and DPC offering the fastest training and inference but the lowest overall savings (30.6%). Third, the PIML model benchmarking confirms that PCICNN is the most control-oriented architecture among the three evaluated models, with its input convexity and non-recurrent structure providing advantages in both optimization-based and agent-based control, despite other models achieving comparable or slightly better raw prediction accuracy. These findings validate the PCICNN as the foundation for scaling to multi-zone applications and motivate the development of the PCGNN and scalable multi-agent RL framework presented in the following chapters.

Chapter 5. Methodology II: PCGNN and Scalable RL for Multiple Zones

This chapter presents the methodology for scaling physics-informed modeling and advanced control from a single zone to multiple thermally coupled zones. The single-zone PCICNN and single-agent control strategies validated in Chapters 3 and 4 demonstrated strong performance for individual zones, but real commercial buildings comprise tens to hundreds of interconnected zones that require coordinated control. As discussed in Section 2.3, extending both the dynamic model and the control framework to this regime introduces fundamental challenges in model architecture, training efficiency, and computational scalability. This chapter addresses these challenges through two methodological innovations. Section 5.1 provides an overview of the integrated framework. Section 5.2 introduces the physics-consistent graph neural network (PCGNN), which replaces the manual adjacency-matrix neuron connections of prior multi-zone PCICNN approaches with group-shared multi-scale causal convolutions and heat diffusion graph layers, enabling scalable and physically interpretable multi-zone temperature prediction. Section 5.3 presents the model-based multi-agent soft actor-critic algorithm with κ-hop neighborhood truncation, which reduces critic input dimensionality from scaling with the total number of zones to scaling with the local neighborhood size. Sections 5.4 and 5.5 describe the two case studies used for validation: a 6-zone simulation study on a commercial building in Boston and an 18-zone real-building deployment at the Syracuse Center of Excellence, respectively.

5.1 Methodology Overview

Figure 45 shows the overview of our proposed physics-informed model-based MARL. It consists of three parts: 1) physics-consistent graph neural network (PCGNN) for multi-zone temperature prediction, 2) model-based MARL with κ-neighborhood truncation for dimension reduction, and 3) real-building case studies for validation. The PCGNN combines two complementary layer types: causal 1-D convolutional layers that capture the temporal effect of zonal inputs (e.g., supply air condition, occupancy, and weather) on indoor temperature, and 2) heat diffusion graph layers that model inter-zone thermal coupling through learned non-negative heat transfer parameters on the building adjacency graph. In model-based MARL, PCGNN is not only used in the environment to generate states and rewards but also used by agents as a differentiable model to optimize their policy and critic networks. The κ-neighborhood truncation makes each agent's critic network receive only the local state from the zone and its κ-hop neighbors rather than the full building state. The proposed PCGNN and model-based MARL with κ-neighborhood truncation is validated in two cases studies using different commercial buildings: a 6-zone case for simulation-based evaluation using a commercial building in Boston (building 1), and a 4-floor 18-zone case for real-world implementation using a commercial building in Syracuse (building 2). The following subsections describe each part of Figure 45 in detail.

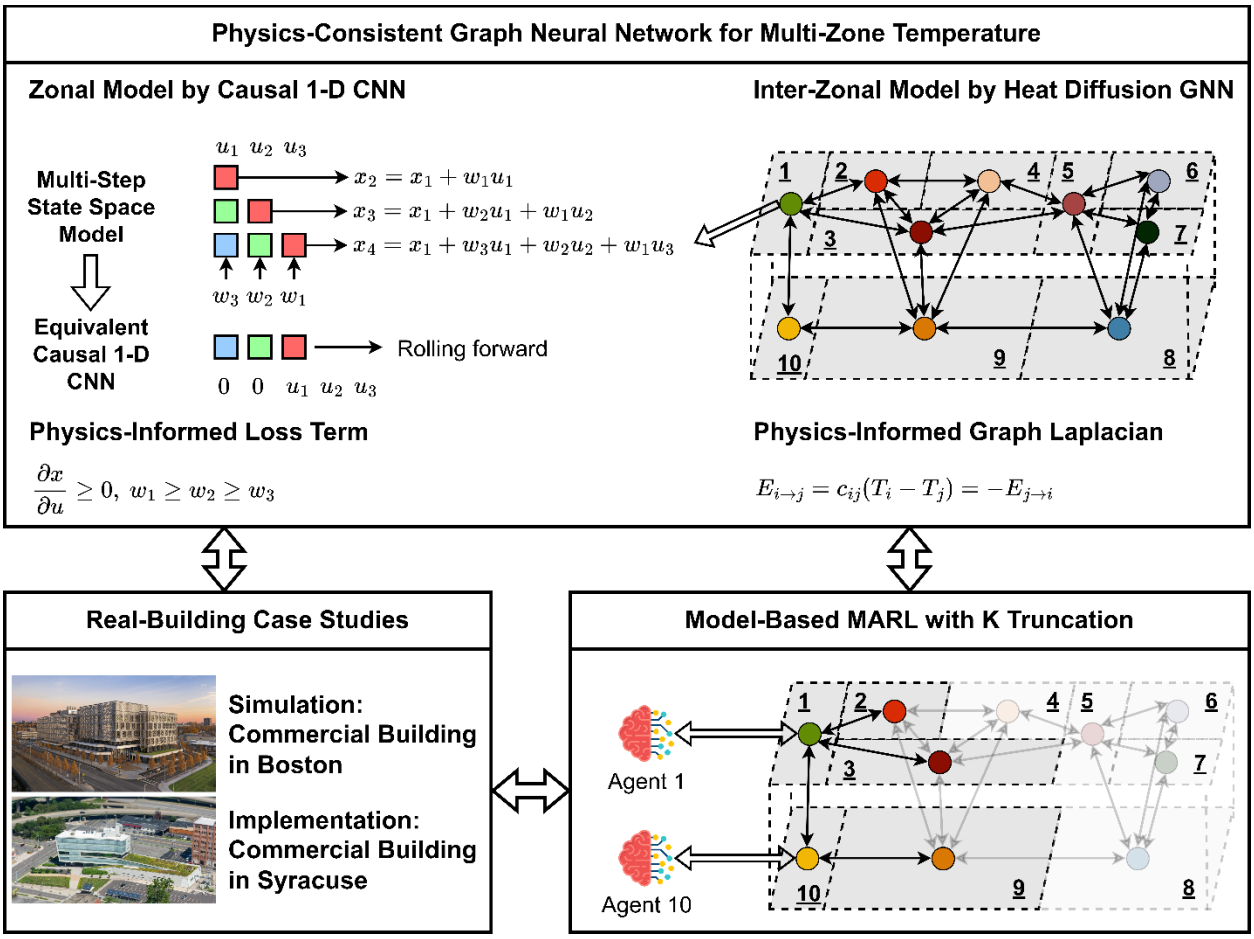


Figure 45. Overview of physics-informed model-based MARL

5.2 Physics-Consistent Graph Neural Network for Multi-Zone Temperature Prediction

In last chapter, we developed physics-consistent input convex neural networks (PCICNN) to model single-zone indoor temperature dynamics. PCICNN was inspired by the lower triangular block Toeplitz structure of coefficient matrices of linear state space model, as shown in multi-step state space model in Figure 45. In this structure, the coefficient matrix is lower triangular and shares the same weights along main and parallel diagonals. Such a structure was encoded into the weight matrices of multi-layer perceptron model, resulting in a partial connection of neurons between different layers and shared weights within the same layer. The extension to multi-zone cases was

through the partial connection of neurons based on the adjacent information [61]. However, such development was highly customized and required significant manual effort to define the weights and neuron connections. Therefore, we proposed a more standardized modeling approach that is more convenient and scale yet achieves equivalent effect as our previous work. As shown in Figure 45, the multi-step state-space model can be equivalent to causal 1-D convolutional layer. The inter-zonal interaction can be modeled by graph layers. The PCGNN alters the two types of layers in a way: M convolutional layers → one graph layer → M convolutional layers → one graph layer → … → M convolutional layers → one graph layer. Considering M convolutional layers and one graph layer as one block, the proposed PCGNN consists of B_{blk} to predict the multi-zone indoor dynamics. The details of each type of layer and PCGNN are described in the following subsections.

Notation and Data Layout

To model multi-zone temperature dynamics, we define batch size B, time length (prediction horizon) T, number of zones (nodes) Z and number of features C_{in}. Then the input sequence has the dimension of:

$$\boldsymbol{X} \in \mathbb{R}^{B\times T\times Z\times C_{in}}, \qquad \boldsymbol{X}_{b,t,z} \in \mathbb{R}^{C_{in}}$$

Where $\boldsymbol{X}$ is model input tensor and $\boldsymbol{X}_{b,t,z}$ is the feature vector of zone z at time t in batch b.

To reduce model parameters, same convolutional layer can be applied to zones that have similar dimensions and functions. We can partition zones into manual groups $\mathcal{R}$, where each group $r \in \mathcal{R}$ contains zone indices $S_r \subset \{1, \dots, Z\}$, and $\cup_r S_r = \{1, \dots, Z\}$, $S_r \cap S_{r'} = \emptyset$. That being said, for

multiple zones, we can manually categorize them into different groups based on their size and intended use, and each zone only belongs to one type.

We denote the hidden representation after some layer as:

$$\boldsymbol{H} \in \mathbb{R}^{B\times T\times Z\times D}$$

Where D is the hidden feature dimension inside the network.

Group-Shared Multi-Scale Causal Convolutional Layer

As shown in Figure 45, the lower triangular block Toeplitz matrix structure of multi-step state space model is mathematically equivalent to causal 1-D convolutional layer, which will be defined in the following. It is also worth noting that shared parameters are used for all zones in each group r to reduce the computational burden.

Causal Convolution Branches

For a group r, take the group slice:

$$\boldsymbol{H}^{(r)}_{b,t,z} = \boldsymbol{H}_{b,t,z}, \qquad z \in S_r, \qquad \boldsymbol{H}^{(r)} \in \mathbb{R}^{B\times T\times |S_r|\times D_{in}}$$

Then we can define two temporal convolutions (short and long) that map D_{in} to D_{out} to capture the short-term and long-term impact of input futures on the zone temperature.

For the short-branch temporal convolution, we have kernel length k_s with dilation 1:

$$\boldsymbol{Y}_s^{(r)}(b,t,z,c) = \sum_{i=0}^{k_s-1} \sum_{m=1}^{D_{in}} W_s^{(r)}(c,m,i)\boldsymbol{H}^{(r)}(b,t-i,z,m) \tag{39}$$

Where $\boldsymbol{Y}_s^{(r)}$ is short-branch convolution output for group r, $W_s^{(r)} \in \mathbb{R}^{D_{out}\times D_{in}\times k_s}$ is short-branch kernel for group g. For the causality between input and output, we use the causal convention $\boldsymbol{H}(b,\tau,\cdot,\cdot) = 0, \forall \tau < 0$, i.e., zero padding as shown in Figure 45. By zero padding, the prediction of time step only uses the current and past information, i.e., inputs in the future steps have no impact of the prediction from the current step. This achieves the same effect of partial connection of MLP's neurons in our previous study but in a more elegant way.

For the long-branch temporal convolution, we have kernel length k_l with dilation d_l:

$$\boldsymbol{Y}_l^{(r)}(b,t,z,c) = \sum_{i=0}^{k_l-1} \sum_{m=1}^{D_{in}} W_l^{(r)}(c,m,i)\boldsymbol{H}^{(r)}(b,t-d_l i,z,m) \tag{40}$$

Where $\boldsymbol{Y}_s^{(r)}$ is long-branch convolution output for group r, $W_l^{(r)} \in \mathbb{R}^{D_{out}\times D_{in}\times k_l}$ is long-branch kernel for group r.

The rationale of short- and long-branch convolutions is to capture short- and long-term temporal relationship between input and output. However, implementing the two branches of convolutions in their naïve form lacks the guarantee of physical consistency. In last chapter, physical consistency was enforced by non-negative weights. Here we define the same mechanism as an option in the following.

Optional Positivity Constraint on Kernels

The indoor temperature should be non-negatively related to its input features, such as cooling/heating load, outdoor temperature, and solar radiation, and such a correlation was guaranteed by enforcing non-negative weights of the neural network in PCICNN. However, non-negative weights are very restrictive and sometimes can limit the expressiveness of the model. Therefore, here we use an optional positivity constraint on kernels:

$$\phi(W) = \begin{cases} W \odot W, & if\ enforce\ positive \\ W, & otherwise \end{cases} \tag{41}$$

$\odot$ is Hadamard product where we take elementwise square of the weight matrix. In the implementation, we will have:

$$W \leftarrow \phi(W)$$

Note that the $\phi(W)$ can be applied to any weights in the PCGNN, not only limited to convolutional layer.

If we enforce non-negative weights, then the input and output are guaranteed to be nonnegatively correlated. When such enforcement is not applied, we will need a loss term to penalize negative gradients of output with respect to input features as described in equation (64) for physical consistency. The choice of the two mechanisms depends on the tradeoff between model expressiveness and the extent of physical consistency required.

Softmax Mixing of the Two Time Scales

As aforementioned, we have two branches of convolution to capture short- and long-term impact of input features on the output. The contribution of the two branches is learned through two logits for each convolutional layer. For every convolutional layer, two learnable logits $\boldsymbol{\eta} = [\eta_s, \eta_l]$ are defined to calculate the mixing weights:

$$\alpha_s, \alpha_l = softmax(\boldsymbol{\eta}), \qquad \alpha_s + \alpha_l = 1, \qquad \alpha_s, \alpha_l \geq 0 \tag{42}$$

Then the mixed output is weighted sum of short and long branches output:

$$\boldsymbol{Y}^{(r)} = \alpha_s \boldsymbol{Y}_s^{(r)} + \alpha_l \boldsymbol{Y}_l^{(r)} \tag{43}$$

Adding a shared bias $\boldsymbol{b} \in \mathbb{R}^{D_{out}}$, we have:

$$\widetilde{\boldsymbol{Y}}^{(r)}(b, t, z, :) = \boldsymbol{Y}^{(r)}(b, t, z, :) + \boldsymbol{b} \tag{44}$$

Applying activation (LeakyReLU) and dropout, we have:

$$\boldsymbol{Z}^r = Dropout\left(\sigma\left(\widetilde{\boldsymbol{Y}}^{(r)}\right)\right), \qquad \sigma(\cdot) = LeakyReLU(\cdot) \tag{45}$$

Residual Connection

Multiple convolution layers could be involved in PCGNN, making it a deep network. Therefore, we apply residual connections to the layers of PCGNN to improve model training.

For a given convolutional layer, if the dimension of input is different from the output, i.e., $D_{in} \neq D_{out}$, we will use linear projection:

$$\boldsymbol{R}^{(r)}(b,t,z,:) = P\boldsymbol{H}^{(r)}(b,t,z,:) + \boldsymbol{p}, \;\; P \in \mathbb{R}^{D_{out} \times D_{in}} \tag{46}$$

Otherwise, we will have $\boldsymbol{R}^{(r)} = \boldsymbol{H}^{(r)}$.

The final output for group g is:

$$\boldsymbol{H'}^{(r)} = \boldsymbol{Z}^{(r)} + \boldsymbol{R}^{(r)} \tag{47}$$

Lastly, we can define the full output $\boldsymbol{H'} \in \mathbb{R}^{B \times T \times Z \times D_{out}}$ by writing each group result into its zone indices:

$$\boldsymbol{H'}_{:,:,S_r:} \leftarrow \boldsymbol{H'}^{r}, \qquad \forall r \in \mathcal{R}$$

The short- and long-branches of convolution, positivity constraints of kernels, learnable mixing of two scales, and residual connections define the 1-D causal convolutional layers of our proposed PCGNN. As shown in Figure 45, convolutions are applied within the node (zone), which aims to capture the temporal impact of features within the zone. To model spatial inter-zonal impact, we use heat diffusion graph layer defined in the following subsection.

Heat Diffusion GNN Layer

As shown in Figure 45, causal convolutional layer only processes features inside the node (zone) over the prediction horizon. The inter-zonal effect is modeled by heat diffusion graph layer. It

operates on each time slice independently, i.e., on each graph instance $g = (b, t)$. Let $\boldsymbol{h}_g \in \mathbb{R}^{Z \times D}$ be the zone embeddings for one (b, t). Let ε_u be the set of undirected edges (i, j) from the adjacency matrix for adjacency between zones i and zone j. The heat diffusion graph layer can be developed based on the heat transfer of adjacent zones.

Nonnegative Heat Transfer Coefficient

Here we assume that the hidden representations denote the temperature of each zone. Each undirected edge (i, j) has a learnable scalar weight $w_{i,j}$ that represents heat transfer coefficient between zones i and j. The heat transfer coefficients are non-negative values. By enforcing nonnegativity using equation (41), we have:

$$\tilde{g}_{i,j} = \phi(w_{ij}) \geq 0$$

To improve training stability, for each graph instance, we define base degree

$$d_i = \sum_{j:(i,j)\in\varepsilon_u} \tilde{g}_{i,j} \tag{48}$$

Let $d_{max} = \max_i d_i$. We cap the step size:

$$\alpha_{cap} = \frac{0.9}{\max(d_{max}, \epsilon)} \tag{49}$$

Where ϵ is a small constant to avoid division by zero, e.g. 10^{-6}.

Then the effective step is:

$$\alpha_{eff} = \delta \cdot \min(\Gamma, \alpha_{cap}) \tag{50}$$

Where a learnable global scale Γ and a gate $\delta \in [0,1]$ are used:

$$\Gamma = \phi(global\ scale) \geq 0, \qquad \delta = \sigma(gate\ logit) \in (0,1)$$

Finally, the effective heat transfer coefficient is:

$$g_{ij} = \alpha_{eff} \tilde{g}_{ij} \geq 0 \tag{51}$$

Diffusion Update

Having the effective heat transfer coefficient, for each undirected edge (i, j), we define the symmetric heat flow between zones:

$$\Delta_{i+} = g_{ij}(\boldsymbol{h}_j - \boldsymbol{h}_i), \qquad \Delta_{j+} = g_{ij}(\boldsymbol{h}_i - \boldsymbol{h}_j) \tag{52}$$

$\Delta Q_{i \leftarrow j}$ represents heat gain of zone i from its adjacencies. Note that $\Delta Q_{i \leftarrow j} = -\Delta Q_{j \leftarrow i}$. It can be interpreted as the heat balance between zone i and zone j, i.e., the heat loss from zone i contributes to the heat gain of zone j, and vice versa.

To translate this heat flow into a temperature change, we must account for the distinct thermal mass of each zone. We introduce a learnable, strictly positive inverse heat parameter $c_i = 1/C_i$ for each node i. Considering all adjacencies of zone i, the node i update is:

$$\boldsymbol{h}_i' = \boldsymbol{h}_i + c_i \sum_{j:(i,j)\in\varepsilon_u} g_{ij}(h_j - h_i) \tag{53}$$

Laplacian Form

We can denote diffusion updates in a compact way with Laplacian form. Let $\boldsymbol{L}$ be the symmetric weighted graph Laplacian constructed from the heat transfer coefficients g_{ij}:

$$\boldsymbol{L}_{ii} = \sum_{j:(i,j)\in\varepsilon_u} g_{ij}\,, \qquad \boldsymbol{L}_{ij} = \begin{cases} -g_{ij}, & (i,j) \in \varepsilon_u \\ 0, & otherwise \end{cases} \tag{54}$$

Let $\boldsymbol{C}^{-1} = diag(c_1, c_2, \dots, c_Z)$ be the diagonal matrix of the learned inverse heat capacities for all Z zones. The full graph layer update becomes:

$$\boldsymbol{h}' = (\boldsymbol{I} - \boldsymbol{C}^{-1}\boldsymbol{L})\boldsymbol{h} \tag{55}$$

Because $\boldsymbol{C}^{-1}$ is applied directly to the nodes, the effective transformation matrix $\boldsymbol{C}^{-1}\boldsymbol{L}$ is asymmetric. At this point, we defined heat diffusion graph layer that obeys heat transfer and energy balance among the zone and its adjacencies. Next subsection assembles causal convolutional and graph layers as our proposed PCGNN.

Full PCGNN

As aforementioned, the PCGNN alters the two types of layers in a way: M convolutional layers → one graph layer → M convolutional layers → one graph layer → … → M convolutional layers → one graph layer. Considering M convolutional layers and one graph layer as one block, the proposed PCGNN consists of B_{blk} to predict the multi-zone indoor dynamics. Let there be B_{blk} blocks, each containing M grouped multis-scale convolutional layers and one heat diffusion graph layer.

We initialize $\boldsymbol{H}^{(0)} = \boldsymbol{X}$, for block $\beta = 1, \dots, B_{blk}$.

For grouped temporal stack, we have M convolutional layers for each zone group. The forward propagation is defined as follows:

$$\boldsymbol{H}^{(\beta,0)} = \boldsymbol{H}^{(\beta-1)} \tag{56}$$

$$\boldsymbol{H}^{(\beta,m)} = GroupedMSConv_{b,m}\left(\boldsymbol{H}^{(b,m-1)}\right), \qquad m = 1, \dots, M \tag{57}$$

Where each $GroupedMSConv$ is the group-assembled operator for convolutional layers.

After grouped temporal convolution stack, we have heat diffusion layer

$$\boldsymbol{H}^{(b)}_{b,t,:,:} = HeatDiffusion_b\left(\boldsymbol{H}^{(b,M)}_{b,t,:,:}\right), \qquad \forall(b,t) \tag{58}$$

After B_{blk} blocks of alternation between convolutional and graph layers, we apply a per-zone linear head:

$$\widehat{\boldsymbol{Y}}_{b,t,z,:} = W_{out} H_{b,t,z,:}^{(B_{blk})} + \boldsymbol{b}_{out} \tag{59}$$

Where $\hat{Y}$ gives us the prediction of all modeled zones over the prediction horizon.

Note that for RL implementation, we only need single-step ahead prediction of multiple zones. Such a single-step model can be easily built based on the trained multi-step model by using 1-step-ahead kernel from multi-scale convolutional layers.

Loss Function

Besides the model architecture of PCGNN, here we define different loss terms for prediction accuracy and physical consistency of the PCGNN. Let predictions $\widehat{\boldsymbol{Y}}$ and targets $\boldsymbol{Y}$ have shape $[B, T, Z, C_{out}]$. We define loss terms as follows.

Per-Zone MSE

$$\mathcal{L}_{mse} = \frac{1}{Z}\sum_{z=1}^{Z} MSE_z\,, \qquad MSE_z = \frac{1}{BTC_{out}}\sum_{b,t,c}\left(\hat{Y}_{b,t,z,c} - Y_{b,t,z,c}\right)^2 \tag{60}$$

Temporal Difference Loss

$$\mathcal{L}_{\Delta} = \frac{1}{B(T-1)ZC_{out}}\sum_{b,t=1}^{T-1}\sum_{z,c}\left[\left(\hat{Y}_{b,t+1,z,c} - \hat{Y}_{b,t,z,c}\right) - \left(Y_{b,t+1,z,c} - Y_{b,t,z,c}\right)\right]^2 \tag{61}$$

This term is to help PCGNN to learn the temperature change between time steps.

Monotonicity Regularization

For a kernel tensor $W \in \mathbb{R}^{D_{out} \times D_{in} \times K}$, we define tap magnitudes by Frobenius norm

$$s_i = \|\phi(W[:,:,i])\|_F, \qquad i = 0, \ldots, K-1 \tag{62}$$

We can regularize the convolutional layer to be near-time-step focus by the loss:

$$\mathcal{L}_{mono} = \sum_{i=0}^{K-2} [\max(0, s_i - s_{i+1})]^2 \tag{63}$$

The rationale of $\mathcal{L}_{mono}$ is that the prediction of a certain step should be largely impacted by its near-term historical inputs rather than distant ones.

Gradient Monotonicity

When non-negative weights are not enforced for all weights, i.e., $\phi(W) = W$ in equation (41), we can guarantee a positive gradient of the model output $\widehat{\boldsymbol{Y}}$ with respect to the input $\boldsymbol{X}$.

$$\mathcal{L}_{grad} = \sum \max(0, -G), \qquad G = \frac{\partial \bar{y}}{\partial \boldsymbol{X}}, \qquad \bar{y} = \frac{1}{BTZC_{out}} \sum_{b,t,z,c} \hat{Y}_{b,t,z,c} \tag{64}$$

Note that it is not necessary to have nonnegative gradients for all input features. For example, we can enforce nonnegative gradients only with respect to control variables.

Bias Penalty

To avoid the prediction heavily relying on bias with vanishing weights, we can also have optional bias penalty.

$$\mathcal{L}_{bias} = \sum_{all\ conv\ layers} \|b\|_1 \tag{65}$$

Total Loss

Based on the loss terms defined above, the total training loss is

$$\mathcal{L} = \mathcal{L}_{mse} + \lambda_{\Delta}\mathcal{L}_{\Delta} + \lambda_{mono}\mathcal{L}_{mono} + \lambda_{grad}\mathcal{L}_{grad} + \lambda_{bias}\mathcal{L}_{bias} \tag{66}$$

The 1-D causal convolutional layer, heat diffusion graph layer, and loss terms for prediction accuracy and physical consistency consist of our proposed PCGNN. Next subsection will define the multi-zone control as a network MDP and describe model-based MARL with κ-neighborhood truncation.

5.3 Model-Based MARL with PCGNN Dynamics and κ-Hop Neighborhood Truncation

Problem Setup as a Network MDP

We consider a multi-zone building control problem modeled as a network Markov decision process (MDP) defined on a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $i \in \mathcal{V}$ represents a thermal zone (agent) and edges $(i, j) \in \mathcal{E}$ represent physical coupling or interaction. At a discrete time t, each agent observes a local zone state $s_t^{(i)} \in \mathbb{R}^{d_s}$ (e.g., zone temperature and related features), and selects a

control action $a_t^{(i)} \in \mathbb{R}^{d_a}$ (e.g., valve/damper/setpoint or other actuator commands). The global state and action are

$$s_t = \left(s_t^{(1)}, \ldots, s_t^{(N)}\right), \qquad a_t = \left(a_t^{(1)}, \ldots, a_t^{(N)}\right)$$

With $N = |\mathcal{V}|$. The objective is to learn decentralized policies $\pi = \{\pi_i\}_{i=1}^{N}$ that maximize the expected discounted return:

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right]$$

Where $r(s_t, a_t)$ encodes thermal comfort, energy usage, and actuation smoothness constraints, and $\gamma \in (0,1)$ is the discount factor.

κ-Hop Truncation for Scalable Local Decision Making

A challenge in MARL is that the joint state-action space grows rapidly with the number of agents. For network-structured systems, theoretical results in network MDPs indicate that localized approximation can well represent global values if the influence of distant nodes on an agent's value can decay exponentially with graph distance [140][141][142]. The temperature of one zone at one step is usually impacted by itself and direct adjacency ($\kappa = 1$), which makes multi-zone temperature dynamics satisfy exponential decay property, and therefore the global state and action values can be well approximated using local information. Following this motivation, we apply κ-hop neighborhood truncation to reduce input dimensionality for scalable learning.

Let $\mathcal{N}_i^{\kappa} \subseteq \mathcal{V}$ denote the set of nodes within graph distance κ from agent i, including i. We define each agent's local observation as the concatenation of states from its neighborhood:

$$o_t^{(i)} = concat\left(\left\{s_t^{(j)} : j \in \mathcal{N}_i^{\kappa_\pi}\right\}\right)$$

Where κ_π is the truncation radius used by the policy. For value estimation we use $\kappa_\pi + 1$ for critic inputs:

$$\tilde{o}_t^{(i)} = concat\left(\left\{s_t^{(j)} : j \in \mathcal{N}_i^{\kappa_\pi+1}\right\}\right)$$

This yields a centralized training and decentralized execution structure: each agent acts using only local information $o_t^{(i)}$ and value learning can incorporate a boarder neighborhood to improve credit assignment and stability.

Note that the reference paper [140] introduces spectral representations as a scalable functional basis for continuous network MDPs. In this work, we adopt the same locality motivation but do not construct spectral embeddings. Instead, we reply on neural function approximation to represent policies and value functions.

Model-Based Soft Actor-Critic

Besides dimension reduction by κ-neighborhood truncation, we also use the learned PCGNN dynamics model as a differentiable world model to improve sample efficiency and stabilize learning. Let f_θ denote the learned one-step PCGNN, we can have state transition defined as follows:

$$s_{t+1} = f_\theta(s_t, a_t) \tag{67}$$

Where θ are model parameters trained separately for PCGNN.

We employ a model-based variant of soft actor-critic (SAC) that leverages the differentiable PCGNN model to reduce the critic complexity. Rather than learning a state-action value function $Q(s, a)$ directly, we learn a state-value function $V(s)$ with a standard MLP and reconstruct the soft Q-value through a one-step model-based lookahead based on Bellman equation:

$$Q(s_t, a_t) \triangleq r(s_t, a_t) + \gamma V(s_{t+1}) = r(s_t, a_t) + \gamma V\big(f_\theta(s_t, a_t)\big) \tag{68}$$

Note that here we decompose the next state s_{t+1} to $f_\theta(s_t, a_t)$ based on the PCGNN. This decomposition shifts the learning burden from approximating $Q: \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ to approximating $V: \mathcal{S} \to \mathbb{R}$, while the transition mapping is captured by the pretrained PCGNN model.

To mitigate overestimation bias, we use to independent value networks $V_{\psi_1}(s)$ and $V_{\psi_2}(s)$ and define a clipped values estimate $V_{min}(s) = \min\left(V_{\psi_1}(s), V_{\psi_2}(s)\right)$, where ψ is the learnable parameter of the value networks.

For each sampled transition (s_t, a_t, r_t), we obtain the next state from the dynamics model $\tilde{s}_{t+1} = f_\theta(s_t, a_t)$. The soft target of critic is defined using target value networks and the policy entropy:

$$y_t = r_t + \gamma\big(V_{min,targ}(\tilde{s}_{t+1}) - \alpha \log \pi_\phi(a_{t+1}|\tilde{s}_{t+1})\big), \qquad a_{t+1} \sim \pi_\phi(\cdot\,|\tilde{s}_{t+1}) \tag{69}$$

Where $\alpha > 0$ is the temperature parameter.

Each value network is trained by minimizing the squared Bellman error:

$$\mathcal{L}_V(\psi_k) = \mathbb{E}\left[\left(V_{\psi_k}(s_t) - y_t\right)^2\right], \qquad k \in \{1,2\} \tag{70}$$

This trains $V(s_t)$ to match the entropy-regularized one-step return under the PCGNN dynamics.

The policy $\pi_\phi(a|s)$ is optimized to maximize expected soft return. Because the objective depends on the differentiable composition $V\left(f_\theta(s,a)\right)$, we can backpropagate through the dynamics model. The objective of policy network is defined as:

$$J(\phi) = \mathbb{E}\left[r(s_t, \tilde{a}_t) + \gamma V_{min}\left(f_\theta(s_t, \tilde{a}_t)\right) - \alpha \log \pi_\phi\left((\tilde{a}_t|s_t)\right)\right] \tag{71}$$

Where $\tilde{a}$ is the action sampled by reparameterization trick.

This explicitly encourages actions that drive the system, through f_θ into high-value next states while maintaining exploration through the entropy term.

Thereby, we have defined our PCGNN and model-based MARL with κ-neighborhood truncation. Next section will define our case studies.

5.4 Simulation Study: Building 1 in Boston

Overview of the Testbed

Figure 46 shows the floor plan and selected zones for building 1 in Boston for simulation verification. The simulation study includes 6 zones: 1 corridor and 5 office rooms. The corridor is conditioned by a dedicated outdoor air system (DOAS) with variable air volume (VAV) boxes,

while the offices are served by both radiant ceiling panel system and DOAS. Each office is equipped with an occupancy presence sensor. The occupancy status determines the operating mode of the radiant panel system: (1) unoccupied mode before 7 AM and after 7 PM, (2) occupied mode when occupancy is detected, and (3) standby mode when the room is unoccupied between 7 AM and 7 PM. The radiant system switches between heating and cooling based on the outdoor air temperature: heating mode when the outdoor temperature is below 55 °F (12.8 °C) and cooling mode when it exceeds 56 °F (13.3 °C). Occupants can also adjust the thermostat setpoint within a predefined range: 72–80°F (22.2-26.7 °C) in cooling mode and 65–74°F (18.3-23.3 °C) in heating mode.

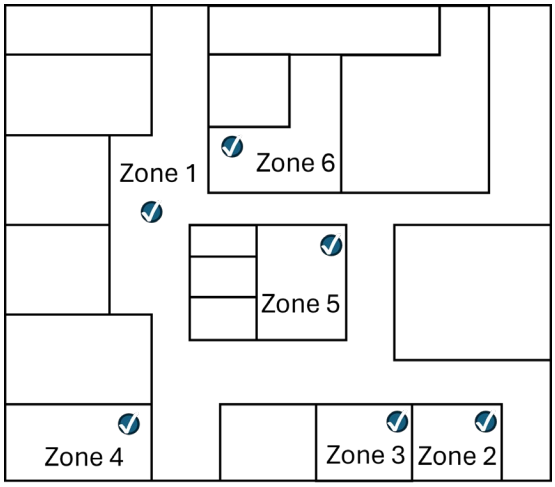


Figure 46. Selected zones of building 1.

PCGNN for Building 1

The input features of the PCGNN for building 1 consist of the current indoor temperature at each modeled zone (a single time step) and sequences of the following variables over the prediction horizon: water-side radiant system load, air-side load of DOAS, outdoor temperature, solar radiation, and occupancy status. The dataset spans from April 15, 2024, to June 20, 2025, with 15-

minute interval, and is split into training, validation, and testing sets with ratios of 0.70, 0.15, and 0.15, respectively. The hyperparameters of the PCGNN are tuned using the Optuna framework [143], and Table 17 lists the best configuration selected based on validation performance.

Table 17. Hyperparameters of PCGNN for building 1.

Hyperparameters	Values
Hidden size	32
Number of blocks B_{blk}	3
Grouped MS conv layers M (each block)	3
Long-branch kernel size	0
Short-branch kernel size	24
Enforce positivity for MS conv	False
Enforce positivity for heat diffusion layer	True
Conv dropout	0.1
Learning rate	1e-3
Batch size	512
Number of epochs	200
Grad clip	0.5
Coefficient λ_{Δ}	1.0
Coefficient λ_{mono}	0
Coefficient λ_{bias}	0
Coefficient λ_{grad}	1.5
Gradient monotonic features	Indoor temperature and radiant load
Early stop patience	20
Shrink factor of learning rate	0.5
Learning rate patience	10
Minimal learning rate	1e-5
Prediction horizon	6 hours

MARL for Building 1

The MARL agents control the thermal load from the radiant ceiling panel systems in the five office zones, while the corridor operates under its default control logic. The DOAS is excluded from MARL control because it serves primarily for indoor air quality and condensation prevention on the radiant panels, contributing only a small fraction of the total heating and cooling load compared

to the radiant ceiling system. The same dataset used for PCGNN training is used for MARL training and evaluation.

The local state of each zone consists of indoor temperature, air-side load from the DOAS, outdoor temperature, solar radiation, occupancy status, and the temperature lower and upper bounds at the current and next 8 steps—34 state variables per zone in total. The reward function is defined as:

$$r(s_t, a_t) = r_1 s_t^{vio} + r_2|a_t - a_{t-1}| + r_3|a_t| + r_4 s_t^{good} \tag{72}$$

Where s_t^{vio} is temperature violation calculated by the deviation from lower and upper bounds; s_t^{good} is a binary indicator of whether the zone is within the comfort bounds (i.e., no temperature violation); and r_1 to r_4 are penalty and reward coefficients for different objectives.

Table 18 lists the MARL hyperparameters for building 1. A 1-hop truncation is used for the policy network, resulting in a maximum of 102 input states for agents. A 2-hop truncation is used for the critic network, which in this case is equivalent to global state concatenation since the 6-zone network has a diameter of 2.

Table 18. Hyperparameters of MARL for building 1.

Hyperparameters	Values
κ-hop truncation for policy network	1
κ-hop truncation for critic network	2
Number of episodes	4e4
Batch size	2048
Hidden dimension	64
Discount factor γ	0.93
Learning rate	1e-4
Target network update rate	0.05
Number of layers of neural networks	10

Penalty for temperature violation r_1	-2
Penalty for non-smoothness of action r_2	1e-3
Penalty for energy usage r_3	0.05
Reward for good IEQ r_4	0.04

5.5 Implementation Study: Building 2 in Syracuse

Overview of the Testbed

Figure 47 shows the floor plan and selected zones of building 2 in Syracuse. The 18 selected zones span 4 floors and include corridors, conference rooms, office rooms, and lobbies. All zones are conditioned by radiant ceiling panels and a DOAS with underfloor air distribution. Building 2 is designed as a living laboratory, allowing the heating and cooling modes to be switched throughout the year. The operating mode is determined by the relationship between indoor temperature and setpoints: heating is provided when the temperature falls below the heating setpoint, and cooling when it exceeds the cooling setpoint. The setpoints are defined based on a daily schedule and room usage type. The building follows an occupied schedule from 6 AM to 6 PM. For frequently occupied rooms such as offices, the heating and cooling setpoints are 70°F (21.1 °C) and 75°F (23.8 °C) during occupied hours, relaxing to 64°F (17.8 °C) and 81°F (27.2 °C) otherwise. For less frequently occupied spaces such as corridors and lobbies, the daytime setpoints are 67°F (19.4 °C) and 78°F (25.6 °C), and the nighttime setpoints are 64°F (17.8 °C) and 81°F (27.2 °C).

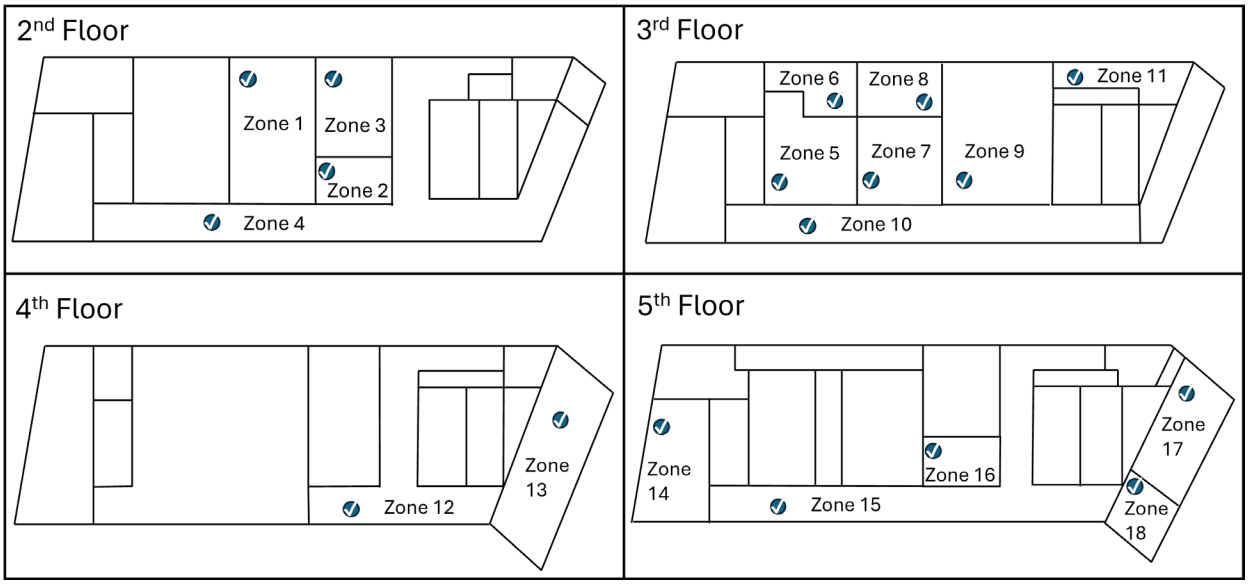


Figure 47. Selected zones of building 2. Dashed lines indicate air wall.

PCGNN for Building 2

The input features of the PCGNN for building 2 include the current indoor temperature at each modeled zone (a single time step) and sequences of the following variables over the prediction horizon: water-side radiant system load, air-side DOAS load, outdoor temperature, solar radiation, and occupancy numbers. The occupancy numbers are either measured by the people counting sensors or estimated based on the daily schedule and normal occupancy numbers. The dataset spans from April 15, 2024, to January 10, 2026, with 15-minute interval, and is split into training, validation, and testing sets with ratios of 0.70, 0.15, and 0.15, respectively. The hyperparameters of the PCGNN are tuned using the Optuna framework [143], and Table 19 lists the best configuration selected based on validation performance.

Table 19. Hyperparameters of PCGNN for building 2.

Hyperparameters	Values
Hidden size	16
Number of blocks B_{blk}	2
Grouped MS conv layers M (each block)	3
Long-branch kernel size	0
Short-branch kernel size	24
Enforce positivity for MS conv	False
Enforce positivity for heat diffusion layer	True
Conv dropout	0.1
Learning rate	1e-3
Batch size	512
Number of epochs	200
Grad clip	0.5
Coefficient λ_{Δ}	0
Coefficient λ_{mono}	0
Coefficient λ_{bias}	0
Coefficient λ_{grad}	2
Gradient monotonic features	Indoor temperature and radiant load
Early stop patience	20
Shrink factor of learning rate	0.5
Learning rate patience	10
Minimal learning rate	1e-5
Prediction horizon	6 hours

MARL for Building 2

As with building 1, the MARL agents control only the radiant ceiling panel systems for building 2, while the DOAS operates under its default control logic. The reward function and hyperparameters follow equation (72) and Table 18, respectively. Unlike the case of building 1, the local state for each zone at building 2 does not include the temperature bounds at current and future time steps. Instead, sine and cosine encodings of the time of day are used, resulting in 7 local state variables per zone. After 1-hop neighborhood truncation, the maximum input dimension of the actor networks is 49, corresponding to zones with up to 7 neighbors (including themselves). For the critic networks with 2-hop truncation, the maximum and minimum input dimensions are

112 and 35, respectively. For reference, the global state dimension without any truncation would be 126, demonstrating that the κ-truncation achieves up to a 72% reduction in critic input dimensionality.

Implementation

MARL agents output the heating or cooling load of the radiant ceiling panel system as their action. To implement this action in the real building, the load must be converted to a controllable valve position. This requires establishing the relationship between the thermal load and the water flow rate through the radiant panels.

We use ϵ-NTU method to convert the load to the water flow rate:

$$\epsilon = \frac{T_s - T_r}{T_s - T_z} = 1 - \exp\left(-\frac{A(G^B + C)}{G}\right) \tag{73}$$

$$Q = \rho \cdot c_p \cdot G \cdot (T_s - T_z)\left(1 - \exp\left(-\frac{A(G^B + C)}{G}\right)\right) \tag{74}$$

Where T represents temperature; the subscripts s, r and z indicate supply water, return water, and zone air. A, B, and C are tunable parameters indicating the heat transfer properties of the radiant ceiling panels; G is the water flow rate; Q is the radiant panel load; ρ and c_p are density and specific heat of water.

Using equation (74), we can convert sampled action Q to the water flow rate G. The flow rate is then mapped to a valve position based on the valve characteristics, enabling direct implementation of the agent's action through the building automation system.

The real-building deployment of MARL in building 2 was conducted from January 12 to February 22, 2026, spanning a total of 42 days. During this period, MARL and the baseline controller alternated on a daily basis to enable direct performance comparison under similar weather conditions.

Chapter 6. Results II: PCGNN and Scalable RL for Multiple Zones

This chapter presents the results corresponding to the multi-zone methodology introduced in Chapter 5, organized by the two case studies and followed by cross-cutting analyses. Section 6.1 reports the simulation results for Building 1 in Boston, including PCGNN prediction performance and MARL control performance across the 6 modeled zones. Section 6.2 presents the real-building implementation results for Building 2 in Syracuse, covering PCGNN prediction accuracy and the 18-zone MARL deployment over the experimental period. Section 6.3 analyzes the training efficiency of both the PCGNN and the κ-neighbor truncated model-based MARL, comparing convergence speed and computational cost against alternative approaches. Section 6.4 examines the impact of measured data quality on MARL performance, identifying how sensor faults, thermostat interactions, and hardware malfunctions propagate through the thermal network and affect control outcomes. Section 6.5 discusses practical considerations for real-building RL deployment, including the role of setpoint information in agent training and the infrastructure-level issues encountered during implementation.

6.1 Simulation Study: Building 1 in Boston

PCGNN Performance

The prediction performance of the PCGNN is first evaluated for building 1. Table 20 shows performance of PCGNN for the 6 modeled zones of building 1 on testing dataset (25 days). The predictions are generated in an autoregressive manner, where each predicted temperature serves as the initial condition for the next time step — the most operationally relevant evaluation mode, as it reflects how the model would be used during MARL rollouts. Table 20 shows that the PCGNN captures the temperature dynamic of all 6 zones accurately, with a mean absolute percentage error (MAPE) below 3% for every zone.

Table 20. Performance of PCGNN on testing dataset of building 1.

Zone	1	2	3	4	5	6
MAPE	0.95%	1.67%	1.43%	2.91%	0.76	0.79%

Figure 48 presents the temperature predictions under perturbed radiant panel loads, maximum cooling and maximum heating, compared to predictions using the measured load. In each row of subplots, only one zone's load is perturbed while all other zones retain their measured load profiles. The first row, for example, shows the response when the corridor (zone 1) is subjected to perturbed conditions. Because all modeled zones are adjacent to the corridor, every zone's prediction is affected, though with varying sensitivity — zone 4 exhibits the strongest response. When zone 2's load is perturbed, its direct neighbor zone 3 is significantly affected, while the remaining zones show only minor changes. Notably, zone 5 also produces a widespread impact across all modeled zones, as it serves as the core zone of the floor; its temperature change propagates through the corridor and subsequently influences other zones. Across all perturbation scenarios, the predicted

temperature consistently increases under maximum heating load and decreases under maximum cooling load. The correct directional response to load perturbations, combined with the spatially decaying influence that follows the adjacency structure, confirms that the proposed PCGNN has learned both the zonal thermal dynamics and the inter-zone heat transfer physics.

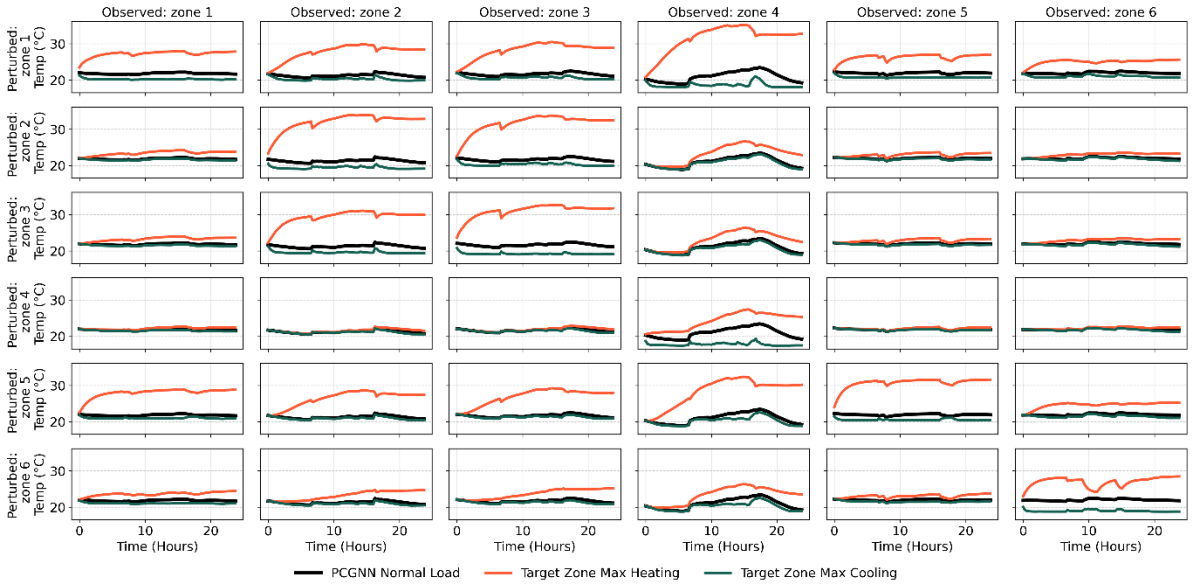


Figure 48. Prediction of PCGNN with perturbed radiant panel load.

MARL Performance

Figure 49 compares the indoor temperature under MARL control with the measured baseline for building 1. Since occupants can adjust their thermostats, the temperature bounds vary across days. For clarity, one representative day is shown. Only five office rooms are presented, as the corridor (zone 1) is not equipped with radiant ceiling panels and is therefore not controlled by MARL.

Under the baseline, zones 2 and 3 exhibit temperature violations. The heating system was activated at 6:00 AM when zones became occupied, but due to the slow response of the hydronic system, room temperatures did not begin to rise until approximately 30 minutes later. In contrast, MARL

provided proactive heating, as evidenced by an earlier rise in room temperature to meet the setpoints. For zone 4, MARL maintained the room temperature within bounds throughout the day, except for brief violations in the early morning, whereas the baseline failed to achieve thermal comfort until the afternoon. Similar performance differences were observed for zones 5 and 6, where MARL consistently maintained indoor temperatures within the comfort range, while the baseline intermittently failed to do so. Over the 114 simulated days, the mean cumulative temperature violation was reduced from 7.57°C under the baseline to 1.60°C under MARL, representing a 79% reduction. Total energy consumption was 26,175 kWh and 22,053 kWh for the baseline and MARL, respectively, corresponding to an energy saving of 15.74%.

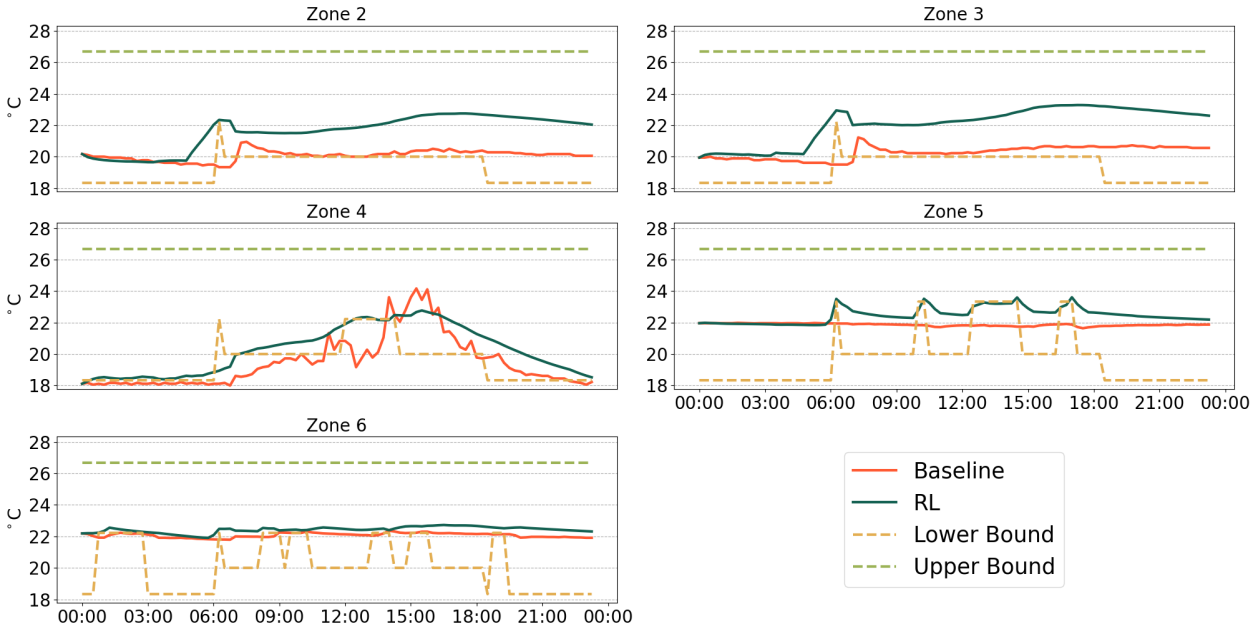


Figure 49. Temperature performance of MARL and baseline for building 1.

In summary, the simulation results on building 1 demonstrate that the proposed framework performs well at both the modeling and control stages. The PCGNN achieves below 3% MAPE across all 6 zones in autoregressive prediction, and the perturbation analysis confirms that the

model correctly captures both zonal thermal responses and inter-zone heat transfer through the building adjacency structure. Built upon the accurate and physically consistent dynamics model, the MARL agents reduce cumulative temperature violations by 79% compared to the baseline while achieving 15.74% energy savings. These results validate the framework in a controlled simulation setting and motivate its deployment into a larger, real-world building.

6.2 Implementation Study: Building 2 in Syracuse

PCGNN Performance

Having validated the physical consistency of the PCGNN through the perturbation analysis on building 1, we now evaluate its prediction performance on building 2 to assess whether the model scales to a larger, more complex multi-zone system. Table 21 reports the autoregressive prediction accuracy on the testing dataset for all 18 modeled zones across 4 floors. Despite the threefold increase in the number of zones and the added complexity of inter-floor thermal coupling, the PCGNN maintains strong prediction accuracy: all zones achieve a MAPE below 3.6%, with the majority under 2.5%. The lowest errors are observed in zones 5 (1.58%) and 6 (1.48%), while the highest are in zones 17 (3.56%) and 18 (3.35%), which are conference rooms where more dynamic occupancy profiles introduce additional variability. These results confirm that the proposed architecture, combining grouped causal convolutions with the heat diffusion graph layer, generalizes effectively from a 6-zone to an 18-zone building without degradation in prediction quality.

Table 21. Performance of PCGNN on testing dataset of building 2.

Zone	1	2	3	4	5	6	7	8	9
MAPE	2.22%	2.05%	2.91%	2.14%	1.58%	1.48%	2.07%	1.77%	2.89%

Zone	10	11	12	13	14	15	16	17	18
MAPE	3.19%	1.94%	2.24%	2.76%	1.70%	2.56%	2.19%	3.56%	3.35%

MARL Performance

During the 42-day real-building deployment, the majority of zones are maintained within the prescribed temperature bounds under MARL most of the time, with the exceptions of rooms 3, 13, 17, and 18. For clarity, we show temperature comparison of representative zones between MARL and baseline in Figure 50 (good performance) and Figure 54 (underperformance). The underperformance in these four zones is not attributable to the PCGNN or the MARL algorithm, but rather to incorrectly commissioned radiant ceiling panels in zones 3 and 13, whose malfunction in turn affected the thermal conditions of neighboring zones 17 and 18. A detailed discussion of these commissioning issues is provided in the next section.

For the remaining zones, MARL exhibits clear predictive control behavior: temperatures begin rising ahead of the occupied period, earlier than under the baseline control, demonstrating that the agents have learned to pre-condition spaces in anticipation of upcoming comfort requirements. This is further confirmed by Figure 51 from the comparison of the radiant panel loads under both controllers for zones 6 and 9. A particularly noteworthy finding from Figure 51. Radiant ceiling panel load comparison between baseline and MARL during implementation for zones 4-9. is the control strategy learned for the third floor: MARL allocates a higher heating load to zone 6, the core zone of the floor, and leverages inter-zone heat transfer to condition its surrounding zones, rooms 4, 5, 7, 8 and 9, with less total energy than the baseline. This emergent strategy, which was not explicitly programmed but learned through multi-agent coordination, validates that the agents exploit the building's thermal network structure to achieve energy-efficient control.

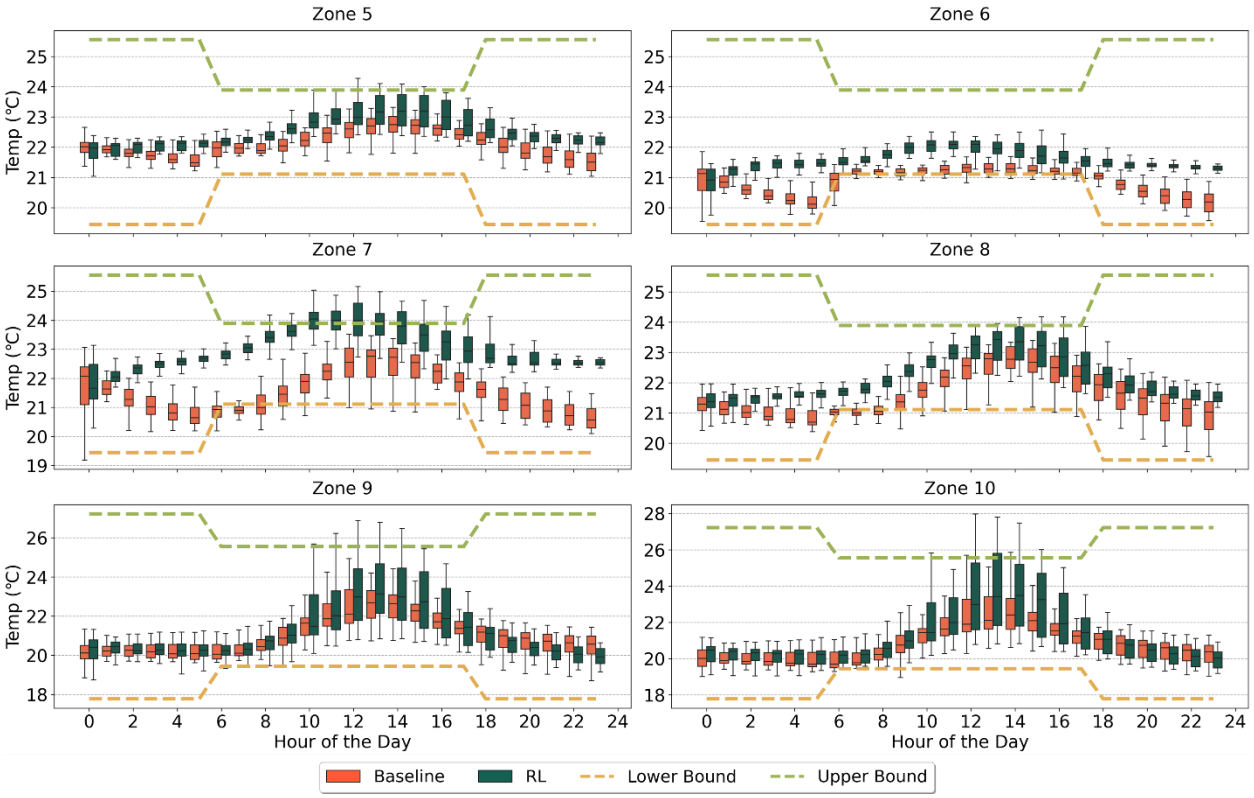


Figure 50. Temperature comparison of zones 4-9 between baseline and MARL during implementation.

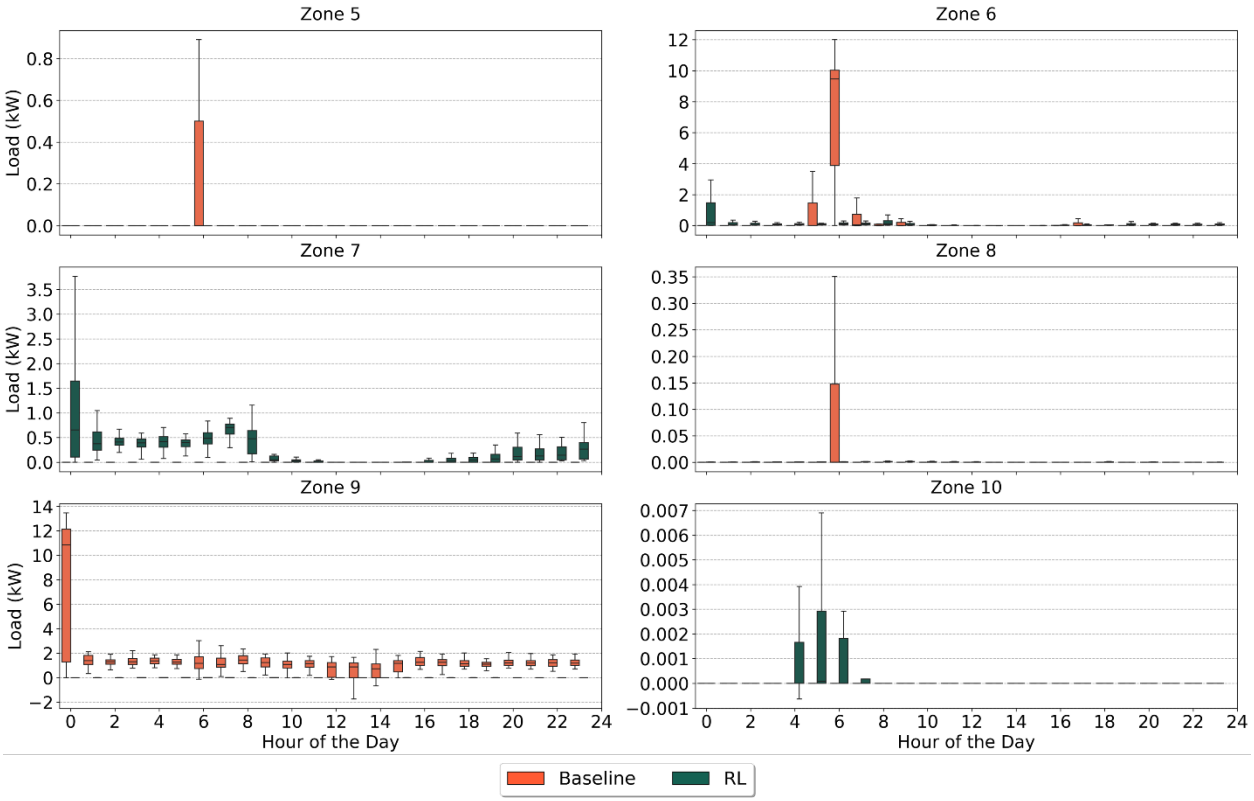


Figure 51. Radiant ceiling panel load comparison between baseline and MARL during implementation for zones 4-9.

The energy consumption and temperature violations under both controllers are quantified in Table 22. To account for varying weather conditions across the alternating deployment days, results are aggregated using a binning method: two bins are defined based on outdoor temperature and solar radiation with intervals of 5°C and 400 W/m², respectively, and the energy usage and temperature violation are averaged within each bin for each controller. The values in parentheses in the MARL load column indicate the percentage energy savings relative to the baseline.

Table 22 shows that the overall temperature violation under MARL is slightly higher than the baseline. This is attributable to two factors. First, as discussed above, the malfunctioning radiant panels in four zones inflate the MARL violation metric. Excluding these zones, the mean

temperature violations are 0.32°C and 0.49°C for the baseline and MARL, respectively, indicating comparable thermal comfort performance between the two controllers. Second, the MARL formulation treats temperature bounds as soft constraints within the reward function. In this setting, the agents inevitably tradeoff between competing objectives — comfort maintenance and energy reduction — leading to occasional violations. This behavior is consistent with findings from last chapter, and enforcing hard constraints in RL remains an active research question in the community.

Despite the slightly higher temperature violations, Table 22 demonstrates that MARL achieves substantial energy savings ranging from 35.06% to 69.67% across weather bins. When the four malfunctioning zones are excluded, the savings increase to 40.89%–86.34%. These significant reductions are driven by two key mechanisms: the proactive pre-conditioning strategy that avoids reactive over-compensation, and the agents learned ability to leverage inter-zone heat transfer for more efficient load distribution across the thermal network. Table 22 also shows uncertainty of energy saving for different scenarios. The uncertainty is quantified based on the standard deviation of energy saving. It shows that the energy saving can be fluctuated for different days within each scenario, but the scalable κ-neighbor truncated MARL always provides energy saving compared to the baseline.

In summary, the 42-day real-building deployment in building 2 demonstrates that the proposed framework scales effectively from simulation to practice. The PCGNN maintains below 3.6% MAPE across all 18 zones, and the MARL agents achieve 35–70% energy savings (41–86% excluding malfunctioning zones) while maintaining comparable thermal comfort to the baseline controller. Beyond raw performance metrics, the deployment reveals that the agents learn physically meaningful control strategies, notably proactive pre-conditioning and exploiting inter-zone heat transfer through core zones without being explicitly programmed to do so. These

emergent behaviors confirm that the combination of a physics-consistent dynamics model and κ-truncated multi-agent learning enables the agents to discover and exploit the building's thermal network structure for energy-efficient control in real-world operating conditions.

Table 22. Energy usage and temperature violation of baseline and MARL control under different weather conditions.

Temp Bin (°C)	Sol Bin (w/m²)	Energy: Baseline (kWh)	Energy: MARL (kWh)	Saving Uncertainty (±%)	Temp Vio: Baseline (°C)	Temp Vio: MARL (°C)
(-17.5, -12.5]	(0, 400]	584.41	327.6 (43.95%)	17.06	0.38	0.81
(-12.5, -7.5]	(0, 400]	621.53	312.44 (49.73%)	5.45	0.395	0.82
	(400, 800]	554.36	359.98 (35.06%)	7.67	0.393	0.86
(-7.5, -2.5]	(0, 400]	507.88	318.02 (37.38%)	11.96	0.408	0.73
	(400, 800]	573.2	272.4 (52.48%)	10.65	0.395	0.877
(-2.5, 2.5]	(0, 400]	669.94	203.39 (69.67%)	26.57	0.803	0.71
(2.5, 7.5]	(0, 400]	281.33	135.49 (51.84%)	31.07	0.55	0.88

6.3 Training Efficiency

PCGNN

As discussed earlier, a key architectural improvement of the proposed PCGNN over our previous work is the use of native PyTorch layers, causal 1-D convolutions and graph-based diffusion operations, to achieve mathematical equivalence to the manually connected neuron structure in [61]. This substitution enables significant acceleration of the training process. In [61], modeling 6 zones (zones 4 to 9) with a 6-hour prediction horizon typically required over 3 hours of training

time. It was infeasible to train the model with more zones and longer prediction horizons. Table 23 compares the training time of the proposed PCGNN across different model scales with our previous model. For consistency, the same zones from [61] are used for the 3-zone and 6-zone configurations: in the 3-zone case, rooms 302, 303, and 303A are aggregated into one zone and rooms 304 and 399C into another, while the 6-zone case models each room individually. The 3-zone and 6-zone models both converge in approximately 30 minutes regardless of the prediction horizon, representing a 6× speedup over the previous approach. The 18-zone model, which was computationally infeasible with the manual connection architecture, requires only 80 minutes for a 6-hour prediction horizon and 96 minutes for a 24-hour horizon. This training efficiency, scaling sub-linearly with the number of zones, is what makes it practical to extend physics-informed multi-zone modeling from a handful of zones to an entire building.

Table 23. Training time of PCGNN and previous model for building 2 with different number of zones and prediction horizons.

Number of Zones	Prediction Horizon	Training Time (PCGNN)	Training Time (Previous Model)
3 zones	6 hours	32 minutes	> 120 minutes
	24 hours	29 minutes	Infeasible
6 zones	6 hours	38 minutes	> 180 minutes
	24 hours	32 minutes	Infeasible
18 zones	6 hours	80 minutes	Infeasible
	24 hours	96 minutes	Infeasible

κ-Neighbor Truncated Model-Based MARL

To demonstrate the effectiveness of the proposed κ-truncated model-based MARL, we compare four configurations on the building 2 case: (1) model-free MARL without truncation (SAC: vanilla), (2) model-free MARL with κ-hop truncation (SAC: κ-hop only), (3) model-based MARL without truncation (SAC: dynamic only), and (4) model-based MARL with both truncation and

the dynamics model (SAC: dynamic & κ-hop). Figure 52 shows the training curves for each individual objective and the total reward.

From the total reward in Figure 52, the model-based approaches converge faster than their model-free counterparts, confirming that leveraging the PCGNN for synthetic rollouts accelerates policy learning. The combined configuration, κ-truncation with the dynamics model, converges to the highest total reward among all four cases. For the two model-free approaches, the agents initially achieve similar total rewards regardless of whether truncation is applied, indicating that the truncated critic can approximate the true value function well even with reduced information. This is consistent with the theoretical result in [140]. which proves that network MDPs with exponential decay properties admit accurate local approximations of the Q-function. However, neither model-free approach reaches the reward level of the model-based configurations.

The decomposition of the total reward into individual objectives is also shown in Figure 52. For energy usage, action smoothness, and indoor environmental quality (IEQ), the model-free approaches struggle to learn a stable balance: improvements in energy reduction come at the cost of deteriorating IEQ performance in the later stages of training, and agents struggle to make smooth action. By contrast, model-based approaches improve both objectives simultaneously, suggesting that access to the dynamics model enables the agents to discover control strategies that reduce energy consumption without sacrificing comfort and action smoothness. With κ-truncation further reducing the critic input dimensionality, and consequently the learning complexity, the proposed combined approach achieves the best overall performance among the four configurations.

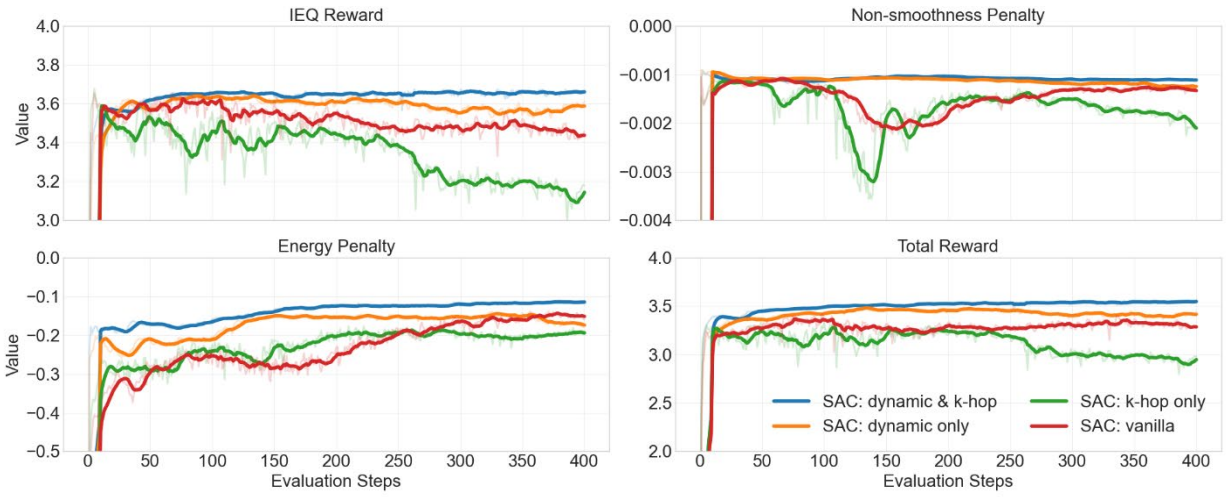


Figure 52. MARL training under different settings for building 2.

6.4 Observations on Measured Data and Its Impact of MARL Performance

Building 1

Building 1 determines the effective setpoint and operating mode based on occupancy status and outdoor temperature. Notably, the threshold for switching between heating and cooling modes has only a 0.5°C difference. This narrow deadband is problematic in practice because outdoor temperature sensors typically have an accuracy range of ±1°C, meaning sensor noise alone can trigger unintended mode switching. As shown in plot (a) of Figure 53, the temperature bounds derived from the effective setpoint and load sign reveal that after 12 PM, the system frequently alternates between heating and cooling modes at the same effective setpoint, resulting in simultaneous heating and cooling throughout the afternoon — a significant source of energy waste.

This issue also has direct implications for MARL training. If the lower and upper temperature bounds are taken directly from the raw measurement data, the agents are trained against inconsistent and physically unrealistic comfort expectations that reflect control system faults rather

than genuine occupant requirements. Under such conditions, the agents fail to learn a meaningful balance among competing objectives. Diagnosing and cleaning the temperature bounds, as shown in plot (b) of Figure 53, is therefore a necessary preprocessing step. With the corrected bounds, the proposed MARL achieves significant improvements in both energy efficiency and indoor temperature regulation, underscoring the importance of data quality for RL-based building control.

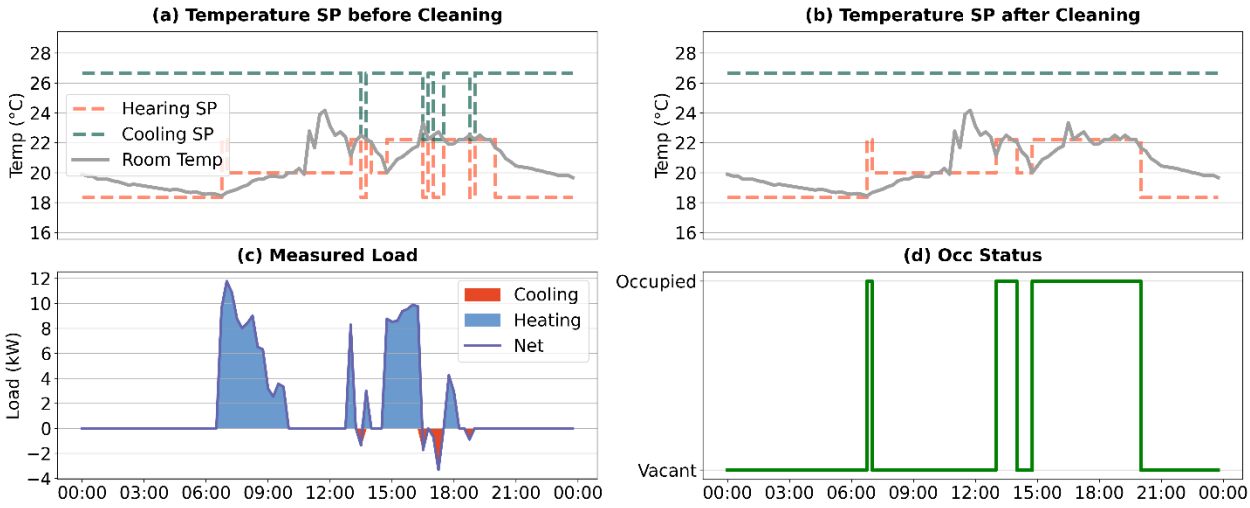


Figure 53. Temperature bounds before (plot 1) and after cleaning (plot 2) and measured load (plot 3), valve position (plot 4), and occupancy (plot 5) for building 1.

Building 2

As noted, four zones, zones 3, 13, 17, and 18, exhibited poor temperature performance under MARL and baseline control (Figure 54). Investigation revealed that this underperformance is attributable to hardware malfunction rather than deficiencies in the PCGNN or MARL algorithm. As shown in Figure 56, the radiant ceiling panels in zones 3 and 13 were commanded to provide heating, indicated by positive valve positions. However, the load feedback shows fluctuations

between cooling and heating even at 100% heating valve opening, suggesting hydraulic short circuits or leakage through the radiant ceiling panels. This malfunction results in an unmet heating demand, explaining the low indoor temperatures observed for zones 3 and 13 in Figure 54.

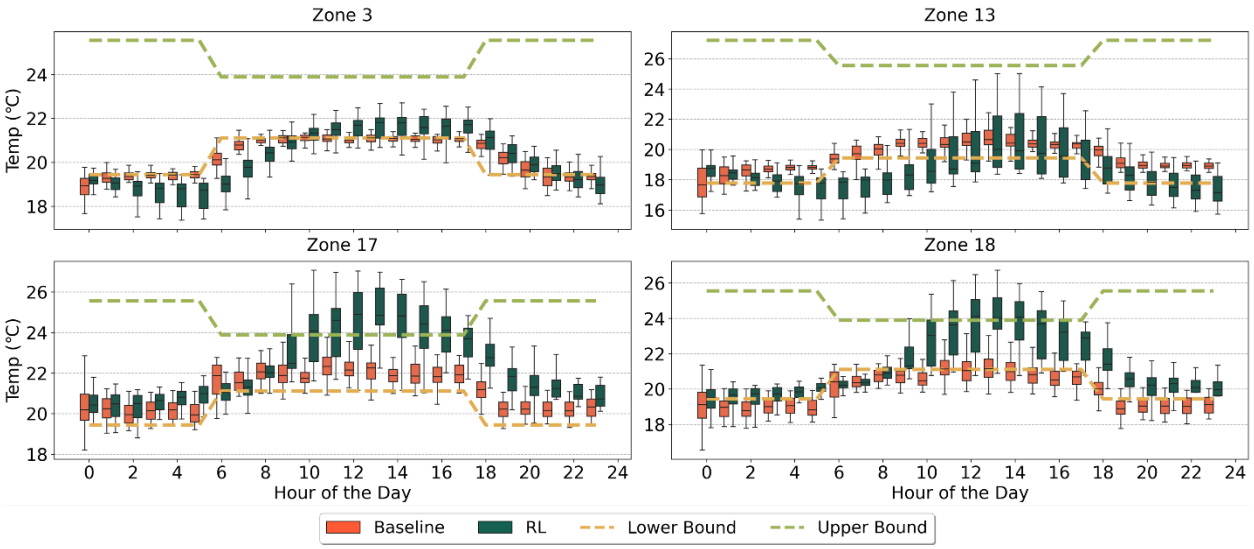


Figure 54. Temperature comparison of zones 3, 13, 17, and 18 between baseline and MARL during implementation.

The consequences propagate through the building's thermal network. Zones 17 and 18 are located directly above zone 13, and the MARL agents, having learned the inter-floor thermal coupling, attempted to compensate for the heating deficit in zone 13 by allocating additional heating to the fifth-floor zones. This compensatory strategy, while physically logical, led to overheating in zones 17 and 18. This finding illustrates both a strength and a limitation of the framework: the agents successfully learned to exploit cross-floor heat transfer, but they cannot fully overcome persistent hardware faults that violate the assumptions of the learned dynamics model.

6.5 Practical Considerations of RL Control

Building 1 is equipped with interactable thermostats, allowing occupants to adjust temperature settings based on their preferences. This dynamic setpoint introduces additional complexity for RL training, as the agents must respond to comfort requirements that change unpredictably throughout the day.

To investigate the impact of setpoint information on agent performance, we compare three state configurations in Figure 55: (1) current and next 8-step setpoints included as states (Current & Future SP), (2) current and previous 8-step setpoints included as states (Current & Previous SP), and (3) current setpoints only (Current SP Only). The results show that including only the current setpoint — the configuration that strictly satisfies the Markov property assumption — leads to poor performance, with the agents unable to learn strategies that adequately meet the temperature requirements. Including future setpoint information yields a significant improvement, with substantially fewer temperature violations. The configuration with past setpoint information performs better than the current-only case but noticeably worse than the future-information case: the agents exhibit delayed reactions to setpoint changes, confirming that historical setpoints serve as a partial but insufficient proxy for anticipating upcoming comfort demands.

These findings have practical implications for real-building deployment. Achieving the best RL performance in building 1 would require forecast models for both occupant thermostat interactions and weather conditions to supply the agents with anticipated future setpoints. This adds implementation complexity and introduces an additional source of error, as forecast inaccuracies could degrade the RL control performance. By contrast, building 2 uses schedule-based setpoints that are known in advance, avoiding this challenge entirely. The comparison highlights that the

complexity of RL deployment depends not only on the algorithm design but also on the controllability and predictability of the building's operational context.

Beyond the simulation findings, the real-building deployment in building 2 revealed that the majority of control failures were not caused by the MARL algorithm itself but by two infrastructure-level issues: sensor failures and BACnet communication errors. Sensor failures occurred due to power outages at individual sensors or disruptions to the cloud service that relays sensor data, both of which prevent the agents from receiving valid state observations and therefore lead to erroneous control decisions. A stable and reliable sensing infrastructure is thus a prerequisite for deploying any advanced control strategy in real buildings. BACnet communication failures arose because the protocol is designed for sequential access; when multiple processes attempt to read or write simultaneously, communication congestion causes errors that interrupt the control loop. Implementing asynchronous access would mitigate this issue in future deployments. These practical challenges underscore that the gap between simulation success and real-building performance is often determined not by the control algorithm but by the reliability of the underlying building automation infrastructure.

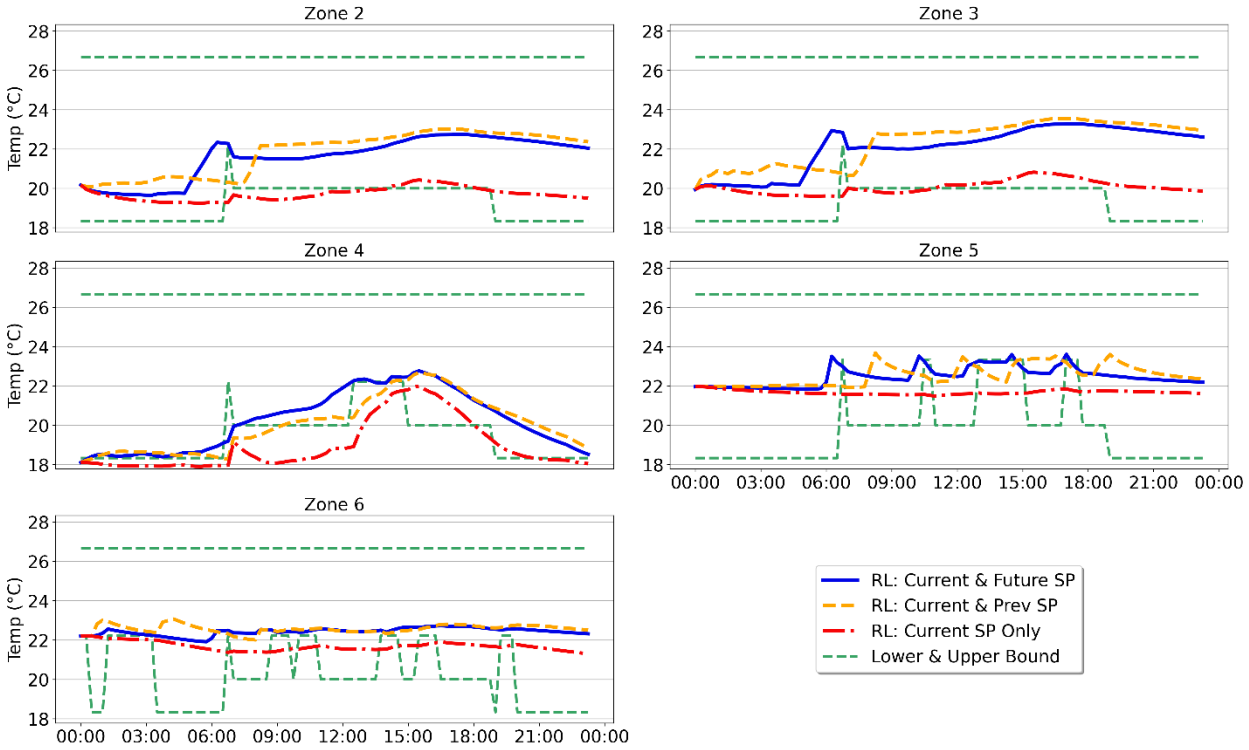


Figure 55. Temperature performance of MARL trained using different states for building 1.

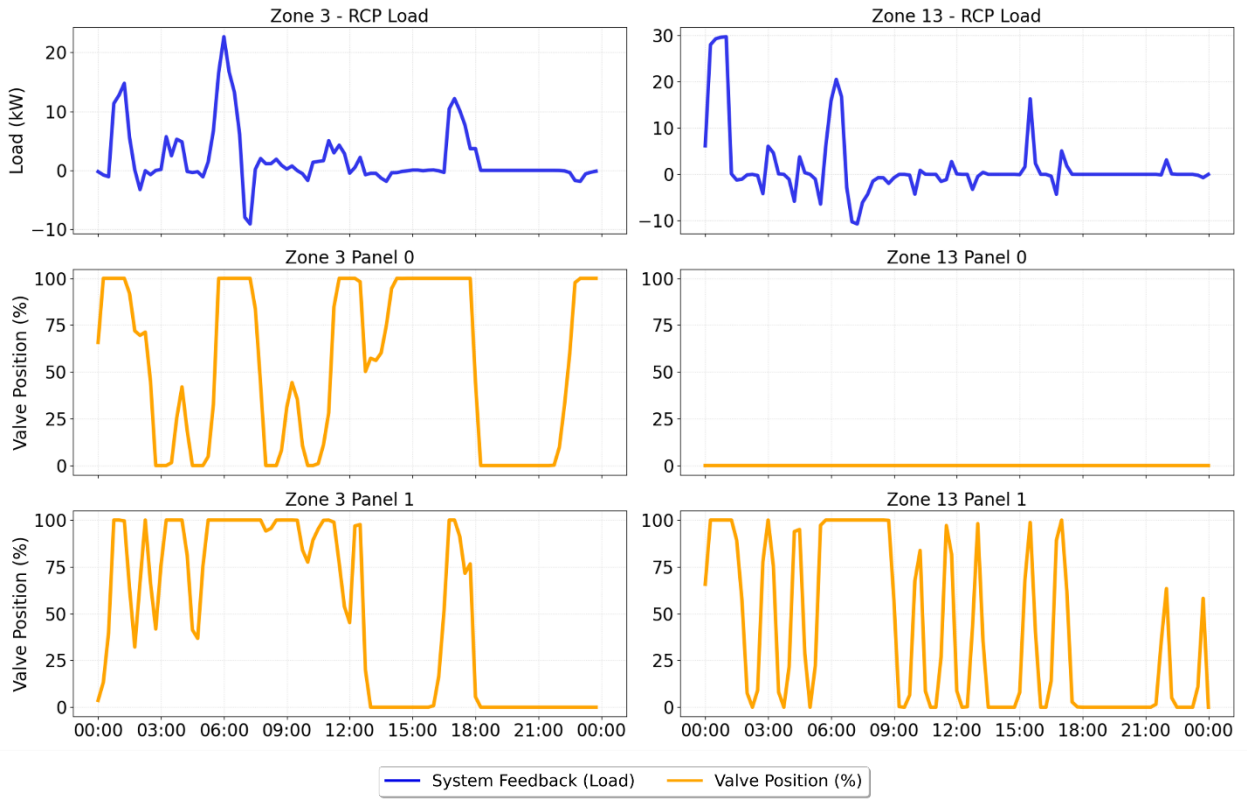


Figure 56. Feedback of radiant panel load and valve positions of zones 3 and 13 of building 2.

In summary, the results presented in this chapter validate the scalability of the proposed PCGNN and κ-neighborhood truncated MARL framework across both simulation and real-building deployment. The PCGNN achieves prediction accuracy comparable to or better than single-zone PCICNN models while scaling to 18 zones with physically consistent responses to perturbed loads and inter-zone thermal coupling. The model-based MARL with κ -hop truncation achieves 35-70% energy savings while maintaining thermal comfort, demonstrating that neighborhood-aware critics can effectively coordinate multi-zone control without the dimensionality burden of fully centralized approaches. The training efficiency analysis confirms that model-based training with PCGNN substantially accelerates agent convergence compared to model-free alternatives, and that the κ-truncation reduces critic input dimensions without sacrificing control performance. However,

the real-building deployment also reveals important practical lessons: data quality issues including sensor faults, inconsistent thermostat setpoints, and hardware malfunctions can degrade MARL performance in ways that are not apparent in simulation, and infrastructure reliability, particularly BACnet communication and cloud-based sensor services, remains critical for sustained advanced control operation. These findings demonstrate that scalable multi-zone PIML-based MARL is feasible for real buildings, while highlighting the data and infrastructure requirements that must be addressed for broader adoption.

Chapter 7. Exploration: Energy and Health Nexus

The COVID-19 pandemic raised the question of whether HVAC systems should also serve as a primary tool for mitigating airborne infection risk in indoor environments. If clean air delivery through ventilation and air cleaning were a dominant mitigation strategy, then infection risk would warrant incorporation as an explicit optimization objective in advanced building control, fundamentally altering the control formulation and potentially increasing energy consumption. This chapter presents an exploratory assessment of that question using WHO's comprehensive airborne infection risk model applied to real measured indoor environment data from the Syracuse Center of Excellence testbed. Section 7.1 presents the two-stage simulation framework. Section 7.2 describes the case study building and measured data. Section 7.3 details the WHO airborne infection risk model. Sections 7.4 and 7.5 describe the methodology for the two simulation stages: Simulation I evaluates mitigation effectiveness under realistic exposure conditions, while Simulation II quantifies the relative contribution of each factor through controlled statistical attribution. Sections 7.6 and 7.7 present the corresponding results. The findings demonstrate that the role of HVAC-based ventilation in infection risk mitigation is conditional: when the presence of an infector is unknown, ventilation is the only continuously available defense; when infector presence is known and behavioral interventions can be activated, ventilation becomes supportive rather than dominant.

7.1 Overview of Simulation Framework

This study uses a two-stage simulation framework to evaluate airborne infection risk mitigation. Simulation 1 assesses effectiveness under realistic conditions; Simulation 2 quantifies relative contributions under controlled baselines. Both stages use consistent model formulation and mitigation variables (Figure 57).

Analysis uses measured data from a single-zone indoor environment. Infection risk follows the WHO-recommended model. Ventilation, filtration, and air cleaning are represented as equivalent clean air delivery, allowing direct comparison across mitigation strategies.

Both simulations consider virus variants, clean air delivery, mask types, and social distance. The stages differ in objectives, assumptions, analytical methods, and sampling strategies.

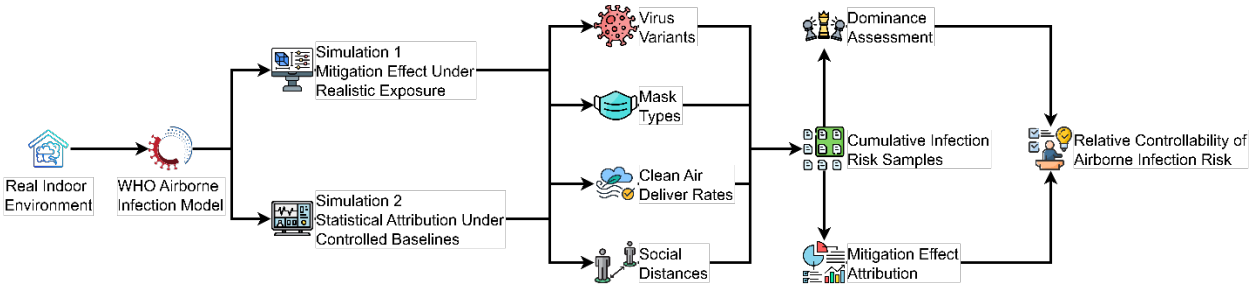


Figure 57. Overall simulation framework for assessing the controllability of airborne infection risk

Simulation 1 evaluates realistic indoor conditions with both short-range and long-range transmission. Measured occupancy, ventilation, temperature, and humidity preserve operational dynamics. Monte Carlo analysis samples uncertainties across biological, behavioral, and environmental variables to assess whether clean air delivery provides substantive risk reduction under realistic conditions.

Simulation 2 quantifies relative contributions under controlled conditions with fixed temperature, humidity, and occupancy. Two baselines are defined: (1) no infector mask with short-range contact, and (2) infector mask without short-range contact. Monte Carlo samples are analyzed statistically attribute risk to individual factors, isolating clean air delivery effects.

This framework combines feasibility-oriented and controlled attribution analyses, grounding conclusions in both realistic exposure conditions and statistically interpretable effects for robust evaluation of HVAC-based infection control.

7.2 Case Study and Measured Indoor Data

The case study uses a student office (Lab 403) at Syracuse University Center of Excellence, representing a typical shared office environment. The single-zone space contains twelve workstations with floor area 120 m^2 and ceiling height 3.0 m (volume = 245 m^3).

HVAC Configuration and Clean Air Delivery

Clean air is delivered via a dedicated AHU-VAV system. The room has two parallel systems: ceiling-based ventilation (conventional HVAC) and displacement ventilation (personalized control). This study considers only ceiling-based ventilation to represent typical centralized operation and avoid confounding effects. Measured outdoor airflow provides baseline clean air delivery; additional delivery from hypothetical air cleaning devices is evaluated parametrically.

Occupancy Measurement and Sensing Infrastructure

The testbed has door-mounted people counting sensor recording occupancy at 15-minute intervals. Real-time occupancy measurements are critical for infection risk assessment, as occupancy influences viral emission potential and ventilation demand. Measured data characterize realistic exposure in Simulation 1 and inform fixed occupancy selection for Simulation 2.

Representative Occupied Day

Figure 58 shows a representative occupied day reflecting typical office usage. The space is unoccupied during early morning cleaning, with student occupancy beginning after 8:30 AM. Peak occupancy occurs around midday (11:30 AM-1:00 PM) during lunch and collaborative periods. Outdoor air ventilation increases with occupancy and exceeds ASHRAE Standard 241 requirements, making this day suitable for evaluating marginal benefits of additional air delivery.

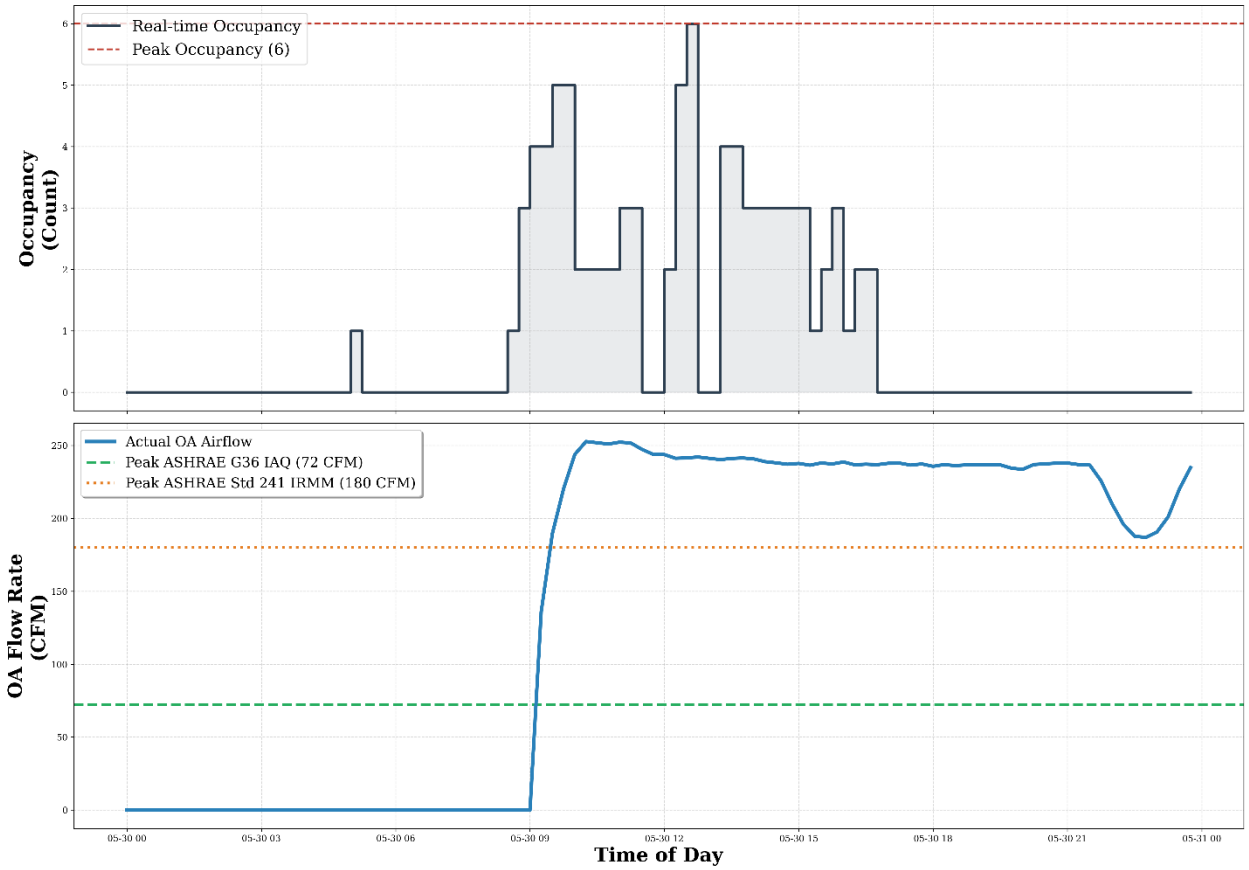


Figure 58. A representative occupied day of the testbed.

Role of Measured Data in Simulation Design

Measured data from the representative day is used in Simulation 1 to preserve realistic temporal patterns of occupancy, ventilation, temperature, and humidity. Simulation 2 uses controlled baseline conditions from the same testbed to isolate mitigation variable effects. This complementary approach supports both feasibility assessment and statistical attribution.

7.3 Airborne Infection Risk Model

The model of individual infection risk follows WHO's recommendations.

The viral load emission rate vR (in RNA copies per hour) is calculated by:

$$vR(D)_j = vl_{in} \cdot E_{c,j}\left(D, f_{amp}, \eta_{out}(D)\right) \cdot BR_k \tag{75}$$

Where vl_{in} is the viral load inside the infected host's respiratory tract (in RNA copies per mL); $E_{c,j}$ is the volumetric particle emission concentration per unit diameter ($mL \cdot m^{-3} \cdot \mu m^{-1}$), for a given expiratory activity j and as a function of the vocalization amplification factor f_{amp} and the size dependent outward mask efficiency $\eta_{out}(D)$; BR_k (m^3/h) is the breathing flow rate for a given physical activity k. The total emission rate can be obtained by integrating (75 over a range of particle diameters.

The volumetric particle emission concentration $E_{c,j}$ in (75) is modeled as:

$$E_{c,j}(D) = N_p(D) \cdot V_p(D) \cdot \left(1 - \eta_{out}(D)\right) \tag{76}$$

Where $N_p(D)$ is the number of particles of size D, and $V_p(D)$ is their volume. $N_p(D)$ is calculated as follows:

$$N_p(D) = \frac{1}{D} \sum_{i \in I(j)} \left[\frac{c_{n,i} \cdot f_{amp,j,i}}{\sqrt{2\pi} \cdot \sigma_{D_i}} \exp\left(- \frac{\left(\ln D - \mu_{D_i}\right)^2}{2\left(\sigma_{D_i}\right)^2} \right) \right] \tag{77}$$

Where $I(j)$ is a subset of $\{B, L, O\}$ determined by the expiratory activity j: for breathing $I(b) = \{B\}$, for speaking or shouting $(sp) = I(sh) = \{B, L, O\}$. This is originally proposed as

Bronchiolar/Laryngeal/Oral (B.L.O) model for viral generation at different locations of lung and respiratory tract of the infector. μ_L and σ_{D_i} are the mean and standard deviation of the natural logarithm of the diameter for each mode (in ln μm); $c_{n,i}$ is the total particle emission concentration for each mode. The amplification factor $f_{amp,i}(i)$ follows:

$$f_{amp,j,i} = \begin{cases} 1 \quad if\ i = B \\ 1 \quad breathing\ and\ speaking\ (i \in \{L, O\}) \\ 2 \quad shouting\ (i \in \{L, O\}) \end{cases} \tag{78}$$

The viral removal rate vRR depends on four aspects: ventilation, settlement, biological decay, and other clean air delivery excluding ventilation.

$$\lambda_{vRR} = \lambda_{ACH} + \lambda_{dep} + \lambda_{bio} + \lambda_{CADR} \tag{79}$$

The removal rate due to ventilation is calculated based on the air change rate per hour (ACH), which depends on the volumetric airflow rate Q_{ACH} and the volume of the space V_r:

$$\lambda_{ACH} = \frac{Q_{ACH}}{V_r} \tag{80}$$

The viral removal rate due to biological decay depends on room temperature and relative humidity and follows:

$$\lambda_{bio} = 0.16030 + 0.04018 \cdot \frac{T - 20.615}{10.585} + 0.02176 \cdot \frac{RH - 45.235}{28.665} - 0.14369 - 0.02636 \cdot \frac{T - 20.615}{10.585} \quad (81)$$

The viral removal rate was originally proposed to be related to indoor temperature, relative humidity, and solar radiation. It is assumed that solar radiation is zero when integrated into the risk assessment model.

The gravitational settlement is calculated based on the settling velocity $v(D)$ and the height h of the month and nose of a person standing. h is approximately 1.5 m.

$$\lambda_{dep} = \frac{v(D)}{h} \quad (82)$$

The settling velocity is calculated using Stokes law:

$$v(D) = \frac{(\rho_p - \rho_{air})(D_{evap} \cdot 10^{-6})^2 g}{18\mu_{air}} \quad (83)$$

Where ρ denotes the density; D_{evap} is the evaporation effect given the particle diameter; μ_{air} is the dynamic viscosity of the air.

The removal effect from the clean air delivery is calculated in the same way as ventilation with the volumetric airflow rate being by the CADR Q_{CADR}. For clarity, we will consider ventilation and

air change rate as the combination of ventilation from the HVAC system and clean air delivery from other air cleaning equipment.

The dynamics of infectious particles that contribute to long-range exposure are described based on the mass balance equation:

$$\frac{dC}{dt} = \frac{vR(D) \cdot N_{inf}}{V_r} - \lambda_{vRR}(D) \cdot C(t, D) \tag{84}$$

Where N_{inf} is the number of infectors.

The short-range concentration of viruses in the expiratory jet at close distance of a given size D, is given by:

$$C_{SR}(t, D) = C(t, D) + \frac{1}{S(x)} \cdot \left(C_{0,SR}(D) - C(t, D)\right) \tag{85}$$

Where $C(t, D)$ is the long-range concentration of viruses. $C_{0,SR}(D)$ is the initial concentration of viruses at mouth/nose defined by:

$$C_{0,SR} = E_{c,j}\left(D, f_{amp}, \eta_{out} = 0\right) \cdot vl_{in} \cdot 10^{-6} \tag{86}$$

The dilution factor $S(x)$ in (85) is described as:

$$S(x) = \begin{cases} \dfrac{2\beta_{r1}(x + x_0)}{D_m} & 0 < x < x^* \\ S(x^*)\left(1 + \dfrac{\beta_{r2}(x - x^*)}{\beta_{r1}(x + x_0)}\right)^3 & x \geq x^* \end{cases} \tag{87}$$

Where β_{r1} and β_{r2} are empirical parameters for radial penetration coefficient and the streamwise penetration coefficient in the jet-like stage of expired jet. x^* is the transition point from jet-like stage to puff-like stage.

Based on the long-range and short-range concentration dynamics, the dose exposure of the susceptible can be calculated as:

$$vD(D) = \sum_{i=1}^{n} \int_{t_i}^{t_{i+1}} [C(t, D) + C_{SR}(t, D)]dt \cdot f_{inf} \cdot BR_k \cdot f_{dep}(D) \cdot (1 - \eta_{in}) \tag{88}$$

Where t_i and t_{i+1} are the start and end time of the exposure event i; n is the total amount of independent exposures; f_{inf} is the viability of the infectious virus; $f_{dep}(D)$ is the deposition fraction in the respiratory tract; and η_{in} is the inward efficiency of the mask.

$$f_{dep}(D) = I_{frac}(D)(0.0587 + \frac{0.911}{1 + e^{4.77 + 1.185 \cdot \ln D_{evap}}} + \frac{0.943}{1 + e^{0.508 - 2.58 \cdot \ln D_{evap}}} \tag{89}$$

$$I_{frac}(D) = 1 - 0.5\left(1 - \frac{1}{1 + 0.00076 \cdot D_{evap}^{2.8}}\right) \tag{90}$$

The total exposure can be calculated by integrating (88) with respect to the particle size D:

$$vD^{total} = \int_0^{D_{max}} vD(D)dD \tag{91}$$

Finally, the probability of airborne transmission can be calculated:

$$P(I|vD^{total}, ID_{50}) = 1 - e^{-\frac{vD^{total}}{\frac{ID_{50}}{\ln 2} \cdot T_{voc} \cdot \frac{1}{1-HI_{exp}}}} \tag{92}$$

Where $P(I|vD^{total}, ID_{50})$ denotes the conditional probability of event I (infection) for given values of the total absorbed and infection doses vD^{total} and ID_{50}. T_{voc} is the reported increase of transmissibility of a variant of concern; HI_{exp} is the host immunity of the exposed occupants, given by the reported vaccine efficiencies.

In the simulation, it is assumed that the indoor air is well-mixed for long-range concentration. The short-range concentration varies according to the social distance. It is also assumed that the outdoor air is clean and pathogen-free.

7.4 Simulation I: Mitigation Effect Under Realistic Exposure

Objective and Scope

Simulation 1 evaluates the effectiveness of airborne infection mitigation strategies under realistic indoor exposure conditions using measured occupancy and environmental data. The simulation preserves temporal dynamics of ventilation, temperature, and occupancy patterns to assess how mitigation measures and behavioral/biological factors affect infection risk under operational

building conditions. The objective is to determine whether clean air delivery provides substantive risk reduction relative to other factors (masking behavior, distancing, virus variant) when each is varied individually from a common baseline scenario.

Measured Data and Temporal Resolution

Episode Selection and Structure

A continuous 24-hour period from the measured dataset is selected for simulation, spanning 96 steps at 15-minute intervals. This episode exhibits typical weekday occupancy patterns with:

- Morning arrival period (approximately 8:30 AM – 10:00 AM)
- Sustained occupancy during working hours
- Midday peak occupancy (11:30 AM – 1:00 PM)
- Gradual departure in late afternoon/evening
- Overnight unoccupied period

Time-Series Inputs from Measurements

The following variables (Table 24) are taken directly from measurements at each 15-minute time step and are not sampled in the Monte Carlo analysis.

Table 24. Time-series inputs from measurements

Variable	Symbol	Unit	Source
Number of occupants	$N_{occ}(t)$	Persons	People counting sensor
Outdoor air ventilation rate	$Q_{ACH}(t)$	CFM	Building automation system
Indoor temperature	$T(t)$	K	Thermostat

These measured inputs preserve realistic temporal variability in environmental conditions, occupancy-driven ventilation behavior, and thermal dynamics that influence viral decay rates throughout the exposure episode.

Exposure Scenario Definition

Both short-range and long-range transmission pathways are explicitly modeled to capture the full spectrum of exposure mechanisms:

1. Long-range (well-mixed) exposure: Occurs continuously throughout the occupied period as virus-laden aerosols are emitted by the infector, mixed throughout the room volume via air circulation, and inhaled by the susceptible at concentrations determined by the room-level mass balance equation. This pathway dominates when individuals are separated by distances greater than approximately 2 meters.
2. Short-range (proximity) exposure: Occurs during defined contact events when a susceptible individual is within close proximity to the infector, experiencing elevated concentrations from the expiratory jet before complete dilution and mixing occur. This pathway can dominate infection risk at distances less than 2 meters, particularly for brief contact durations.

Short-Range Contact Event

A single short-range contact event is defined to represent collaborative work or persistent close seating arrangements common in shared office environments:

- Start time: 2:00 PM on a typical workday
- Duration: 15 minutes (0.25 hours), which is half of the duration from [144].

- Social distance: Varied parametrically across scenarios: 0.5m (intimate distance), 1.0m, 1.8m (6-feet social distance).
- Activity and expiration: Independently sampled for the short-range interaction (see Monte Carlo sampling)

The choice of a 15-minute contact event reflects typical durations for collaborative tasks (e.g., discussing a document, reviewing shared work) while being conservative enough to capture substantial short-range exposure without assuming unrealistically prolonged close contact.

Infector and Susceptible Configuration

- Number of infectors: 1 individual assumed infectious throughout the episode
- Number of the susceptible evaluated: 1 individual assessed for infection risk when the number of occupants is greater than 1, i.e., when there is a susceptible.
- Viral shedding assumption: The viral load is sampled from empirically derived distributions. Large particles that contribute to short-range contact are assumed to be 0 when the infector wears a mask. In other words, the short-range exposure is not considered if the infector is wearing a mask. This assumption is made because the mask will block the emission jet from the infector, as used in the WHO's infection risk calculator.
- Exposure continuity: Susceptible experiences continuous long-range exposure during all occupied periods, plus elevated short-range exposure during a 15-minute contact event when the infector is not wearing a mask.
- Host immunity: Fixed at zero (no prior immunity or vaccination) to represent a conservative worst-case scenario.

Baseline Scenario Definition

All parametric evaluations are conducted relative to a common baseline scenario that represents a reference condition. In the baseline scenario, the infector is assumed not wearing mask to allow short-range exposure and represent unmitigated emission. Based on the same rationale, the susceptible is also assumed not to wear masks. The social distance is set as 1.0 m to represent proximity for office collaboration. The clean air delivery is based on the measurement data as shown in Figure 58. Original SARS-CoV-2 strain is assumed as the reference strain.

In this baseline scenario, both long-range and short-range exposures contribute to infection risk, and no additional mitigation measures beyond measured baseline ventilation are applied.

Monte Carlo Sampling Strategy

A Monte Carlo experiment with 100,000 independent samples is conducted for each parametric scenario to ensure robust statistical characterization of infection risk uncertainty. Each sample represents a unique combination of biological and behavioral parameters drawn from their respective probability distributions. The large sample size ensures reliable estimation of both central tendencies and tail risks.

Sampled Variables and Distributions

For all scenarios, the following parameters are sampled independently to represent inter-individual variability, as shown in Table 25.

Table 25. Sampled variables and distribution

Variable	Distribution	Parameters	Reference
Viral load in sputum (vl_{in})	Gaussian kernel	Empirical from symptomatic patients	[146]
Infectious dose (ID_{50})	Uniform	10-100 RNA copies	[145][147]
Viable-to-RNA ratio (r_{inf})	Uniform	0.01-0.6	[145]
Population activity (breathing rate)	Log-normal (seated)	$\mu = -0.687, \sigma = 0.105$	[145]
Short-range activity (breathing rate)	Log-normal (standing)	$\mu = -0.574, \sigma = 0.094$	[145]

Note that viable-to-RNA ratio r_{inf} is used to calculate the viability of the infectious virus $f_{inf} = r_{inf}(1 - HI_{inf})$ in (88), where HI_{inf} is the infected host immunity.

These sampled parameters represent biological and physiological uncertainty that exists regardless of which mitigation measures or behavioral factors are being evaluated.

Parametric Scenario Groups

To assess the relative impact of different factors on infection risk, four scenario groups are evaluated. In each group, one parameter is varied while all others remain at baseline values, as shown in Table 26. For the clean air delivery augmentation, reductions of 170 CFM and 80 CFM from the measured clean air delivery represents the ventilation rate based on Guideline 36 and Standard 241, as shown in Figure 58. Increases of 100 and 300 CFM are considered as equivalent to clean air delivery increase by portable HEPA filter and multiple air cleaners.

Table 26. Parametric scenario groups. Bold represents baseline conditions.

Variable	Levels
Clean air delivery augmentation (CFM)	[-170, -80, **0**, +100, +300]

Social distance (m)	[0.5, 0.8, **1.0**, 1.2, 1.5 1.8]
Mask types	[**No mask**, cloth, Type I/Surgical, FFP2/N95]
Virus variant	[**Original**, Alpha, Beta, Gamma, Delta, Omicron]

Table 27 shows the filtration efficiencies of different mask types.

Table 28 shows the transmission factors of different virus variants. It is worth noting that the transmission factor T_{voc} is used in the dominator in the exponent of (92). Therefore, lower values of T_{voc} indicate increases of viral transmissibility of the variant compared to the original strain of SARS-CoV-2.

Table 27. Mask types and filtration efficiencies

Mask Type	Inward Efficiency	Outward Efficiency	Source
No Mask	0.0	0.0	-
Cloth	0.225	0.35	[148]
Type I	0.5	0.5	[148]
FFP2 (N95)	0.865	0.865	[148]

Table 28. Transmission factors of different virus variants.

Virus variant	Transmission factor (T_{voc})	Source
Original	1.0	Baseline
Alpha	0.78	[149]
Beta	0.80	[149]
Gamma	0.72	[149]
Delta	0.51	[149]
Omicron	0.20	[150]

Simulation Outputs

For each Monte Carlo sample and each time step throughout the 24-hour episode, the simulation outputs:

1. Viral concentration [RNA copies/m^3]: this is a summation of both long-range and short-range concentrations.
2. Incremental deposited exposure
3. Cumulative infection probability [%]: time-evolved risk accounting for all exposure intervals up to the current time.

7.5 Simulation II: Statistical Attribution of Mitigation Effects

Objective and Scope

Simulation II quantifies the marginal contribution of each factor to infection risk through systematic sensitivity analysis and statistical modeling under controlled baseline conditions. Unlike Simulation I, which evaluates mitigation effectiveness under realistic temporal dynamics, Simulation II isolates the effect of each variable by fixing environmental and occupancy conditions. The objective is to rigorously determine whether clean air delivery provides a dominant, moderate, or minor contribution to risk reduction compared to other factors, using standardized effect sizes derived from logistic regression.

Controlled Baseline Conditions

To enable statistical attribution and eliminate confounding variability from temporal dynamics, Table 29 shows fixed conditions across all Monte Carlo samples.

Table 29. Controlled baseline conditions

Parameter	Fixed Value	Rationale
Room volume	245 m^3	Lab 403 geometry
Exposure duration	11 hours (7:00AM – 6:00PM)	Maximum stay
Number of occupants	2	1 infector and 1 susceptible

		person
Indoor temperature	Fixed at 22 ℃	Removes temporal temperature variation
Indoor relative humidity	Fixed at 50%	Removes humidity-driven decay variation
Number of infectors	1	Single-source scenario
Number of susceptible people	1	Individual risk assessment

Unlike Simulation I, which used measured time-varying data, these fixed conditions ensure that observed variability in infection risk is attributable solely to the sampled variables rather than environmental or occupancy fluctuations.

Two-Scenario Design

Simulation II evaluates two distinct scenarios based on infector mask status, which determines whether short-range transmission occurs.

Scenario 1: No Mask for Infector with Short-Range Contact

When the infector does not wear a mask, the expiratory jet remains intact, creating localized high-concentration zones near the infector. This allows short-range transmission to occur during close-proximity interactions. When both long-range and short-range occur, clean air delivery rate, social distance, mask wearing of the susceptible, short contact duration, and both short-range and population activity will impact the infection risk of individuals. Therefore, aforementioned variables are sampled based on their typical range for the tested room as shown in Table 30.

Table 30. Sampled variables for scenario 1.

Variable	Type	Range	Sampling
Clean air delivery	Continuous	50-600 CFM (50-	Discrete uniform

rate		CFM interval)	
Social distance	Continuous	0.5-1.8m (0.3-m interval)	Discrete uniform
Short-range contact duration	Continuous	1-30 min (5-min interval)	Discrete uniform
Susceptible mask type	Categorical	None / Cloth / Type I/FFP2	Uniform
Virus variant	Categorical	Original / Alpha / Beta / Gamma / Delta / Omicron	Uniform
Short-range activity	Categorical	Seated/Standing	Uniform
Short-range expiration	Categorical	Breathing/Speaking	Uniform
Population activity	Categorical	Seated/Standing	Uniform
Population expiration	Categorical	Breathing/Speaking	Uniform

The purpose of scenario 1 is to quantify the marginal effect of each factor when both short-range and long-range transmission pathways contribute to infection risk.

Scenario 2: Mask for Infector without Short-Range Contact

When the infector wears a mask, the expiratory jet is disrupted and rapidly dispersed. The mask acts as a physical barrier that breaks up the momentum and directionality of exhaled air, causing rapid lateral mixing with ambient room air. As a result, the coherent jet structure that maintains elevated concentrations at close distance is eliminated, and short-range concentration enhancement does not occur. All transmission occurs via the long-range, well-mixed pathway.

For scenario 2, the sampled variables are mostly the same as scenario 1 shown in Table 30. Exceptions are that short-range related variables are not considered. In addition, mask types of the infector are sampled with range cloth, type I, and FFP2, based on uniform categorical distribution.

The purpose of scenario 2 is to quantify the marginal effect of each factor when only long-range transmission occurs.

For each of the two scenarios, 100,000 parameter combinations are sampled to ensure comprehensive coverage of uncertainty space. The sampling strategy results in a total of 774,144 unique combinations when accounting for all discrete parameter levels within the continuous sampling framework.

For each of the 100,000 samples per scenario, the individual infection probability is computed over the 11-hour exposure period using the airborne infection risk assessment model. This represents the probability that the susceptible individual becomes infected given the specific combination of sampled parameters under fixed environmental conditions.

Statistical Modeling and Feature Importance Analysis

To quantify the marginal contribution of each factor to infection risk and assess the relative importance of clean air delivery compared to other mitigation strategies, three complementary analytical approaches are employed: generalized linear modeling (GLM) for statistical inference of marginal effects, machine learning-based feature importance ranking to capture nonlinear relationships and interactions, and moment-independent global sensitivity analysis to characterize distributional drifts. These methods address different aspects of the research question and provide converging evidence regarding ventilation effectiveness.

Generalized Linear Model (GLM)

The relationship between infection risk and all sampled variables is quantified using a binomial generalized linear model with logit link function.

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \boldsymbol{\beta X} \tag{93}$$

Where p is the infection probability (response variable); $\boldsymbol{X}$ is the matrix of the predictor variables including all sampled factors; β_0 is the intercept; $\boldsymbol{\beta}$ are regression coefficients representing the marginal effect of each predictor on the log-odds of infection.

The logit transformation ensures predicted probabilities remain bounded between 0 and 1, and the model assumes that infection events follow a Bernoulli distribution. For all continuous predictors (ventilation rate, social distance, short-range contact duration), coefficients represent the change in log-odds per unit increase in the predictor. For all categorical variables (mask types, virus variants, activity levels, expiration types), dummy coding is used with a defined reference case, and coefficients represent the log-odds difference relative to the reference level.

For all categorical variables, the reference case is defined as: no mask for the susceptible person, original strain for virus variant, seated for activity level, and breathing for expiration type. All regression coefficients represent the change in log-odds of infection relative to this reference case. To facilitate interpretation, coefficients are converted to odds ratios ($OR = e^{\beta}$) and percentage change in odds.

$$Percentage\ change\ in\ odds = \left(e^{\beta} - 1\right) \times 100\% \tag{94}$$

Negative coefficients ($\beta < 0$) indicate risk reduction (odds decrease), while positive coefficients indicate risk increase (odds increase). Statistical significance is assessed using Wald Z-tests, with Z-statistics and corresponding p-values reported for each coefficient. Additionally, 95% confidence intervals are reported for all coefficients to quantify estimation uncertainty.

Model fit is evaluated using:

- Null deviance D_{null}: Measures the deviance of an intercept-only model (no predictors), representing baseline variability in infection risk.
- Residual deviance D_{res}: Measures the deviance of the fitted model with all predictors, representing unexplained variability after accounting all sample factors.
- Pseudo-R^2 (McFadden's): Calculated as $R^2 = 1 - \frac{D_{res}}{D_{null}}$. This measures the proportional reduction in deviance due to including predictors.

The GLM approach provides interpretable marginal effects under the assumption that predictors influence risk independently (i.e., no interaction terms unless explicitly specified). This is valuable for understanding how each factor influences risk when others are held constant at their mean values. However, GLM assumes linearity on the logit scale and does not automatically capture nonlinear relationships or higher-order interactions among predictors unless interaction terms are explicitly included in the model specification.

Machine Learning Feature Importance (XGBoost + SHAP)

To account for nonlinear relationships and interactions among predictors, a gradient boosting model (XGBoost) is trained on the Monte Carlo simulation results, and feature importance is quantified using SHAP (Shapley Additive explanations) values.

XGBoost builds an ensemble of decision trees sequentially, with each tree correcting errors from the previous ensemble. Model performance is assessed using R^2. Since we have an exhaustive search result of the variables, we directly train the model on all data.

SHAP values decompose each prediction into individual feature contributions based on cooperative game theory. For each feature j and sample i, the SHAP value $\phi_j^{(i)}$ represents that feature's contribution to the prediction.

$$\hat{y}_i = \phi_0 + \sum_{j=1}^{M} \phi_j^{(i)} \tag{95}$$

Where ϕ_0 is the baseline (mean) prediction and M is the number of features.

Features are ranked by mean absolute SHAP value.

$$Importance = \frac{1}{n}\sum_{i=1}^{n}\left|\phi_j^{(i)}\right| \tag{96}$$

Higher values indicate greater influence on predictions across all scenarios. Unlike GLM coefficients, SHAP values inherently account for all interactions and nonlinearities, providing a global measure of predictive importance.

Delta Sensitivity Analysis

To assess how the entire infection risk distribution changes when each variable is fixed, the Delta index (Borgonovo's measure) is calculated. Unlike variance-based sensitivity indices (e.g., Sobol indices) that only capture changes in output variance, the Delta index measures shift in the entire probability density function (PDF), capturing changes in mean, variance, skewness, and tail behavior simultaneously.

The Delta sensitivity index for variable X_i is defined as.

$$\delta_i = \frac{1}{2} E_{X_i}\left[\int \left| f_Y(y) - f_{Y|X_i(y)} \right| dy\right] \tag{97}$$

Where $f_Y(y)$ is the unconditional PDF of infection risk across all scenarios, $f_{Y|X_i(y)}$ is the conditional PDF when X_i is fixed, and the integral measures the total variation distance between these distributions.

The unconditional PDF is estimated using kernel density estimation (KDE) with Gaussian kernel across all samples. Conditional PDFs are estimated using KDE for each stratum. Delta index is computed as expected total variation distance that is calculated numerically for each condition.

For reference, first-order Sobol indices (S_1) are also calculated.

$$S_1^i = \frac{Var(E[Y|X_i])}{Var(Y)} \tag{98}$$

To distinguish direct effects from combined effects, Spearman rank correlations (ρ_{ij}) are calculated between distributional shifts induced by each pair of variables. For each variable pair, conditional mean risks are computed, and shifts from the unconditional mean are correlated across all samples.

$$\rho_{ij} = 1 - \frac{6 \sum_{k=1}^{n} d_k^2}{n(n^2 - 1)} \tag{99}$$

Where d_k is the rank difference between shifts induced by variables X_i and X_j for sample k.

Synthesis Across Analytical Methods

The three analytical methods provide complementary perspectives on the relationship between mitigation factors and infection risk:

GLM quantifies independent marginal effects with statistical significance testing, assuming linearity and independence. It answers: "What is the expected change in log-odds from a unit change in each variable?"

XGBoost + SHAP ranks overall predictive importance accounting for nonlinearities and interactions automatically. It answers: "Which variables are most important for predicting risk across all scenarios?"

Delta Analysis measures distributional shifts beyond variance, distinguishing direct versus combined effects through correlation. It answers: "How does the entire risk distribution change with each variable, and does it work independently or in combination?"

Convergence across all three methods provides robust evidence independent of methodological assumptions. Divergence reveals nuances: a variable may be statistically significant (GLM) but have low practical importance (SHAP) or may have moderate distributional impact (Delta) but work primarily through combination with other factors (high correlations) rather than independently. This multi-method framework enables comprehensive characterization of clean air delivery's role—distinguishing whether it is a dominant standalone control, a supportive measure requiring combination with other interventions, or a marginal factor with limited practical impact under realistic conditions.

7.6 Results of Simulation I: Mitigation Effect Under Realistic Exposure

Simulation I evaluates infection risk under realistic temporal dynamics with measured occupancy and ventilation patterns. Results are presented for parametric variations in clean air delivery, social distance, infector mask type, susceptible mask type, and virus variant, all relative to a common baseline scenario where neither infector nor susceptible wear masks, social distance is 1.0m, clean air delivery rate follow measured baseline ventilation, and the virus is the original SARS-CoV-2 strain.

Time-Resolved Infection Risk Dynamics

For better readability, we only show the time-series results with different clean air delivery rates in Figure 59 and Figure 60 with different short-range expiration activity. Figure 59 and Figure 60 represent breathing and speaking as expiration activity during the 15-minute short-range contact period, respectively. The exposure in the two figures indicates instantaneous exposure rather than cumulative exposure. The instantaneous exposure is calculated based on the virus concentration (both short-range and long-range) at the given time step, integrated by the interval of between time

steps. The risk in the two figures represents cumulative infection probability of susceptible individuals. The colored shaded area in the risk plot represents 5%~95% uncertainty interval, with the colored lines indicating the averaged infection risk under different clean air delivery rates. Red shared areas indicate short-range contact periods.

Result with breathing as short-range expiration

It is observed that when short-range expiration only involves breathing, the virus concentrations under different clean air delivery rates vary. A higher clean air delivery rate results in an earlier peak of the virus concentration with a lower peak value and a shorter decaying rate when the room is unoccupied. The temporal evolution reveals three distinct phases. During the accumulation phase from morning arrival until the short-range contact event, viral concentration increases gradually as the infector emits aerosols during occupied periods. Higher clean air delivery rates maintain lower peak concentrations throughout this period. The short-range contact event at 2:00 PM causes a sharp spike in both concentration and exposure, though with breathing-only expiration, this spike remains relatively modest. Following occupant departure, the decay phase demonstrates that higher ventilation rates produce steeper concentration decay curves, with the 550 CFM case (3.81 ACH) reaching negligible concentrations approximately one hour faster than the 80 CFM case (0.55 ACH).

Compared to the baseline condition that has 250 CFM clean air delivery rate (approximately 1.73 ACH), the ventilation following Guideline 36 (0.55 ACH) has 18.53% higher infection risk, and the ventilation based on Standard 241 (1.12 ACH) has 7.06% higher infection risk. For the two cases with higher clean air delivery rate, the infection risk is lowered by 7.27% when additional 100 CFM clean air is supplied (resulting in 2.43 ACH) and 16.34% when additional 300 CFM

clean air is supplied (resulting in 3.81 ACH). The 5th-95th percentile uncertainty bands illustrated in the shaded regions show substantial variability in infection outcomes driven by sampled biological parameters including viral load, infectious dose, and viable-to-RNA ratio. While the bands overlap considerably across ventilation scenarios, indicating that biological uncertainty often exceeds the effect of ventilation changes, the mean risk reductions remain statistically discernible. Under this conservative scenario when both infectors and susceptible individuals do not wear masks and expiration activity is light, i.e., breathing only, clean air delivery can significantly impact the individual infection probability.

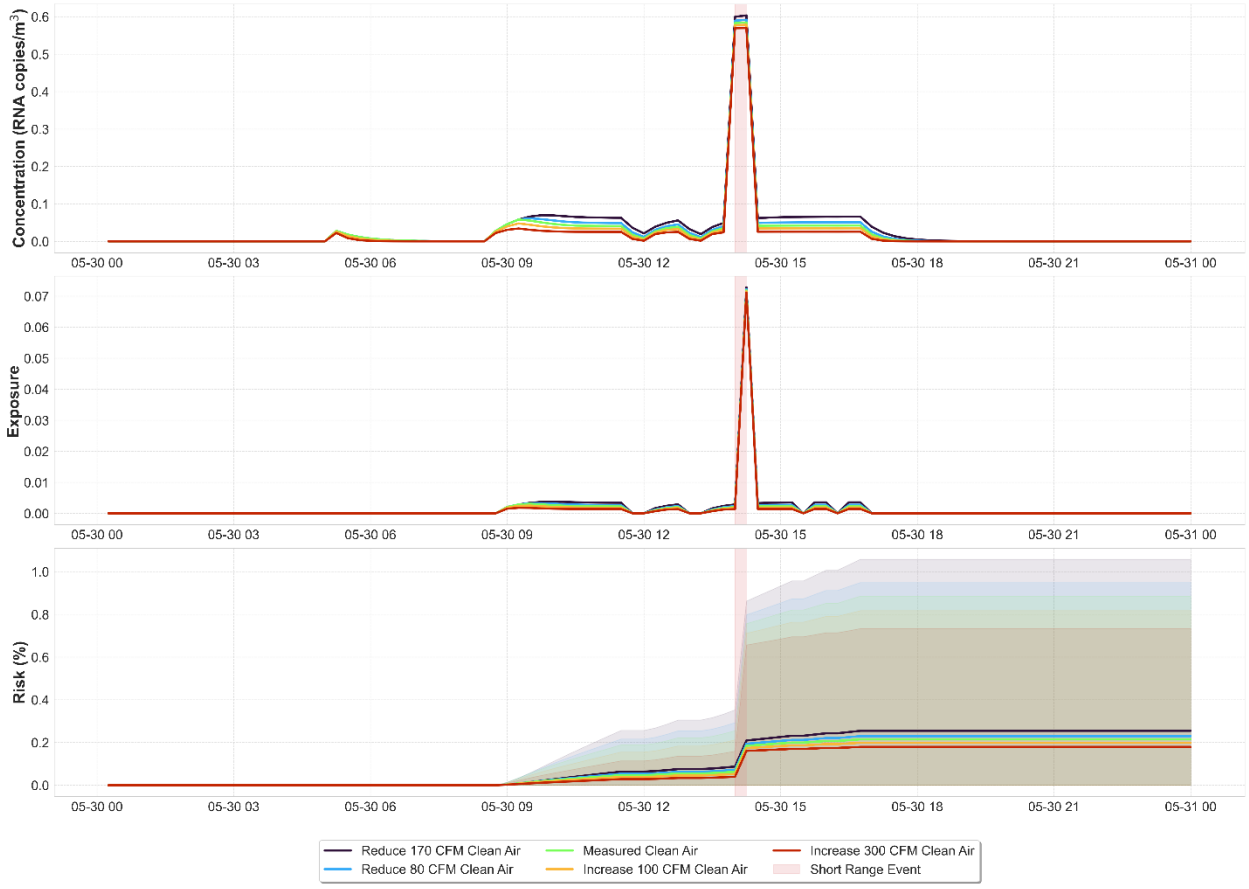


Figure 59. Time-series results of different clean air delivery rate with breathing as short-range expiration.

Result with speaking as short-range expiration

Despite infection risk reduction by clean air delivery rate under light expiration activity, it is unrealistic that short-range contact only involves breathing as expiration activity. It is more likely that speaking will occur during short-range contact such as collaborative works. If speaking is assumed as the expiration type, it is shown in Figure 60 that short-range transmission will dominate the virus concentration, exposure, and therefore infection risk. The 15-minute speaking event produces a concentration spike that completely dominates the infection risk profile, with the sharp increase in exposure during this brief period accounting for the majority of total cumulative exposure by day's end. In this case, the relative changes of infection risk under different clean air delivery rates are 0.35%, 0.13%, -0.14%, and -0.30% compared to the baseline case, respectively, for -170 CFM, -80 CFM, +100 CFM, and +300 CFM clean air augmentations. These relative changes are smaller than the statistical noise in the Monte Carlo samples and fall well within the overlapping uncertainty bands, indicating that they are operationally negligible.

The stark ineffectiveness of ventilation under speaking conditions can be attributed to a fundamental timescale mismatch between expiratory jet dynamics and room-level air mixing. During the 15-minute contact event at 1.0 m distance, short-range concentrations are determined by momentum-driven jet penetration rather than by the room's steady-state ventilation rate. By the time exhaled aerosols fully mix into room air, where ventilation could theoretically dilute them, substantial inhaled dose has already occurred through the short-range pathway. The remaining long-range exposure throughout the day contributes relatively little to total risk when short-range emissions are vigorous. This finding indicates that under more realistic conditions, clean air delivery alone cannot provide substantial mitigation effect on the individual infection risk, especially when there is short-range transmission with moderate and heavy expiration.



Figure 60. Time-series results of different clean air delivery rate with speaking as short-range expiration.

Comparative Effectiveness Across All Factors

For the rest of the variables, including mask types of infectors and susceptible people, virus variants, and social distances, aggregated results are shown in Figure 61 and Figure 62 for breathing and speaking short-range expirations, respectively. The infection risk for different values of variables and the relative change compared to the baseline conditions are shown from Table 31 to Table 35.

Under the breathing-only short-range expiration scenario shown in Figure 61, a clear hierarchy of intervention effectiveness emerges. Source control through infector masking proves most effective, with FFP2 masks providing the most significant mitigation, followed by surgical masks (Type I) and cloth masks. Receptor protection via susceptible masking demonstrates high effectiveness as well. Notably, the cloth mask worn by the susceptible provides risk reduction comparable to the most aggressive ventilation augmentation tested (+300 CFM, which achieves 16.34% reduction). Social distancing also shows significant effectiveness, with increasing separation from 1.0m to 1.8m reducing risk by 44.95%, while decreasing to 0.5m increases risk by 323.62%. Virus variant biological variability produces substantial risk changes, with Omicron being most infectious.

It can be found that the infection risk mitigation from 300 CFM clean air delivery augmentation with light expiration is equivalent to the condition if the susceptible wear cloth masks. This equivalence highlights the limited leverage of HVAC-based interventions compared to simple behavioral measures. Source control through infector masking provides three to five times greater risk reduction than maximum tested ventilation augmentation, while social distancing from 1.0 m to 1.8 m provides approximately two times greater risk reduction than adding 300 CFM of clean air delivery.

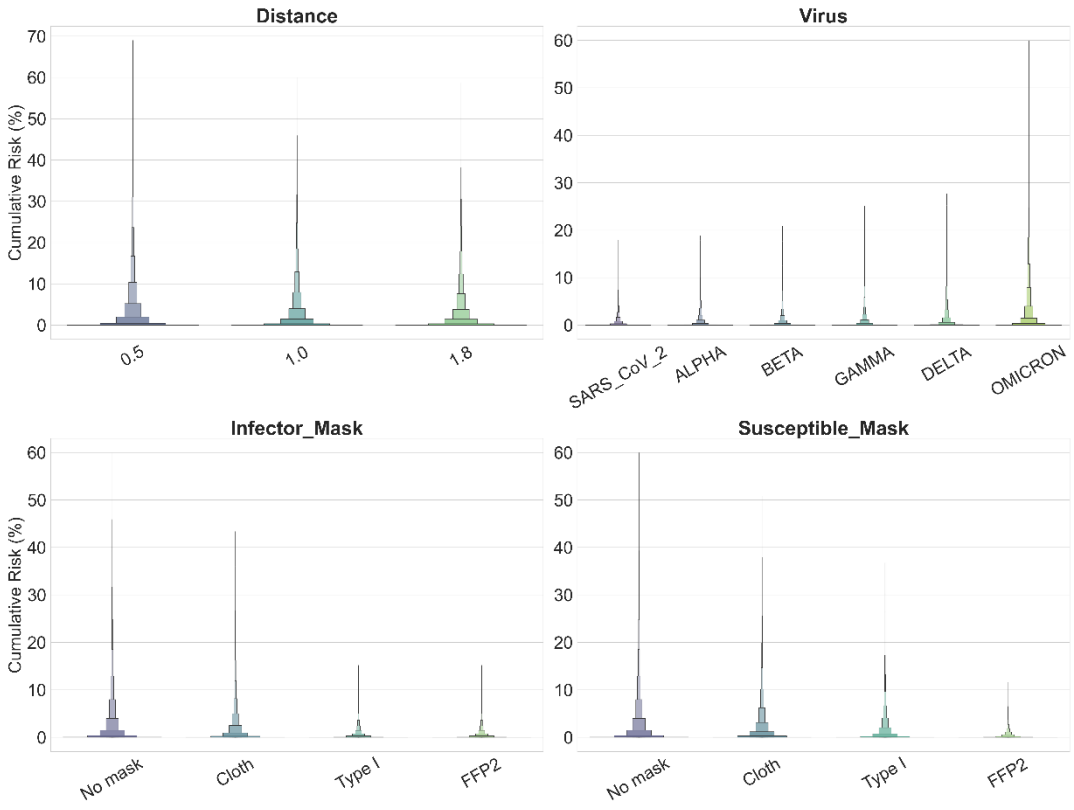


Figure 61. Infection risk boxplot of all samples with breathing as short-range expiration.

When speaking occurs during short-range contact, hierarchical effectiveness changes dramatically, with short-range transmission dominating the final cumulative infection risk (Table 31). Under these conditions, infector masking becomes overwhelmingly dominant (>95% risk reduction across all mask types in Table 33), while social distancing demonstrates strong effectiveness and susceptible masking remains beneficial though less impactful (Table 32). Virus variant biological variability produces substantial effects (Table 35), while clean air delivery variation produces negligible effects (Table 31).

The exceptional effectiveness of infector masks derives from a dual mechanism rather than filtration alone. First, masks filter emitted aerosols according to their outward efficiency. Second, and critically, masks disrupt the expiratory jet that maintains high concentration at close distances,

effectively eliminating the short-range pathway and forcing transmission through the long-range route where other controls can influence outcomes. This dual mechanism explains why even cloth masks provide substantial risk reduction far exceeding what filtration efficiency alone would predict.

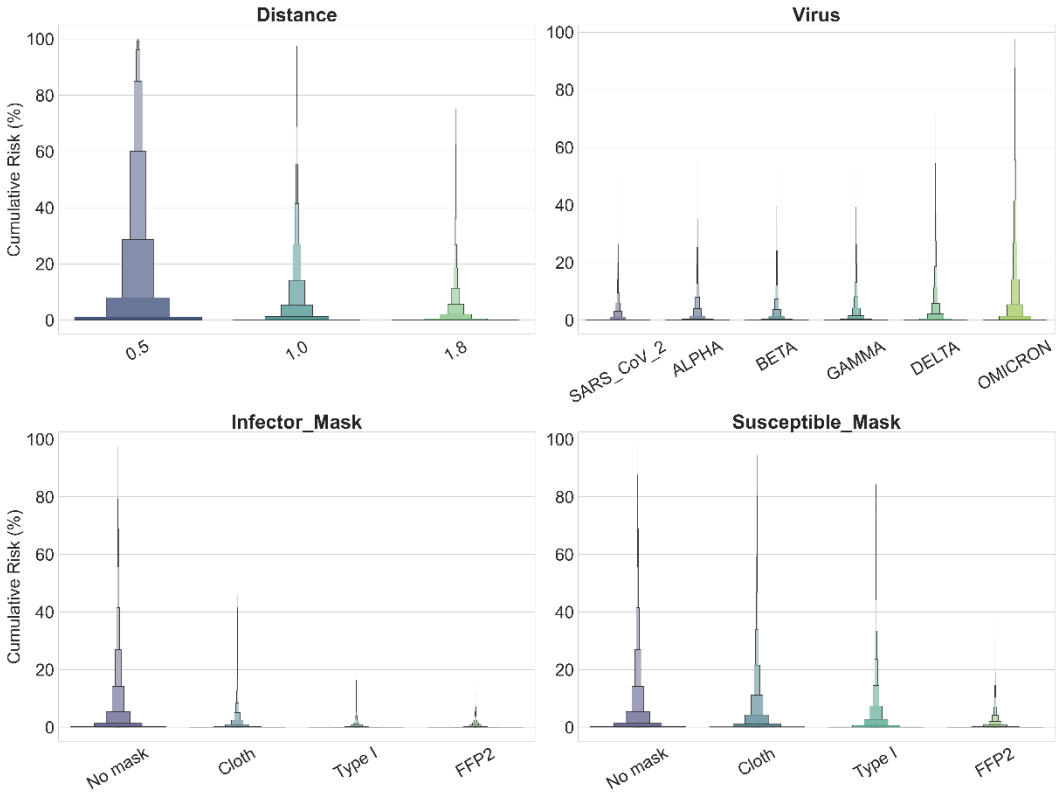


Figure 62. Infection risk boxplot of all samples with speaking as short-range expiration.

Table 31. Individual infection risk of different clean air augmentation and relative change (in the parathesis) compared to the baseline conditions.

Short-range expiration	-170 CFM	-80 CFM	0 CFM	+100 CFM	+300 CFM
Breathing	0.26%	0.24%	0.22% (-)	0.20%	0.19%

	(18.53%)	(7.06%)		(-7.27%)	(-16.34%)
Speaking	4.273% (0.35%)	4.264% (0.13%)	4.258% (-)	4.252% (-0.14%)	4.245% (-0.30%)

Table 32. Individual infection risk of different social distances and relative change (in the parathesis) compared to the baseline conditions.

Short-range expiration	0.5m	1.0m	1.8m
Breathing	0.95% (323.62%)	0.22% (-)	0.12% (-44.95%)
Speaking	12.01% (182.15%)	4.26% (-)	1.20% (-71.75%)

Table 33. Individual infection risk of different mask types for infectors and relative change (in the parathesis) compared to the baseline conditions.

Short-range expiration	No Mask	Cloth	Type I	FFP2
Breathing	0.22% (-)	0.06% (-70.31%)	0.02% (-91.34%)	0.02% (-91.34%)
Speaking	4.26% (-)	0.06% (-98.51%)	0.02% (-99.57%)	0.02 (-99.57%)

Table 34. Individual infection risk of different mask types for the susceptible and relative change (in the parathesis) compared to the baseline conditions.

Short-range expiration	No Mask	Cloth	Type I	FFP2
Breathing	0.22% (-)	0.18% (-21.89%)	0.11% (-49.12%)	0.03% (-86.08%)
Speaking	4.26% (-)	3.61% (-15.28%)	2.66% (-37.63%)	0.92 (-78.41%)

Table 35. Individual infection risk of different virus variants and relative change (in the parathesis) compared to the baseline conditions

Short-range expiration	Original	Alpha	Beta	Gamma	Delta	Omicron
Breathing	0.22% (-)	0.27%	0.26%	0.30%	0.42%	0.98%

		(21.07%)	(16.10%)	(33.68%)	(88.07%)	(337.16%)
Speaking	4.26% (-)	4.89% (14.95%)	4.84% (13.83%)	5.23% (22.98%)	6.26% (47.08%)	10.03% (135.76%)

7.7 Simulation II: Statistical Attribution of Mitigation Effects

Simulation II quantifies the marginal contribution of each factor to infection risk under controlled baseline conditions using three complementary analytical approaches. Results are presented separately for two scenarios that differ in transmission pathway: Scenario 1 includes both short-range and long-range transmission, while Scenario 2 includes only long-range transmission.

Scenario 1: No Mask for Infector with Short-Range Contact

When the infector does not wear a mask, short-range transmission during the close proximity contact period can occur alongside continuous long-range exposure. Under these conditions, all three analytical methods converge on consistent findings regarding the relative importance of different factors.

Generalized Linear Model Results

Table 36GLM results (Table 36) show all predictors significantly affect infection risk ($p < 0.05$, pseudo-$R^2 = 0.7609$), confirming each factor reliably influences outcomes when others are held constant.

Population expiration type shows the strongest effect, with speaking increasing infection odds by 186% due to higher aerosol emission rates. Virus variant demonstrates substantial variability: Omicron increases odds by 145% relative to the original strain, followed by Delta, Gamma, Alpha, and Beta, reflecting significant transmissibility differences across strains.

Among controllable measures, FFP2 masks reduce infection odds by 71%, surgical masks by 36%, and cloth masks by 14%. Each additional meter of social distance reduces odds by 49%, while each minute of close contact increases odds by 1.8%.

Ventilation shows statistically significant but modest effects: 1 ACH increase reduces infection odds by 13%. While reliable, this effect is small compared to other interventions—achieving the same risk reduction as a surgical mask would require approximately 2.5 ACH increase, necessitating doubling or tripling typical ventilation rates with corresponding energy penalties.

Table 36. GLM results for scenario 1: No Mask for Infector with Short-Range Contact.

	Coef β	Odd Change	Std Err	Z (Coef/Std)	P > |Z|	[0.025	0.975]
Intercept	-2.7377	-	0.060	45.877	0.000	-2.855	-2.621
Cloth mask for susceptible	-0.1415	-13.19%	0.029	-4.955	0.000	-0.198	-0.086
FFP2 mask for susceptible	-1.2510	-71.38%	0.039	-31.776	0.000	-1.328	-1.174
Type I mask for susceptible	-0.4385	-35.50%	0.031	-14.295	0.000	-0.499	-0.378
Alpha variant	0.1504	16.23%	0.045	3.378	0.001	0.063	0.238
Beta variant	0.1354	14.5%	0.045	3.031	0.002	0.048	0.223
Delta variant	0.3976	48.82%	0.042	9.359	0	0.314	0.481
Gamma variant	0.1978	21.87%	0.044	4.484	0	0.111	0.284
Omicron variant	0.8991	145.74%	0.039	22.805	0	0.822	0.976
Short range activity: standing	0.0535	5.5%	0.023	2.326	0.02	0.008	0.099
Short	0.7195	105.34%	0.024	29.84	0	0.672	0.767

range expiration: speaking							
Population activity: standing	0.0766	7.96%	0.023	3.33	0.001	0.031	0.122
Population expiration: speaking	1.0322	180.72%	0.025	40.635	0	0.982	1.082
Ventilation (ACH)	-0.1363	-12.74%	0.01	-14.099	0	-0.155	-0.117
Distance	-0.6947	-50.08%	0.028	-25.159	0	-0.749	-0.641
Short-range contact duration	0.0187	1.89%	0.001	15.865	0	0.016	0.021

XGBoost + SHAP Feature Importance

Figure 63 presents the SHAP feature importance ranking for Scenario 1. The XGBoost model achieves excellent predictive performance with $R^2 > 0.90$, confirming its ability to accurately predict infection risk across the diverse parameter combinations.

The SHAP analysis reveals that population expiration type is the dominant predictor, with the highest mean absolute SHAP values, substantially exceeding all other features. This confirms the GLM finding that whether occupants speak versus breath during exposure period is the single most important determinant of infection risk. Short-range expiration type ranks second, indicating that vocalization specifically during the 15-minute close-proximity event is nearly as critical as population-level expiration. The high importance of both expiration variables underscores that emission intensity, whether during sustained exposure or brief close contact, is the fundamental driver of risk variability.

Susceptible mask type ranks third, demonstrating that receptor protection provides substantial predictive values even when short-range transmission occurs. Virus variant rank fourth, reflecting

biological variability in transmissibility. Social distance and short-range contact duration occupy middle positions, indicating they meaningfully affect predictions but are less critical than expiration intensity and masking.

Ventilation ranks seventh out of nine features, placing it in the lower tier importance. While GLM showed ventilation has statistically significant marginal effect, SHAP analysis reveals it contributes relatively little to overall infection risk when all factors vary. This divergence indicates that ventilation has a reliable but small effect. It consistently reduces risk in the expected direction, but the magnitude is insufficient to substantially explain risk variability across scenarios.

The SHAP values distributions provide additional insights. For ventilation, the narrow distribution of SHAP values around zero and substantial overlap between low and high feature values confirms its limited impact – changing ventilation produces small infection risk changes regardless of other conditions.

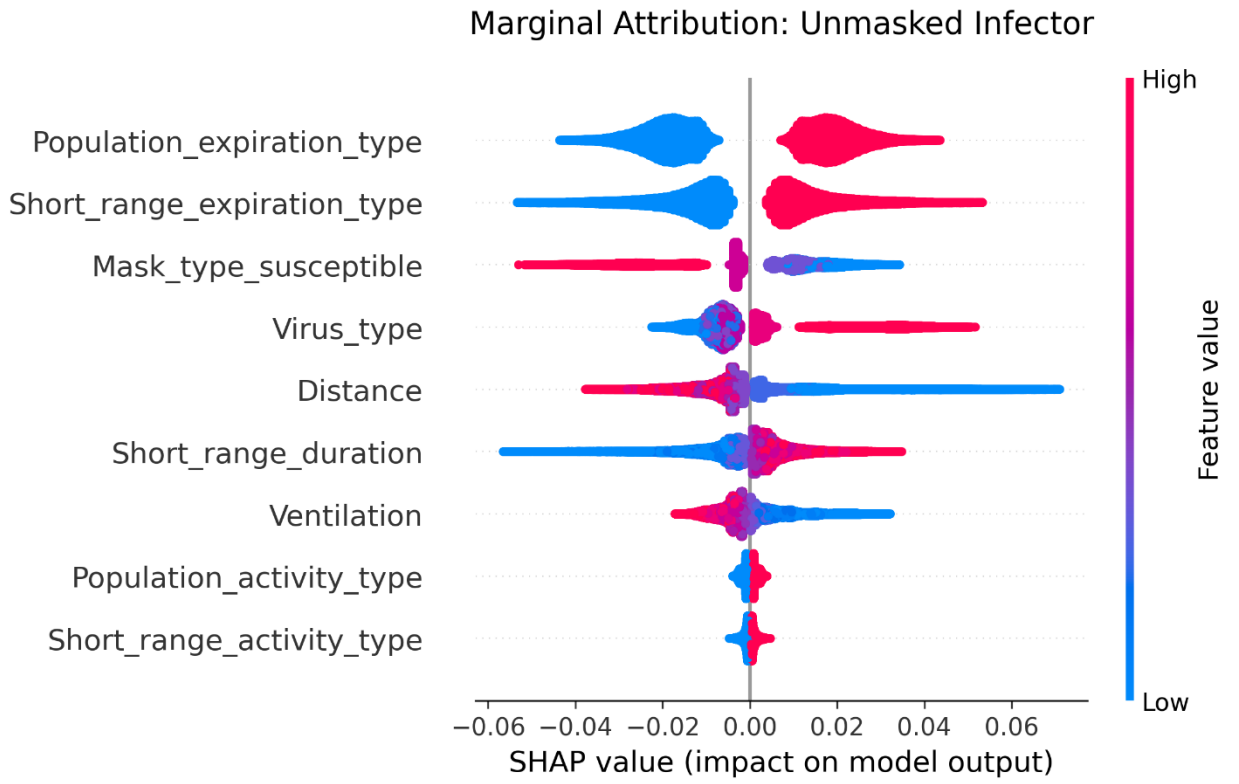


Figure 63. SHAP analysis for scenario 1: No Mask for Infector with Short-Range Contact.

Delta Sensitivity Analysis

Table 37 presents the Delta indices and first-order Sobol indices for all variables in Scenario 1. The Delta analysis provides a moment-independent assessment of how much variable affects the entire risk distribution, capturing beyond variance alone.

The Delta indices of population expiration type, short-range expiration type, virus variant are consistent with GLM and SHAP analysis. Ventilation demonstrates low-to-moderate Delta with low Sobol. The low Sobol index confirms that ventilation explains only a small fraction of variance in infection risk, consistent with GLM and SHAP findings. The Delta index, while comparable to

Sobol, remains in the lower tier among all variables, indicating that ventilation produces modes shifts in the overall risk distribution.

Figure 64 presents the Spearman rank correlation matrix for Scenario 1, revealing the structure of interactions between variables in the order of 10^{-5} (effectively zero), which remains negligible.

This finding indicates that variables act largely independently rather than synergistically or antagonistically. When one variable shifts infection risk in a particular direction, other variables' effects are essentially unaffected. The weak correlation structure has several important implications:

1. **Additive effects**: The combined effect of multiple interventions can be estimated by summing their individual contributions without accounting for interaction terms.
2. **Robust recommendations**: Each mitigation measure provides similar benefit regardless of whether other interventions are in place, simplifying implementation guidance.
3. **No hidden synergies**: Ventilation does not exhibit strong synergistic effects with masks, distancing, or other measures. Its modest contribution remains modest regardless of other factors.

The absence of strong correlations also confirms that the limited effectiveness of ventilation observed across all three methods is not due to antagonistic interactions with other variables that might mask its true potential. Rather, ventilation's modest contribution reflects its fundamental limited leverage for infection risk control when biological variability (virus variant, viral load) and behavioral factors (expiration type, masking) dominate risk outcomes.

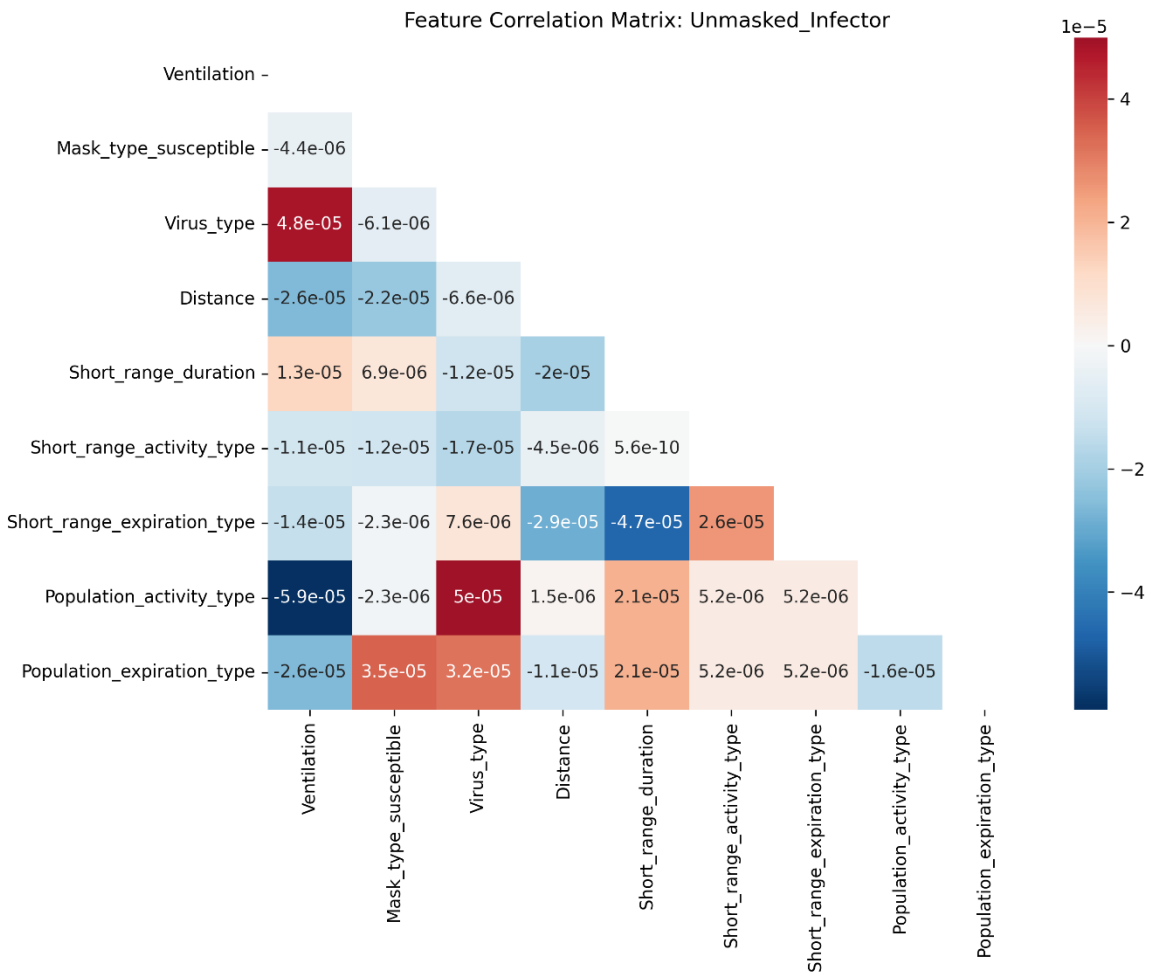


Figure 64. Spearman correlation for scenario 1: No Mask for Infector with Short-Range Contact.

Table 37. Delta analysis for scenario 1: No Mask for Infector with Short-Range Contact.

Feature	Delta (Total)	S1 (Direct)
Population expiration type	0.237969027	0.211899532
Mask type of susceptible	0.177949973	0.142356059
Short-range expiration type	0.127230332	0.109352275
Virus type	0.103467726	0.102273906
Ventilation	0.099117011	0.027131787

Distance	0.098731196	0.096835757
Short-range duration	0.074280954	0.030717331
Population activity type	0.057159343	0.00182637
Short range activity type	0.049336641	0.001009704

Scenario 2: Mask for Infector without Short-Range Contact

When the infector wears mask, the expiratory jet is disrupted and short-range transmission is eliminated, leaving only long-range exposure via well-mixed room air. This scenario represents conditions most favorable to ventilation effectiveness, as all transmission occurs through the pathway that room-level dilution can influence.

Generalized Linear Model Results

Table 38 presents the GLM regression results for Scenario. The model achieves exceptional fit with pseudo-R^2 of 0.9895, explaining 98.95% of deviance, which is substantially higher than Scenario 1. This indicates that infection risk is far more predictable when only long-range transmission occurs, as the complex fluid dynamics of expiratory jet are eliminated.

Population expiration type remains the dominant factor, with speaking increasing infection odds by 1863%. While speaking continues to dominate infection risk, the coefficient magnitude cannot be directly compared to Scenario 1 due to different baseline conditions. The critical finding is that expiration type remains the most influential variable regardless of transmission pathway. Virus variant demonstrates similar trends to Scenario 1.

Infector masking demonstrates source control effects, with both FFP2 and Type I masks reducing odds by 59%, while cloth masks reduce odds by 16%. Susceptible masking shows FFP2 reducing odds by 79%, Type I by 42%, and cloth by 16%. Population activity shows modest effects, increasing odds by 17%.

Ventilation demonstrates improved effectiveness, with each 1 ACH increase reducing infection odds by 25%. This represents approximately 1.9-fold improvement over Scenario 1's 13% per ACH effect, confirming that ventilation's leverage increases when short-range is eliminated. However, the effect remains modest compared to FFP2 susceptible masks (79% reduction) or speaking versus breathing (1863% increase).

Table 38. GLM results for scenario 2: Mask for Infector without Short-Range Contact.

	Coef β	Odd Change	Std Err	Z (Coef/Std)	P > |Z|	[0.025	0.975]
Intercept	-5.6418	-	0.074	-76.296	0	-5.787	-5.497
FFP2 mask for Infector	-0.8866	-58.79%	0.029	-30.618	0	-0.943	-0.83
Type I mask for Infector	-0.8865	-58.79%	0.029	-30.615	0	-0.943	-0.83
Cloth mask for susceptible	-0.1751	-16.06%	0.028	-6.194	0	-0.231	-0.12
FFP2 mask for susceptible	-1.5376	-78.51%	0.044	-34.93	0	-1.624	-1.451
Type I mask for susceptible	-0.5371	-41.56%	0.031	-17.242	0	-0.598	-0.476
Alpha variant	0.1863	20.48%	0.047	3.926	0	0.093	0.279
Beta variant	0.1676	18.25%	0.048	3.518	0	0.074	0.261
Delta variant	0.4926	63.66%	0.045	11.025	0	0.405	0.58
Gamma variant	0.2453	27.8%	0.047	5.234	0	0.153	0.337
Omicron variant	1.1137	204.56%	0.041	27.273	0	1.034	1.194
Population activity: standing	0.1575	17.06%	0.023	6.724	0	0.112	0.203
Population	2.9774	1863.67%	0.052	57.014	0	2.875	3.08

expiration: speaking							
Ventilation (ACH)	-0.2848	-24.78%	0.01	-28.285	0	-0.305	-0.265

XGBoost + SHAP Feature Importance

Figure 65 presents the SHAP feature importance ranking for Scenario 2. Population expiration type retains the highest importance, followed by infector mask type in second place, susceptible mask type in third, and virus variant in fourth. Ventilation ranks fifth, showing improved position compared to seventh in Scenario 1. It confirms that increased predictive importance of ventilation when only long-range transmission occurs, though it remains below expiration type, masking, and virus variant.

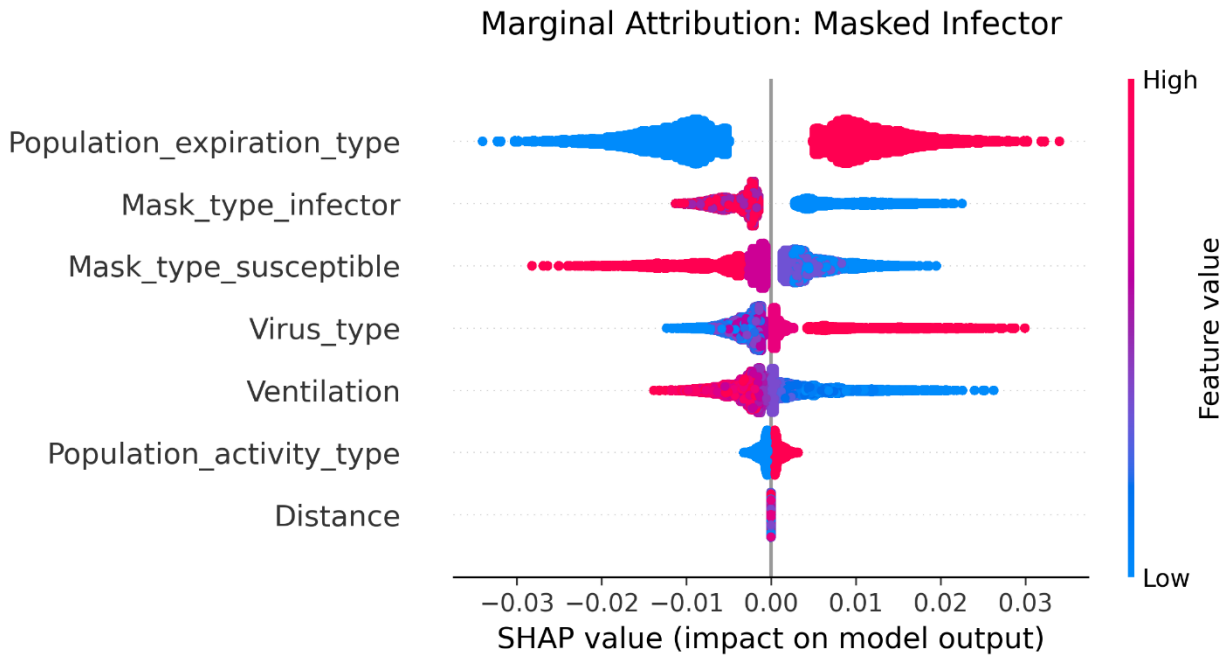


Figure 65. SHAP analysis for scenario 2: Mask for Infector without Short-Range Contact.

Delta Sensitivity Analysis

Table 39 presents the Delta indices and Sobol indices for Scenario 2. Population expiration exhibits the highest Delta and highest Sobol with nearly identical values indicating proportional effects on all distributional moments. Susceptible mask type shows moderate Delta with lower Sobol, suggesting masks shift mean risk substantially while producing smaller variance changes. Infector mask type shows similar patterns.

Ventilation demonstrates notably increased Delta and Sobol compared to Scenario, confirming greater distributional impact when short-range is absent. The pattern of $\delta > S_1$ (0.119 vs 0.054, ratio of 2.2:1) indicates that ventilation shifts the distribution location (mean risk) more than it changes the spread (variance), consistent with a variable that provides modest but consistent risk reduction across scenarios. However, ventilation's Delta index remains below population expiration (0.376), susceptible masks (0.154), and infector masks (0.127), confirming its secondary role even under favorable long-range-only conditions.

Virus variant ($\delta = 0.109$, $S_1 = 0.083$) and population activity type ($\delta = 0.090$, $S_1 = 0.003$) show moderate and low effects, respectively. Population activity's pattern of high δ but very low S_1 (ratio of 31:1) suggests it shifts mean risk without changing variance—likely due to small but consistent breathing rate differences between seated and standing.

Figure 66 presents the Spearman correlation matrix for Scenario 2. All correlations remain extremely weak (order of 10^{-6} to 10^{-5}), with the highest being 1.9×10^{-5} between population activity and distance. This confirms that variables continue to act independently in Scenario 2, indicating that ventilation's improved effectiveness is due to the changed transmission pathway (long-range only) rather than synergistic interactions with other variables.

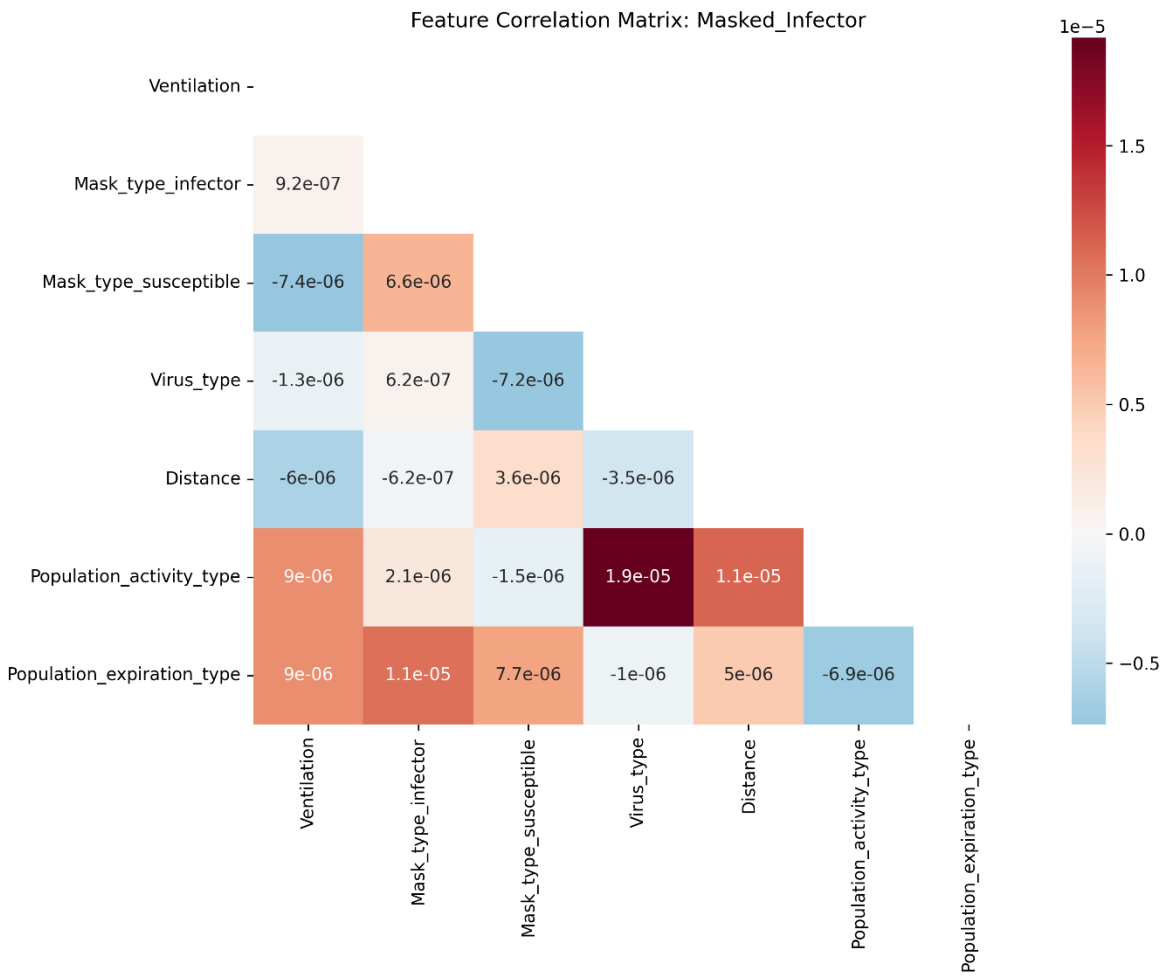


Figure 66. Spearman correlation for scenario 2: Mask for Infector without Short-Range Contact.

Table 39. Delta analysis for scenario 2: Mask for Infector without Short-Range Contact.

Feature	Delta (Total)	S1 (Direct)
Population expiration type	0.375812	0.372545
Mask type susceptible	0.154465	0.089158
Mask type infector	0.127105	0.089427
Ventilation	0.118604	0.053747
Virus type	0.108541	0.083023
Population activity type	0.08995	0.002859

Cross-Scenario Comparison

Ventilation improvement with transmission pathway change: Ventilation shows consistent improvement across all three methods when moving from Scenario 1 to Scenario 2. These improvements confirm that ventilation has greater leverage when all transmission occurs through the well-mixed pathway. However, ventilation consistently ranks below population expiration, masking, and virus variant across all methods in both scenarios.

Persistent dominance of expiration and masking: Population expiration maintains top position in GLM, SHAP, and Delta across both scenarios. Making consistently occupies top-tier positions. This demonstrates that emission intensity and protective equipment are fundamental drivers of risk regardless of transmission pathway.

Improved predictability without short-range: Pseudo-R^2 increases from Scenario 1 to 2 points improvement in model fit. The near perfect fit in Scenario 2 suggests that when short-range jet dynamics are eliminated, the sample variables explain virtually all infection risk variability.

Independence of effects: Weak Spearman correlations (order of 10^{-5} to 10^{-6}) in both scenarios indicate that within each scenario, variables act independently. In Scenario 1, this confirms that ventilation, susceptible masking, distance, and other factors provide additive effects without strong synergies or antagonisms. In Scenario 2, the weak correlations indicate that among the variables that remain active (ventilation, both mask types, virus variant, activity), their effects are also independent.

The improved effectiveness of ventilation in Scenario 2 compared to Scenario 1 is not due to synergistic interactions with other variables (as evidenced by weak correlations) but rather reflects the structural difference between scenarios: in Scenario 2, infector masking eliminates short-range transmission entirely by disrupting the expiratory jet, forcing all exposure to occur through the

long-range pathway where ventilation operates. This is a scenario design feature rather than a variable interaction. When the dominant short-range pathway is removed by design (through infector masking), the remaining long-range pathway, where ventilation has its primary mechanism of action, becomes the sole route of transmission, allowing ventilation's effect to manifest more fully. However, even under these favorable conditions, ventilation's effect remains modest compared to other factors that continue to vary within the scenario.

Convergence Across Methods: Ventilation as Supportive Measure

All three analytical approaches converge on consistent conclusions across both scenarios: when the presence of an infector is known and targeted behavioral interventions can be deployed, ventilation is a supportive mitigation strategy rather than a dominant control. While ventilation demonstrates statistically significant effects ($p < 0.001$ in both scenarios), improved effectiveness when short-range is eliminated (GLM coefficient approximately doubles, SHAP ranking improves from 7th to 5th, Delta increases), and measurable distributional impacts (Delta = 0.119 in Scenario 2), it consistently ranks below population expiration, masking, and virus variant. The weak correlations across all variable pairs in both scenarios confirm that ventilation works independently, providing modest additive risk reduction without strong synergies or antagonisms with other measures. However, it is important to recognize that the above comparison assumes that the presence of an infector is known, which enables the deployment of targeted behavioral interventions such as masking, social distancing, and reduced vocalization. In practice, building occupants and operators often do not know whether an infector is present. Under such conditions of unknown infection status, ventilation through the HVAC system becomes the only continuously operable mitigation strategy, as it does not require awareness of any specific infector's presence. This distinction highlights that ventilation's role is conditional on the operational context: when

infector presence is unknown, ventilation is the primary available defense; when infector presence is known and behavioral interventions are activated, ventilation becomes a supportive complement to more effective targeted measures.

In summary, the results of both simulations converge on a consistent conclusion: when the presence of an infector is known and targeted behavioral countermeasures can be activated, HVAC-based clean air delivery is a supportive rather than dominant strategy for airborne infection risk mitigation. Under realistic exposure conditions in Simulation I, ventilation provides statistically significant but modest risk reduction. In the controlled attribution analysis of Simulation II, all three analytical methods — generalized linear modeling, XGBoost with SHAP values, and Delta sensitivity analysis — consistently rank ventilation below behavioral and biological factors including respiratory activity type, masking, virus variant, and social distancing. Even under the most favorable conditions for ventilation (Scenario 2, where infector masking eliminates short-range transmission and all exposure occurs through the well-mixed pathway), ventilation's effectiveness approximately doubles but remains substantially less impactful than masking or respiratory activity patterns. Critically, the weak Spearman correlations across all variables confirm that these effects are largely additive and independent, meaning ventilation's modest contribution is not masked by antagonistic interactions with other factors but rather reflects its fundamental limited leverage over infection risk when biological variability and behavioral factors vary simultaneously. Nevertheless, in the realistic and common scenario where the presence of an infector is unknown to building occupants and operators, ventilation is the only continuously available mitigation measure that does not require identification of an infector. Under such conditions, maintaining adequate ventilation through the HVAC system represents the primary line of defense against airborne infection risk. These findings carry a direct implication

for advanced building control design: HVAC systems should maintain adequate ventilation as a baseline measure to provide continuous protection under conditions of unknown infection status, while recognizing that once infector presence is identified and behavioral interventions become deployable, ventilation's relative contribution becomes supportive rather than dominant.

Chapter 8. Conclusion

This chapter concludes the dissertation by synthesizing the findings across all preceding chapters. Section 8.1 summarizes the key results from single-zone modeling and control (Chapters 3–4), multi-zone modeling and control (Chapters 5–6), and the energy-health nexus exploration (Chapter 7). Section 8.2 revisits the six contributions of this dissertation in light of the results obtained. Section 8.3 discusses the limitations of the current work and outlines directions for future research.

8.1 Summary of Dissertation

This dissertation investigated physics-informed machine learning models and advanced control strategies for building HVAC systems, progressing from single-zone simulation to scalable multi-zone real-building deployment, with a complementary exploration of the energy-health nexus. The research was motivated by three interconnected barriers to the widespread adoption of advanced building controls: the lack of scalable and control-oriented dynamic models, the absence of consensus on control strategy selection, and the gap between single-zone research and multi-zone deployment.

For single-zone modeling and control, the physics-consistent input convex neural network (PCICNN) was developed as a control-oriented dynamic model that embeds temporal causality through partial connectivity, ensures input convexity through non-negative weight constraints, and enforces physical consistency through both architecture design and loss function augmentation. The PCICNN achieved prediction accuracy within sensor error bounds (MAPE of approximately 1% for temperature and 5% for CO2) while maintaining physically consistent responses to perturbed inputs, validating its suitability for advanced control. The subsequent long-term real-building comparison of HDDPC, DPC, and RL revealed that RL provided the most robust and consistent savings (up to 48%) across varying outdoor temperatures, benefiting from its model-free deployment and ability to implicitly learn the coupling between room-level and system-level dynamics. HDDPC achieved the highest peak energy savings (over 50%) but was vulnerable to information gaps between its hierarchical control levels and non-optimal decisions from online optimization. DPC offered the fastest training and inference but delivered the lowest overall savings (30.6%), with the most variable performance. The dual-phase PIML model benchmarking further confirmed that PCICNN is the most control-oriented architecture among representative

PIML models, with its input convexity and non-recurrent structure providing advantages for both optimization-based and agent-based control paradigms.

For multi-zone modeling and control, the physics-consistent graph neural network (PCGNN) was developed to overcome the scalability limitations encountered when extending PCICNN to multiple zones through manual adjacency-matrix connections. The PCGNN replaces neuron-level manual connections with group-shared multi-scale causal convolutions and heat diffusion graph layers, achieving scalable and physically interpretable multi-zone temperature prediction. The κ-neighborhood truncated critic design for multi-agent soft actor-critic exploits the exponential decay of thermal coupling with distance, reducing critic input dimensionality from scaling with the total number of zones to scaling with the local neighborhood size. Combined with model-based training using the PCGNN, the framework was validated at two scales: a 6-zone simulation on a commercial building in Boston and an 18-zone real-building deployment at the Syracuse Center of Excellence, achieving 35–70% energy savings while maintaining thermal comfort. This represents one of the largest real-building MARL deployments for HVAC control in literature.

As a complementary exploration, the WHO-model-based feasibility assessment demonstrated that the role of HVAC-based clean air delivery in airborne infection risk mitigation is conditional on the operational context. When the presence of an infector is known and targeted behavioral interventions such as masking and social distancing can be deployed, ventilation serves a supportive rather than dominant role, as three complementary analytical methods consistently ranked it below behavioral factors across both transmission scenarios examined. However, when the presence of an infector is unknown—the more common condition in routine building operation—ventilation through the HVAC system is the only continuously available mitigation strategy, making it the primary line of defense against airborne infection risk under uncertainty.

8.2 Contributions

The specific contributions of this dissertation, as outlined in Section 2.5 and substantiated by the results in Chapters 4, 6, and 7, are as follows:

1. A control-oriented PIML architecture for single-zone building dynamics. The PCICNN embeds temporal causality through partial connectivity inspired by the lower-triangular structure of multi-step state-space models, ensures input convexity through non-negative weight constraints, and enforces physical consistency through architecture design and loss function augmentation. The model achieved prediction accuracy within sensor error bounds while providing physically consistent multi-step predictions without recurrence, serving as the foundation for all three advanced control strategies and for the subsequent multi-zone extension.

2. The first long-term real-building comparison of MPC, RL, and DPC. The five-month deployment of HDDPC, DPC, and RL in the same real building using the same PCICNN dynamic model provides the first experiment-based, head-to-head comparison under consistent conditions. The results reveal practical trade-offs that simulation-based comparisons cannot capture, including the impact of information gaps in hierarchical MPC, the sensitivity of RL performance to training episode length, and the variability of DPC across operating conditions.

3. The first dual-phase benchmarking framework for PIML models. The systematic evaluation framework jointly assesses PIML models in both open-loop prediction and closed-loop real-building control, benchmarking three representative architectures (PCNN, PCICNN, and ModNN). The results establish that prediction accuracy alone is insufficient

for model selection in control applications, and that PCICNN's input convexity and non-recurrent structure provide distinct advantages in closed-loop settings.

4. A scalable PIML architecture for multi-zone building dynamics. The PCGNN replaces manual adjacency-matrix neuron connections with group-shared multi-scale causal convolutions and heat diffusion graph mechanisms, enabling scalable multi-zone temperature prediction beyond the 6-zone limit of prior PIML models. The causal convolutional structure is mathematically equivalent to the lower-triangular Toeplitz matrix of multi-step state-space models but is significantly more parameter-efficient.

5. A scalable model-based multi-agent RL framework with neighborhood-truncated critics. The κ-neighborhood truncated critic design reduces critic input dimensionality from scaling with the total number of zones to scaling with the local neighborhood size, exploiting the exponential decay of thermal coupling with distance. Combined with model-based training using the PCGNN, the framework achieves 35–70% energy savings across both simulation and real-building deployment at up to 18 zones, representing one of the largest real-building MARL deployments for HVAC control.

6. Exploratory assessment of HVAC's role in airborne infection risk mitigation. As a complementary exploration, WHO's comprehensive airborne infection risk model is applied to real measured indoor environment data through Monte Carlo simulation, generalized linear modeling, and machine learning-based sensitivity analysis. The study demonstrates that when the presence of an infector is known, behavioral interventions such as masking outperform HVAC-based clean air delivery by factors of 3–10. However, when

infector presence is unknown, ventilation is the only continuously available mitigation measure, serving as the primary line of defense under conditions of uncertainty.

8.3 Limitations and Future Work

While this dissertation advances the state of the art in physics-informed modeling and advanced control for building HVAC systems, several limitations and opportunities for future research are identified.

First, the PCGNN has been validated for up to 18 thermally coupled zones, but its performance for significantly larger buildings with hundreds of zones remains to be demonstrated. The computational scaling of both model training and MARL coordination at that scale warrants further investigation. Techniques such as hierarchical MARL, where agents are organized at multiple spatial scales, may be necessary to extend the framework to campus-level or district-level applications.

Second, the real-building deployments in both single-zone and multi-zone studies revealed that infrastructure reliability — particularly BACnet communication, cloud-based sensor services, and hardware integrity — is often the binding constraint on sustained advanced control operations. Future work should develop robust fault detection and fallback mechanisms that enable graceful degradation rather than complete control failure when sensor or communication faults occur.

Third, the transferability of trained PCGNN models and MARL policies across buildings with different layouts and HVAC configurations has not been explored. Each new building currently requires data collection and model retraining. Transfer learning and domain adaptation techniques

could significantly reduce the data requirements and deployment time for new buildings, which is essential for commercial scalability.

Fourth, while this dissertation established that airborne infection risk need not be a primary HVAC control objective, the integration of other health-related metrics — such as long-term exposure to particulate matter, volatile organic compounds, or thermal stress indices — may warrant investigation as complementary objectives that do not carry the same energy penalty as aggressive ventilation strategies.

Finally, the rapid advancement of large language models and foundation models presents opportunities for building energy management. Potential applications include natural language interfaces for building operators, automated fault diagnosis from sensor data streams, and LLM-assisted control policy tuning. Integrating such capabilities into the advanced control pipeline could lower the barrier to adoption for building operators without specialized technical expertise, accelerating the transition from research demonstrations to widespread deployment.

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EDUCATION

Ph.D. in Mechanical and Aerospace Engineering August 2021 – May 2026 (Expected)
Syracuse University *Syracuse, NY*

- Specialization: Reinforcement Learning, Model Predictive Control, Differentiable Predictive Control, Physics-Informed Machine Learning, Quantum Computing
- Advisor: Dr. Bing Dong

Visiting Fellow in Engineering and Applied Sciences September 2025 – December 2025
Harvard University *Boston, MA*

- Specialization: Scalable Reinforcement Learning, Physics-Informed Machine Learning, Building Control Implementation
- Advisor: Dr. Bing Dong, Dr. Na Li

M.S. in Mechanical and Aerospace Engineering August 2019 – May 2021
Syracuse University *Syracuse, NY*

- Specialization: Thermal and Fluid Dynamics, Control and Optimization, Advanced Heat Transfer
- Advisor: Dr. Bing Dong

B.E. in Mechanical Engineering September 2015 – June 2019
Zhengzhou University *Zhengzhou, China*

- Specialization: Mechanical Design, Manufacturing, Control, and Optimization

ACADEMIC DISTINCTIONS

- 2019 - Syracuse University College of Engineering & Computer Science Graduate Student Grant
- 2021 - Syracuse University Graduate Fellowship
- 2023 - Student Features, Center of Excellence (CoE), *Syracuse University* [LINK]
- 2024 – ASHRAE Graduate Student Grand-in-Aid Award [LINK]

RESEARCH AREA

- Model Predictive Control
- Physics-Informed Machine Learning for Building Modelling
- Quantum Computing for Future Building Control
- Real-Building Control Deployment
- Reinforcement Learning
- Machine Learning in Building Control
- Building Sensing and Energy monitoring
- Differentiable Predictive Control
- Large Language Model for buildings

TECHNICAL SKILLS

- *Programming languages and tools:* Python, MATLAB, R
- *Skills:* Building Sensing and Simulation, Control and Optimization, Machine Learning, Quantum Computing, Large-Language Model

RESEARCH PROJECTS

ARPA-E: Quantification of HVAC Energy Savings for Occupancy Sensing in Buildings through An Innovative

Testing Methodology

Advanced research Projects Agency - Energy, 2021 – 2022

- Setup sensing networks for commercial buildings
- Assist in developing frameworks of occupant-centric control for target buildings
- Mentor new Ph.D. student on the project

Honeywell Syracuse Next Generation IAQ Research

Honeywell International Inc., 2021-2024

- Design and develop database for building management system operation
- Develop both physics-based model and physics-informed machine learning model for campus office buildings
- Develop advanced control framework for the buildings to balance indoor environmental quality and energy efficiency
- Implement model predictive control, differentiable predictive control, and reinforcement learning in real buildings to evaluate and compare long-term performances of different advanced controls
- Develop and conduct experiments to test and analyze human's productivity under different indoor environmental qualities

CoE Whole Building Testbed Research

Syracuse Center of Excellence in Environmental Energy Systems, 2023-present

- Design and develop sensing frameworks for detailed building energy usage
- Design and develop databases for building management system operation, energy usage, and indoor environmental quality
- Modelling multi-zone office floors by physics-informed machine learning
- Design, develop, and implement reinforcement learning to control radiant cooling/heating system in the building

NYSERDA Intelligent Buildings: Optimizing Across Multiple Siloed Energy Assets

New York State Energy Research and Development Authority, 2024-present

- Design and develop sensing frameworks for detailed building energy usage
- Design and develop databases for building management system operation, energy usage, and indoor environmental quality
- Modelling multi-zone office floors by physics-informed machine learning
- Design, develop, and implement advanced control for whole building with multiple energy assets including TES, BES, HVAC, and PV.
- Mentor new Ph.D. student on the project

PROFESSIONAL EXPERIENCE

Research Assistant August 2020 – Present

BEST Lab, Syracuse University *Syracuse, NY*

- Oversaw research project timelines and authored comprehensive reports
- Disseminated findings through scholarly journal publications
- Enhanced building models by physics-informed machine learning
- Created and developed advanced HVAC control algorithms based on model predictive control and reinforcement learning for indoor environmental quality and energy efficiency
- Assisted in developing large-scale model predictive control for rooftop units using quantum computing
- Assisted in developing large-scale control framework for EV-to-Grid-to-Building control for flexibility using quantum computing
- Assisted in developing control framework for EV charging behavior and customized electricity price plan for households to improve grid flexibility, profit, and carbon emissions
- Collaborated with global researchers on personalized environmental control system (IEA EBC Annex

87 2023-present)
- Developed various advanced data visualization techniques to present research findings
- Mentor new Ph.D. students and visiting students
- Developing LLM-assisted advanced building control frameworks

Co-Lecturer Jan – May 2022 & Aug – Dec 2023
Dept. of Mechanical and Aerospace Engineering, Syracuse University *Syracuse, NY*
- Building modelling by thermal resistance and capacitance networks
- Model predictive control for building energy systems
- Convex optimization
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Research Engineer Intern May – Aug 2025
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- VOLTTRON-HomeAssistant integration
- Non-intrusive load monitoring development

PUBLICATION

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Presentations

A Predictive Control Strategy to Maximize Energy Savings While Maintaining Indoor Air Quality in Commercial Buildings October 2022

The 5th International Conference on Building Energy and Environment *Montreal, Quebec, Canada and Online*

Development of a data-driven predictive control based on a novel physics-informed neural network

September 2023

Building Simulation 2023: 18th Conference of IBPSA *Shanghai, China and Online*

Building HVAC Optimal Control: A Practical Reinforcement Learning Framework Based on Physics-Informed Machine Learning

February 2025

2025 ASHRAE Winter Conference Orlando, Florida and In Person

Seminar 58: AI Enabled Intelligent Building Simulation and Operation: A Long-Term Implementation of AI-Driven Control in an Office Building

June 2025

2025 ASHRAE Annual Conference Arizona, Phoenix and In Person

SERVICE

Invited Reviewer, *Journal of International Journal of Ventilation* 2021

Invited Reviewer, *Journal of Building and Environment* 2023~2025

Invited Reviewer, *Journal of Building Simulation* 2023~2025

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